rag_evaluation

February 7, 2026

1 Evaluación del Sistema RAG

Este notebook evalúa el sistema RAG (Retrieval-Augmented Generation) del agente periodístico.

1.1 Métricas a evaluar:

1.1.1 Retrieval (Recuperación)

- **Precision@K**: De los K documentos recuperados, ¿cuántos son relevantes?
- **Hit Rate**: ¿Al menos un documento es relevante?
- **MRR (Mean Reciprocal Rank)**: ¿En qué posición está el primer resultado relevante?

1.1.2 Generation (Generación)

- **Faithfulness**: ¿La respuesta está basada en los documentos? (no alucina)
- **Answer Relevancy**: ¿La respuesta es relevante a la pregunta?
- **Citation Accuracy**: ¿Las citas [1], [2] son correctas?

[1]:
```
# Imports
import sys
sys.path.insert(0, '..')

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from datetime import datetime
from typing import Optional

# Nuestros módulos
from src.storage.database import get_db
from src.storage.repositories.article import ArticleRepository
from src.rag.engine import RAGEngine
from src.processing.embeddings import get_embedding_generator
from src.llm.factory import get_available_client

# Configuración de visualización
plt.style.use('seaborn-v0_8-whitegrid')
pd.set_option('display.max_colwidth', 100)
```

```
print("  Imports completados")
```

```
Imports completados
```

1.2 1. Estado de la Base de Datos

[2]:
```
# Verificar estado de la DB
from sqlalchemy import func
from src.storage.models import Article, ProcessedArticle, AgentSession

with get_db() as db:
    total_articles = db.query(func.count(Article.id)).scalar()
    with_embeddings = db.query(func.count(Article.id)).filter(Article.embedding.
 ↪isnot(None)).scalar()
    processed = db.query(func.count(ProcessedArticle.id)).scalar()
    sessions = db.query(func.count(AgentSession.id)).scalar()

    # Por categoría
    categories = db.query(
        Article.category,
        func.count(Article.id)
    ).group_by(Article.category).all()

print("=" * 50)
print("ESTADO DE LA BASE DE DATOS")
print("=" * 50)
print(f"Total artículos:        {total_articles}")
print(f"Con embeddings:         {with_embeddings}")
print(f"Procesados (resumen):   {processed}")
print(f"Sesiones del agente:    {sessions}")
print()
print("Por categoría:")
for cat, count in sorted(categories, key=lambda x: x[1], reverse=True):
    print(f"  {cat or 'Sin categoría'}: {count}")
```

```
==================================================
ESTADO DE LA BASE DE DATOS
==================================================
Total artículos:        1688
Con embeddings:         1688
Procesados (resumen):   48
Sesiones del agente:    23

Por categoría:
  sports: 295
  science: 271
  general: 264
```

```
technology: 252
health: 244
business: 174
entertainment: 78
Sin categoría: 46
world: 34
nation: 24
politics: 4
Otras Ciudades: 1
Medellín: 1
```

1.3 2. Dataset de Evaluación

Creamos un conjunto de preguntas de prueba para evaluar el sistema.

[3]:
```
# Dataset de evaluación AMPLIADO
# 25 preguntas cubriendo diferentes categorías y niveles de especificidad

EVALUATION_QUESTIONS = [
    # === ECONOMÍA / BUSINESS (6 preguntas) ===
    {
        "question": "¿Qué está pasando con la economía?",
        "category": "business",
        "expected_topics": ["economía", "PIB", "inflación", "mercado"],
    },
    {
        "question": "¿Cuáles son las perspectivas económicas para 2026?",
        "category": "business",
        "expected_topics": ["economía", "crecimiento", "proyección", "2026"],
    },
    {
        "question": "¿Cómo está la inflación actualmente?",
        "category": "business",
        "expected_topics": ["inflación", "precios", "aumento", "porcentaje"],
    },
    {
        "question": "¿Qué está pasando con los salarios?",
        "category": "business",
        "expected_topics": ["salario", "aumento", "trabajador", "empresa"],
    },
    {
        "question": "¿Hay noticias sobre inversiones?",
        "category": "business",
        "expected_topics": ["inversión", "capital", "mercado", "fondo"],
    },
    {
        "question": "¿Qué dicen los analistas económicos?",
        "category": "business",
```

```
        "expected_topics": ["analista", "economía", "expectativa",␣
↪"pronóstico"],
    },

    # === TECNOLOGÍA (5 preguntas) ===
    {
        "question": "¿Cuáles son las últimas noticias de tecnología?",
        "category": "technology",
        "expected_topics": ["tecnología", "inteligencia artificial",␣
↪"software", "apps"],
    },
    {
        "question": "¿Qué avances hay en inteligencia artificial?",
        "category": "technology",
        "expected_topics": ["inteligencia artificial", "IA", "modelo",␣
↪"aprendizaje"],
    },
    {
        "question": "¿Hay noticias sobre redes sociales?",
        "category": "technology",
        "expected_topics": ["red social", "usuario", "plataforma", "app"],
    },
    {
        "question": "¿Qué está pasando con las empresas tecnológicas?",
        "category": "technology",
        "expected_topics": ["empresa", "tecnología", "startup", "innovación"],
    },
    {
        "question": "¿Hay noticias sobre ciberseguridad?",
        "category": "technology",
        "expected_topics": ["seguridad", "hacker", "datos", "protección"],
    },

    # === SALUD (4 preguntas) ===
    {
        "question": "¿Qué noticias hay sobre salud?",
        "category": "health",
        "expected_topics": ["salud", "medicina", "hospital", "enfermedad"],
    },
    {
        "question": "¿Hay noticias sobre vacunas?",
        "category": "health",
        "expected_topics": ["vacuna", "inmunización", "dosis", "campaña"],
    },
    {
        "question": "¿Qué dicen los expertos sobre salud mental?",
        "category": "health",
```

```
        "expected_topics": ["salud mental", "depresión", "ansiedad",␣
↪"bienestar"],
    },
    {
        "question": "¿Hay avances médicos recientes?",
        "category": "health",
        "expected_topics": ["médico", "tratamiento", "estudio",␣
↪"investigación"],
    },

    # === DEPORTES (4 preguntas) ===
    {
        "question": "¿Qué está pasando en el mundo del deporte?",
        "category": "sports",
        "expected_topics": ["deporte", "fútbol", "jugador", "equipo"],
    },
    {
        "question": "¿Cuáles son los resultados deportivos recientes?",
        "category": "sports",
        "expected_topics": ["resultado", "partido", "victoria", "derrota"],
    },
    {
        "question": "¿Hay noticias de fútbol?",
        "category": "sports",
        "expected_topics": ["fútbol", "gol", "equipo", "liga"],
    },
    {
        "question": "¿Qué fichajes o traspasos hay en el fútbol?",
        "category": "sports",
        "expected_topics": ["fichaje", "traspaso", "jugador", "club"],
    },

    # === CIENCIA (3 preguntas) ===
    {
        "question": "¿Hay noticias sobre ciencia?",
        "category": "science",
        "expected_topics": ["ciencia", "investigación", "estudio",␣
↪"descubrimiento"],
    },
    {
        "question": "¿Qué descubrimientos científicos hay?",
        "category": "science",
        "expected_topics": ["descubrimiento", "científico", "estudio",␣
↪"hallazgo"],
    },
    {
        "question": "¿Hay noticias sobre el espacio o astronomía?",
```

```
        "category": "science",
        "expected_topics": ["espacio", "NASA", "planeta", "universo"],
    },

    # === ENTRETENIMIENTO (2 preguntas) ===
    {
        "question": "¿Qué noticias hay de entretenimiento?",
        "category": "entertainment",
        "expected_topics": ["entretenimiento", "película", "serie", "música"],
    },
    {
        "question": "¿Hay estrenos de cine o series?",
        "category": "entertainment",
        "expected_topics": ["estreno", "película", "serie", "cine"],
    },

    # === MUNDO / GENERAL (1 pregunta) ===
    {
        "question": "¿Cuáles son las noticias internacionales más importantes?",
        "category": "world",
        "expected_topics": ["internacional", "país", "gobierno", "mundo"],
    },
]

print(f"Dataset de evaluación: {len(EVALUATION_QUESTIONS)} preguntas")
print("\nDistribución por categoría:")
from collections import Counter
cat_counts = Counter(q['category'] for q in EVALUATION_QUESTIONS)
for cat, count in cat_counts.most_common():
    print(f"  {cat}: {count} preguntas")
```

```
Dataset de evaluación: 25 preguntas

Distribución por categoría:
  business: 6 preguntas
  technology: 5 preguntas
  health: 4 preguntas
  sports: 4 preguntas
  science: 3 preguntas
  entertainment: 2 preguntas
  world: 1 preguntas
```

1.4 3. Métricas de Retrieval

[4]:
```
def evaluate_retrieval(
    question: str,
    expected_category: str,
    k: int = 5,
```

```python
    threshold: float = 0.5
) -> dict:
    """
    Evalúa la calidad del retrieval para una pregunta.

    Returns:
        dict con métricas de retrieval
    """
    with get_db() as db:
        engine = RAGEngine(db, use_reranking=False)

        # Obtener documentos recuperados
        articles = engine.get_context_only(
            question=question,
            max_sources=k,
        )

        if not articles:
            return {
                "question": question,
                "k": k,
                "retrieved": 0,
                "relevant": 0,
                "precision_at_k": 0.0,
                "hit_rate": 0.0,
                "mrr": 0.0,
                "avg_similarity": 0.0,
                "similarities": [],
                "categories": [],
            }

        # Calcular métricas
        similarities = [score for _, score in articles]
        categories = [art.category for art, _ in articles]

        # Un documento es "relevante" si:
        # 1. Su categoría coincide con la esperada, O
        # 2. Su similarity score es > threshold
        relevant_count = sum(
            1 for art, score in articles
            if art.category == expected_category or score > threshold
        )

        # Precision@K
        precision_at_k = relevant_count / len(articles)

        # Hit Rate (al menos uno relevante)
```

```
        hit_rate = 1.0 if relevant_count > 0 else 0.0

        # MRR (Mean Reciprocal Rank)
        mrr = 0.0
        for i, (art, score) in enumerate(articles, 1):
            if art.category == expected_category or score > threshold:
                mrr = 1.0 / i
                break

        return {
            "question": question,
            "k": k,
            "retrieved": len(articles),
            "relevant": relevant_count,
            "precision_at_k": precision_at_k,
            "hit_rate": hit_rate,
            "mrr": mrr,
            "avg_similarity": np.mean(similarities),
            "similarities": similarities,
            "categories": categories,
        }

print("  Función de evaluación de retrieval definida")
```

```
 Función de evaluación de retrieval definida
```

[5]:
```
# Evaluar retrieval para todas las preguntas
retrieval_results = []

for q in EVALUATION_QUESTIONS:
    print(f"Evaluando: {q['question'][:40]}...")
    result = evaluate_retrieval(
        question=q["question"],
        expected_category=q["category"],
        k=5
    )
    retrieval_results.append(result)

# Convertir a DataFrame
df_retrieval = pd.DataFrame(retrieval_results)
df_retrieval
```

```
Evaluando: ¿Qué está pasando con la economía?…
2026-02-07 18:37:49 [info     ] Using LM Studio (local)


2026-02-07 18:37:49 [info     ] Initialized
```

LMStudioEmbeddingGenerator

```
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:37:49 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 18:37:49 [info     ] Searching for articles

 limit=5 query=¿Qué está pasando con la

economía? threshold=0.5
2026-02-07 18:37:49 [info     ] Search completed

 results_count=5
Evaluando: ¿Cuáles son las perspectivas económicas …
2026-02-07 18:37:49 [info     ] Using LM Studio (local)


2026-02-07 18:37:49 [info     ] Initialized

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 llm_available=True use_reranking=False
2026-02-07 18:37:49 [info     ] Searching for articles

 limit=5 query=¿Cuáles son las perspectivas

económicas para 2026? threshold=0.5
2026-02-07 18:37:49 [info     ] Search completed

 results_count=5
Evaluando: ¿Cómo está la inflación actualmente?…
2026-02-07 18:37:49 [info     ] Using LM Studio (local)


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 llm_available=True use_reranking=False
2026-02-07 18:37:49 [info     ] Searching for articles

 limit=5 query=¿Cómo está la inflación

actualmente? threshold=0.5
2026-02-07 18:37:50 [info     ] Search completed

 results_count=5
Evaluando: ¿Qué está pasando con los salarios?…
```

```
2026-02-07 18:37:50 [info     ] Using LM Studio (local)


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2026-02-07 18:37:50 [info     ] Searching for articles

 limit=5 query=¿Qué está pasando con los

salarios? threshold=0.5
2026-02-07 18:37:50 [info     ] Search completed

 results_count=5
Evaluando: ¿Hay noticias sobre inversiones?…
2026-02-07 18:37:50 [info     ] Using LM Studio (local)


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 llm_available=True use_reranking=False
2026-02-07 18:37:50 [info     ] Searching for articles

 limit=5 query=¿Hay noticias sobre

inversiones? threshold=0.5
2026-02-07 18:37:50 [info     ] Search completed

 results_count=5
Evaluando: ¿Qué dicen los analistas económicos?…
2026-02-07 18:37:50 [info     ] Using LM Studio (local)


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 limit=5 query=¿Qué dicen los analistas
económicos? threshold=0.5
2026-02-07 18:37:50 [info     ] Search completed
 results_count=5
Evaluando: ¿Cuáles son las últimas noticias de tecn…
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2026-02-07 18:37:50 [info     ] Searching for articles
 limit=5 query=¿Cuáles son las últimas
noticias de tecnología? threshold=0.5
2026-02-07 18:37:50 [info     ] Search completed
 results_count=5
Evaluando: ¿Qué avances hay en inteligencia artific…
2026-02-07 18:37:50 [info     ] Using LM Studio (local)

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2026-02-07 18:37:50 [info     ] Searching for articles
 limit=5 query=¿Qué avances hay en
inteligencia artificial? threshold=0.5
2026-02-07 18:37:50 [info     ] Search completed
 results_count=5
Evaluando: ¿Hay noticias sobre redes sociales?…
2026-02-07 18:37:50 [info     ] Using LM Studio (local)

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LMStudioEmbeddingGenerator
```

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2026-02-07 18:37:50 [info     ] Searching for articles

 limit=5 query=¿Hay noticias sobre redes

sociales? threshold=0.5
2026-02-07 18:37:50 [info     ] Search completed

 results_count=5
Evaluando: ¿Qué está pasando con las empresas tecno…
2026-02-07 18:37:51 [info     ] Using LM Studio (local)


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 llm_available=True use_reranking=False
2026-02-07 18:37:51 [info     ] Searching for articles

 limit=5 query=¿Qué está pasando con las

empresas tecnológicas? threshold=0.5
2026-02-07 18:37:51 [info     ] Search completed

 results_count=5
Evaluando: ¿Hay noticias sobre ciberseguridad?…
2026-02-07 18:37:51 [info     ] Using LM Studio (local)


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 limit=5 query=¿Hay noticias sobre

ciberseguridad? threshold=0.5
2026-02-07 18:37:51 [info     ] Search completed

 results_count=5
Evaluando: ¿Qué noticias hay sobre salud?…
```

```
2026-02-07 18:37:51 [info     ] Using LM Studio (local)

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salud? threshold=0.5
2026-02-07 18:37:51 [info     ] Search completed
 results_count=5
Evaluando: ¿Hay noticias sobre vacunas?…
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 limit=5 query=¿Hay noticias sobre
vacunas? threshold=0.5
2026-02-07 18:37:52 [info     ] Search completed
 results_count=5
Evaluando: ¿Qué dicen los expertos sobre salud ment…
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```

```
2026-02-07 18:37:52 [info     ] Searching for articles
 limit=5 query=¿Qué dicen los expertos sobre
salud mental? threshold=0.5
2026-02-07 18:37:52 [info     ] Search completed
 results_count=5
Evaluando: ¿Hay avances médicos recientes?…
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 limit=5 query=¿Hay avances médicos
recientes? threshold=0.5
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 results_count=5
Evaluando: ¿Qué está pasando en el mundo del deport…
2026-02-07 18:37:53 [info     ] Using LM Studio (local)

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 llm_available=True use_reranking=False
2026-02-07 18:37:53 [info     ] Searching for articles
 limit=5 query=¿Qué está pasando en el mundo
del deporte? threshold=0.5
2026-02-07 18:37:53 [info     ] Search completed
 results_count=5
Evaluando: ¿Cuáles son los resultados deportivos re…
2026-02-07 18:37:53 [info     ] Using LM Studio (local)

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 llm_available=True use_reranking=False
2026-02-07 18:37:53 [info     ] Searching for articles

 limit=5 query=¿Cuáles son los resultados

deportivos recientes? threshold=0.5
2026-02-07 18:37:54 [info     ] Search completed

 results_count=5
Evaluando: ¿Hay noticias de fútbol?…
2026-02-07 18:37:54 [info     ] Using LM Studio (local)


2026-02-07 18:37:54 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
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2026-02-07 18:37:54 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 18:37:54 [info     ] Searching for articles

 limit=5 query=¿Hay noticias de fútbol?
threshold=0.5
2026-02-07 18:37:54 [info     ] Search completed

 results_count=5
Evaluando: ¿Qué fichajes o traspasos hay en el fútb…
2026-02-07 18:37:54 [info     ] Using LM Studio (local)


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2026-02-07 18:37:54 [info     ] RAG engine initialized

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2026-02-07 18:37:54 [info     ] Searching for articles

 limit=5 query=¿Qué fichajes o traspasos hay

en el fútbol? threshold=0.5
2026-02-07 18:37:55 [info     ] Search completed

 results_count=5
Evaluando: ¿Hay noticias sobre ciencia?…
```

```
2026-02-07 18:37:55 [info     ] Using LM Studio (local)

2026-02-07 18:37:55 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:37:55 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:37:55 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
ciencia? threshold=0.5
2026-02-07 18:37:55 [info     ] Search completed
 results_count=5
Evaluando: ¿Qué descubrimientos científicos hay?…
2026-02-07 18:37:56 [info     ] Using LM Studio (local)

2026-02-07 18:37:56 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:37:56 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:37:56 [info     ] Searching for articles
 limit=5 query=¿Qué descubrimientos
científicos hay? threshold=0.5
2026-02-07 18:37:56 [info     ] Search completed
 results_count=5
Evaluando: ¿Hay noticias sobre el espacio o astrono…
2026-02-07 18:37:56 [info     ] Using LM Studio (local)

2026-02-07 18:37:56 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:37:56 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
```

```
2026-02-07 18:37:56 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre el
espacio o astronomía? threshold=0.5
2026-02-07 18:37:56 [info     ] Search completed
 results_count=5
Evaluando: ¿Qué noticias hay de entretenimiento?…
2026-02-07 18:37:56 [info     ] Using LM Studio (local)

2026-02-07 18:37:56 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:37:56 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:37:56 [info     ] Searching for articles
 limit=5 query=¿Qué noticias hay de
entretenimiento? threshold=0.5
2026-02-07 18:37:57 [info     ] Search completed
 results_count=5
Evaluando: ¿Hay estrenos de cine o series?…
2026-02-07 18:37:57 [info     ] Using LM Studio (local)

2026-02-07 18:37:57 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:37:57 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:37:57 [info     ] Searching for articles
 limit=5 query=¿Hay estrenos de cine o
series? threshold=0.5
2026-02-07 18:37:57 [info     ] Search completed
 results_count=5
Evaluando: ¿Cuáles son las noticias internacionales…
2026-02-07 18:37:57 [info     ] Using LM Studio (local)

2026-02-07 18:37:57 [info     ] Initialized
LMStudioEmbeddingGenerator
```

```
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:37:57 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 18:37:57 [info     ] Searching for articles

 limit=5 query=¿Cuáles son las noticias

internacionales más impor threshold=0.5
2026-02-07 18:37:57 [info     ] Search completed

 results_count=5
```

[5]:

	question	k	retrieved	\
0	¿Qué está pasando con la economía?	5	5	
1	¿Cuáles son las perspectivas económicas para 2026?	5	5	
2	¿Cómo está la inflación actualmente?	5	5	
3	¿Qué está pasando con los salarios?	5	5	
4	¿Hay noticias sobre inversiones?	5	5	
5	¿Qué dicen los analistas económicos?	5	5	
6	¿Cuáles son las últimas noticias de tecnología?	5	5	
7	¿Qué avances hay en inteligencia artificial?	5	5	
8	¿Hay noticias sobre redes sociales?	5	5	
9	¿Qué está pasando con las empresas tecnológicas?	5	5	
10	¿Hay noticias sobre ciberseguridad?	5	5	
11	¿Qué noticias hay sobre salud?	5	5	
12	¿Hay noticias sobre vacunas?	5	5	
13	¿Qué dicen los expertos sobre salud mental?	5	5	
14	¿Hay avances médicos recientes?	5	5	
15	¿Qué está pasando en el mundo del deporte?	5	5	
16	¿Cuáles son los resultados deportivos recientes?	5	5	
17	¿Hay noticias de fútbol?	5	5	
18	¿Qué fichajes o traspasos hay en el fútbol?	5	5	
19	¿Hay noticias sobre ciencia?	5	5	
20	¿Qué descubrimientos científicos hay?	5	5	
21	¿Hay noticias sobre el espacio o astronomía?	5	5	
22	¿Qué noticias hay de entretenimiento?	5	5	
23	¿Hay estrenos de cine o series?	5	5	
24	¿Cuáles son las noticias internacionales más importantes?	5	5	

	relevant	precision_at_k	hit_rate	mrr	avg_similarity	\
0	5	1.0	1.0	1.0	0.703291	
1	5	1.0	1.0	1.0	0.779847	
2	5	1.0	1.0	1.0	0.694293	
3	5	1.0	1.0	1.0	0.665026	
4	5	1.0	1.0	1.0	0.649381	
5	5	1.0	1.0	1.0	0.716177	
6	5	1.0	1.0	1.0	0.695031	

7	5	1.0	1.0	1.0	0.793396
8	5	1.0	1.0	1.0	0.685344
9	5	1.0	1.0	1.0	0.724724
10	5	1.0	1.0	1.0	0.652272
11	5	1.0	1.0	1.0	0.670216
12	5	1.0	1.0	1.0	0.689010
13	5	1.0	1.0	1.0	0.771481
14	5	1.0	1.0	1.0	0.706181
15	5	1.0	1.0	1.0	0.725907
16	5	1.0	1.0	1.0	0.689005
17	5	1.0	1.0	1.0	0.717969
18	5	1.0	1.0	1.0	0.715293
19	5	1.0	1.0	1.0	0.680628
20	5	1.0	1.0	1.0	0.697583
21	5	1.0	1.0	1.0	0.721734
22	5	1.0	1.0	1.0	0.654737
23	5	1.0	1.0	1.0	0.683852
24	5	1.0	1.0	1.0	0.668834

```
           similarities  \
0   [0.708400710213602, 0.7076221465793159, 0.7046728690094256,
0.7031199950695121, 0.6926411047128278]
1   [0.7920783374566635, 0.7891624537594829, 0.7855018845500815,
0.7700354689507776, 0.7624575525421…
2   [0.7288017985114708, 0.7135637824839622, 0.6815833652492715,
0.6757481332041785, 0.6717672156941…
3   [0.6923042444068034, 0.6607639773749197, 0.6603538405263937,
0.6561797419290761, 0.6555295361351…
4   [0.6646349233824295, 0.6591065336339025, 0.6435108184814499,
0.6411540271571322, 0.6384965181350…
5   [0.7561173955097227, 0.7129108740410114, 0.7097581028938496,
0.7073710524406167, 0.694725793285746]
6   [0.7094207156395512, 0.6982863252501822, 0.6982863252501822,
0.6869722363708965, 0.6821910493996…
7   [0.8132605421608494, 0.8092758073323681, 0.7995699762913595,
0.777683698465943, 0.7671891957054537]
8   [0.6973772797916564, 0.6858360562969457, 0.6843352898059603,
0.6824160639071587, 0.6767544948087…
9    [0.7282612271931713, 0.7271964387414833, 0.7271964387414833,
0.72546757844711, 0.7154971957206929]
10  [0.6632155383803211, 0.6588143082193058, 0.6489953513123292,
0.6451795504934374, 0.6451571179973…
11  [0.6928171379130228, 0.6676093340762881, 0.664320171089181,
0.6638248154438462, 0.6625084083524965]
12  [0.7086220482296879, 0.6934606567715081, 0.6915821616898691,
0.6835596357715266, 0.667826433907365]
13  [0.8060757517814865, 0.7928029647830739, 0.7558418512344575,
```

```
0.7555832717170402, 0.7471002785623…
14  [0.7165983826044229, 0.7070397385406987, 0.7051476943140809,
0.701130562818557, 0.7009907008399212]
15  [0.7457690461209135, 0.7313948478118364, 0.7311527408543568,
0.7109018126375459, 0.7103143757769…
16  [0.6949398441774445, 0.6937788955901749, 0.6907761594911729,
0.6829063325356888, 0.6826216971092…
17  [0.7286035852793582, 0.7226822476415925, 0.7215275843730409,
0.7104182611536585, 0.7066139371938…
18  [0.7235122170564935, 0.7178218624379483, 0.7135388467770475,
0.7118761327303637, 0.7097134974684…
19  [0.7044266496086133, 0.6825979730370125, 0.6818413179091419,
0.6739918790457216, 0.6602807056402…
20  [0.7350962416627455, 0.6903903995548446, 0.6893718531742845,
0.6891751659900536, 0.683881653119839]
21  [0.7355230327505529, 0.7214056302373704, 0.7206750915020838,
0.7155711902466168, 0.7154975276592…
22  [0.6638834466517388, 0.6612980942600122, 0.6517574928059404,
0.6492346896581123, 0.6475099676871…
23   [0.7199110317018926, 0.67960606558579, 0.6753792391358127,
0.6745063094519723, 0.6698549001977133]
24  [0.6860548819205485, 0.6789380333908964, 0.6642694669091114,
0.6584824571119184, 0.6564248176760…

                                                        categories
0             [business, business, business, business, business]
1             [business, business, business, technology, sports]
2               [business, business, technology, health, health]
3                 [business, science, sports, business, general]
4             [business, business, technology, business, sports]
5             [business, business, business, business, business]
6   [technology, technology, technology, technology, technology]
7             [business, technology, health, health, technology]
8                [general, general, general, nation, technology]
9     [technology, technology, technology, business, technology]
10                 [nation, technology, general, sports, health]
11                      [health, health, health, health, health]
12                      [health, health, health, health, health]
13                    [science, health, health, health, science]
14                     [science, health, health, health, health]
15                      [sports, sports, sports, sports, sports]
16                      [sports, sports, sports, sports, sports]
17                      [sports, sports, sports, sports, sports]
18                      [sports, sports, sports, sports, sports]
19                 [science, science, science, science, science]
20                 [science, science, science, science, science]
21                    [health, science, health, health, science]
```

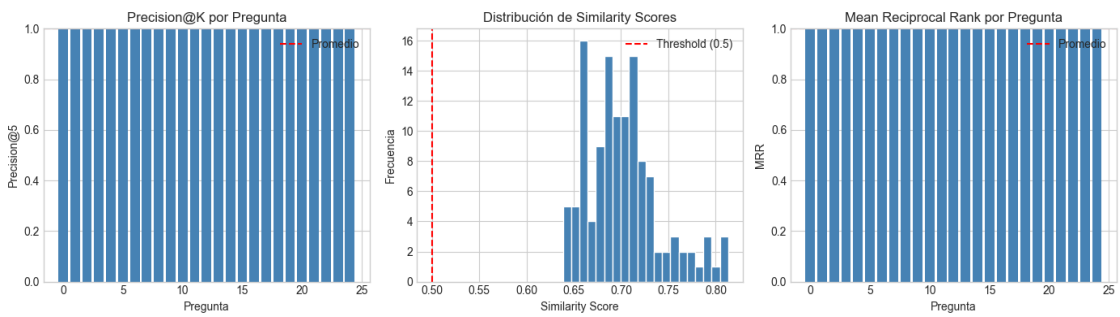
```
22                [business, sports, technology, sports, health]
23           [science, general, entertainment, science, general]
24               [business, technology, sports, sports, general]
```

[6]:
```python
# Resumen de métricas de Retrieval
print("=" * 50)
print("MÉTRICAS DE RETRIEVAL")
print("=" * 50)
print(f"Precision@5 promedio:  {df_retrieval['precision_at_k'].mean():.2%}")
print(f"Hit Rate promedio:     {df_retrieval['hit_rate'].mean():.2%}")
print(f"MRR promedio:          {df_retrieval['mrr'].mean():.3f}")
print(f"Similarity promedio:   {df_retrieval['avg_similarity'].mean():.3f}")
```

```
==================================================
MÉTRICAS DE RETRIEVAL
==================================================
Precision@5 promedio:  100.00%
Hit Rate promedio:     100.00%
MRR promedio:          1.000
Similarity promedio:   0.702
```

[7]:
```python
# Visualización de métricas de Retrieval
fig, axes = plt.subplots(1, 3, figsize=(14, 4))

# Precision@K
axes[0].bar(range(len(df_retrieval)), df_retrieval['precision_at_k'],␣
 ↪color='steelblue')
axes[0].axhline(y=df_retrieval['precision_at_k'].mean(), color='red',␣
 ↪linestyle='--', label='Promedio')
axes[0].set_xlabel('Pregunta')
axes[0].set_ylabel('Precision@5')
axes[0].set_title('Precision@K por Pregunta')
axes[0].set_ylim(0, 1)
axes[0].legend()

# Similarity distribution
all_similarities = [s for sims in df_retrieval['similarities'] for s in sims]
axes[1].hist(all_similarities, bins=20, color='steelblue', edgecolor='white')
axes[1].axvline(x=0.5, color='red', linestyle='--', label='Threshold (0.5)')
axes[1].set_xlabel('Similarity Score')
axes[1].set_ylabel('Frecuencia')
axes[1].set_title('Distribución de Similarity Scores')
axes[1].legend()

# MRR
axes[2].bar(range(len(df_retrieval)), df_retrieval['mrr'], color='steelblue')
```

```
axes[2].axhline(y=df_retrieval['mrr'].mean(), color='red', linestyle='--',␣
↪label='Promedio')
axes[2].set_xlabel('Pregunta')
axes[2].set_ylabel('MRR')
axes[2].set_title('Mean Reciprocal Rank por Pregunta')
axes[2].set_ylim(0, 1)
axes[2].legend()

plt.tight_layout()
plt.savefig('../docs/images/retrieval_metrics.png', dpi=150,␣
↪bbox_inches='tight')
plt.show()
```



1.5 4. Métricas de Generation (RAG completo)

[8]:
```
def evaluate_generation(
    question: str,
    expected_topics: list[str],
    max_sources: int = 5,
    use_reranking: bool = False,
) -> dict:
    """
    Evalúa la calidad de la generación RAG.

    Métricas:
    - Citation count: ¿Cuántas fuentes citó?
    - Topic coverage: ¿Menciona los temas esperados?
    - Answer length: Longitud de la respuesta
    - Confidence: Score de confianza del sistema
    """
    with get_db() as db:
        engine = RAGEngine(db, use_reranking=use_reranking)

        response = engine.ask(
            question=question,
```

```
            max_sources=max_sources,
        )

        # Contar citas en la respuesta
        import re
        citations_in_answer = len(re.findall(r'\[\d+\]', response.answer))

        # Verificar cobertura de temas
        answer_lower = response.answer.lower()
        topics_found = [
            topic for topic in expected_topics
            if topic.lower() in answer_lower
        ]
        topic_coverage = len(topics_found) / len(expected_topics) if␣
 ↪expected_topics else 0

        return {
            "question": question,
            "answer": response.answer,
            "sources_used": len(response.sources),
            "citations_in_answer": citations_in_answer,
            "confidence": response.confidence,
            "answer_length": len(response.answer),
            "topic_coverage": topic_coverage,
            "topics_found": topics_found,
            "use_reranking": use_reranking,
        }

print("  Función de evaluación de generación definida")
```

```
 Función de evaluación de generación definida
```

[9]:
```
# Evaluar generación para todas las preguntas
generation_results = []

for q in EVALUATION_QUESTIONS:
    print(f"Evaluando RAG: {q['question'][:40]}...")
    result = evaluate_generation(
        question=q["question"],
        expected_topics=q["expected_topics"],
        max_sources=5,
        use_reranking=False,
    )
    generation_results.append(result)

# Convertir a DataFrame
df_generation = pd.DataFrame(generation_results)
```

```
df_generation[['question', 'sources_used', 'citations_in_answer', 'confidence',␣
↪'topic_coverage']]
```

```
Evaluando RAG: ¿Qué está pasando con la economía?…
2026-02-07 18:37:58 [info     ] Using LM Studio (local)


2026-02-07 18:37:58 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:37:58 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 18:37:58 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Qué está pasando con la economía?
2026-02-07 18:37:58 [info     ] Searching for articles

 limit=5 query=¿Qué está pasando con la

economía? threshold=0.5
2026-02-07 18:37:58 [info     ] Search completed

 results_count=5
2026-02-07 18:37:58 [info     ] Generating RAG response

 articles=5 query=¿Qué está pasando con la

economía?
2026-02-07 18:38:27 [info     ] RAG response generated

 answer_length=1435
confidence=0.861276190124162 sources_used=3
2026-02-07 18:38:27 [info     ] RAG query completed

 confidence=0.861276190124162
sources_cited=3 sources_found=5
Evaluando RAG: ¿Cuáles son las perspectivas económicas …
2026-02-07 18:38:27 [info     ] Using LM Studio (local)


2026-02-07 18:38:27 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:38:27 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
```

```
2026-02-07 18:38:27 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Cuáles son las perspectivas económicas para 2026?
2026-02-07 18:38:27 [info     ] Searching for articles
 limit=5 query=¿Cuáles son las perspectivas
económicas para 2026? threshold=0.5
2026-02-07 18:38:28 [info     ] Search completed
 results_count=5
2026-02-07 18:38:28 [info     ] Generating RAG response
 articles=5 query=¿Cuáles son las
perspectivas económicas para 2026?
2026-02-07 18:38:47 [info     ] RAG response generated
 answer_length=1561
confidence=0.8777828450510818 sources_used=3
2026-02-07 18:38:47 [info     ] RAG query completed
 confidence=0.8777828450510818
sources_cited=3 sources_found=5
Evaluando RAG: ¿Cómo está la inflación actualmente?…
2026-02-07 18:38:47 [info     ] Using LM Studio (local)

2026-02-07 18:38:47 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:38:47 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:38:47 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Cómo está la inflación actualmente?
2026-02-07 18:38:47 [info     ] Searching for articles
 limit=5 query=¿Cómo está la inflación
actualmente? threshold=0.5
2026-02-07 18:38:49 [info     ] Search completed
 results_count=5
2026-02-07 18:38:49 [info     ] Generating RAG response
 articles=5 query=¿Cómo está la inflación
actualmente?
```

```
2026-02-07 18:39:10 [info     ] RAG response generated
 answer_length=594
confidence=0.8242365580995433 sources_used=2
2026-02-07 18:39:10 [info     ] RAG query completed
 confidence=0.8242365580995433
sources_cited=2 sources_found=5
Evaluando RAG: ¿Qué está pasando con los salarios?…
2026-02-07 18:39:10 [info     ] Using LM Studio (local)

2026-02-07 18:39:10 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:39:10 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:39:10 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué está pasando con los salarios?
2026-02-07 18:39:10 [info     ] Searching for articles
 limit=5 query=¿Qué está pasando con los
salarios? threshold=0.5
2026-02-07 18:39:12 [info     ] Search completed
 results_count=5
2026-02-07 18:39:12 [info     ] Generating RAG response
 articles=5 query=¿Qué está pasando con los
salarios?
2026-02-07 18:39:29 [info     ] RAG response generated
 answer_length=363
confidence=0.6784608488813607 sources_used=1
2026-02-07 18:39:29 [info     ] RAG query completed
 confidence=0.6784608488813607
sources_cited=1 sources_found=5
Evaluando RAG: ¿Hay noticias sobre inversiones?…
2026-02-07 18:39:29 [info     ] Using LM Studio (local)

2026-02-07 18:39:29 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
```

```
2026-02-07 18:39:29 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:39:29 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Hay noticias sobre inversiones?
2026-02-07 18:39:29 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
inversiones? threshold=0.5
2026-02-07 18:39:30 [info     ] Search completed
 results_count=5
2026-02-07 18:39:30 [info     ] Generating RAG response
 articles=5 query=¿Hay noticias sobre
inversiones?
2026-02-07 18:39:49 [info     ] RAG response generated
 answer_length=582
confidence=0.8511501516998521 sources_used=3
2026-02-07 18:39:49 [info     ] RAG query completed
 confidence=0.8511501516998521
sources_cited=3 sources_found=5
Evaluando RAG: ¿Qué dicen los analistas económicos?…
2026-02-07 18:39:49 [info     ] Using LM Studio (local)

2026-02-07 18:39:49 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:39:49 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:39:49 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué dicen los analistas económicos?
2026-02-07 18:39:49 [info     ] Searching for articles
 limit=5 query=¿Qué dicen los analistas
económicos? threshold=0.5
2026-02-07 18:39:50 [info     ] Search completed
 results_count=5
```

```
2026-02-07 18:39:50 [info     ] Generating RAG response
 articles=5 query=¿Qué dicen los analistas
económicos?
2026-02-07 18:40:10 [info     ] RAG response generated
 answer_length=1011
confidence=0.8652524248296389 sources_used=3
2026-02-07 18:40:10 [info     ] RAG query completed
 confidence=0.8652524248296389
sources_cited=3 sources_found=5
Evaluando RAG: ¿Cuáles son las últimas noticias de tecn…
2026-02-07 18:40:10 [info     ] Using LM Studio (local)


2026-02-07 18:40:10 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:40:10 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:40:10 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Cuáles son las últimas noticias de tecnología?
2026-02-07 18:40:10 [info     ] Searching for articles
 limit=5 query=¿Cuáles son las últimas
noticias de tecnología? threshold=0.5
2026-02-07 18:40:11 [info     ] Search completed
 results_count=5
2026-02-07 18:40:11 [info     ] Generating RAG response
 articles=5 query=¿Cuáles son las últimas
noticias de tecnología?
2026-02-07 18:40:33 [info     ] RAG response generated
 answer_length=1864
confidence=0.9390062660764192 sources_used=5
2026-02-07 18:40:33 [info     ] RAG query completed
 confidence=0.9390062660764192
sources_cited=5 sources_found=5
Evaluando RAG: ¿Qué avances hay en inteligencia artific…
2026-02-07 18:40:33 [info     ] Using LM Studio (local)
```

```
2026-02-07 18:40:33 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:40:33 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:40:33 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué avances hay en inteligencia artificial?
2026-02-07 18:40:33 [info     ] Searching for articles
 limit=5 query=¿Qué avances hay en
inteligencia artificial? threshold=0.5
2026-02-07 18:40:34 [info     ] Search completed
 results_count=5
2026-02-07 18:40:34 [info     ] Generating RAG response
 articles=5 query=¿Qué avances hay en
inteligencia artificial?
2026-02-07 18:40:53 [info     ] RAG response generated
 answer_length=1533
confidence=0.9586791687982389 sources_used=5
2026-02-07 18:40:53 [info     ] RAG query completed
 confidence=0.9586791687982389
sources_cited=5 sources_found=5
Evaluando RAG: ¿Hay noticias sobre redes sociales?…
2026-02-07 18:40:53 [info     ] Using LM Studio (local)

2026-02-07 18:40:53 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:40:53 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:40:53 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Hay noticias sobre redes sociales?
2026-02-07 18:40:53 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre redes
sociales? threshold=0.5
```

```
2026-02-07 18:40:54 [info     ] Search completed
 results_count=5
2026-02-07 18:40:54 [info     ] Generating RAG response
 articles=5 query=¿Hay noticias sobre redes
sociales?
2026-02-07 18:41:13 [info     ] RAG response generated
 answer_length=801
confidence=0.9370687673844187 sources_used=5
2026-02-07 18:41:13 [info     ] RAG query completed
 confidence=0.9370687673844187
sources_cited=5 sources_found=5
Evaluando RAG: ¿Qué está pasando con las empresas tecno…
2026-02-07 18:41:13 [info     ] Using LM Studio (local)

2026-02-07 18:41:13 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:41:13 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:41:13 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué está pasando con las empresas tecnológicas?
2026-02-07 18:41:13 [info     ] Searching for articles
 limit=5 query=¿Qué está pasando con las
empresas tecnológicas? threshold=0.5
2026-02-07 18:41:14 [info     ] Search completed
 results_count=5
2026-02-07 18:41:14 [info     ] Generating RAG response
 articles=5 query=¿Qué está pasando con las
empresas tecnológicas?
2026-02-07 18:41:37 [info     ] RAG response generated
 answer_length=1214
confidence=0.8655102736450758 sources_used=3
2026-02-07 18:41:37 [info     ] RAG query completed
 confidence=0.8655102736450758
sources_cited=3 sources_found=5
Evaluando RAG: ¿Hay noticias sobre ciberseguridad?…
```

```
2026-02-07 18:41:37 [info     ] Using LM Studio (local)


2026-02-07 18:41:37 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:41:37 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:41:37 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Hay noticias sobre ciberseguridad?
2026-02-07 18:41:37 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
ciberseguridad? threshold=0.5
2026-02-07 18:41:38 [info     ] Search completed
 results_count=5
2026-02-07 18:41:38 [info     ] Generating RAG response
 articles=5 query=¿Hay noticias sobre
ciberseguridad?
2026-02-07 18:41:56 [info     ] RAG response generated
 answer_length=433
confidence=0.6717628616438612 sources_used=1
2026-02-07 18:41:56 [info     ] RAG query completed
 confidence=0.6717628616438612
sources_cited=1 sources_found=5
Evaluando RAG: ¿Qué noticias hay sobre salud?…
2026-02-07 18:41:57 [info     ] Using LM Studio (local)


2026-02-07 18:41:57 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:41:57 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:41:57 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué noticias hay sobre salud?
```

```
2026-02-07 18:41:57 [info     ] Searching for articles
 limit=5 query=¿Qué noticias hay sobre
salud? threshold=0.5
2026-02-07 18:41:58 [info     ] Search completed
 results_count=5
2026-02-07 18:41:58 [info     ] Generating RAG response
 articles=5 query=¿Qué noticias hay sobre
salud?
2026-02-07 18:42:21 [info     ] RAG response generated
 answer_length=1578
confidence=0.9340431946749933 sources_used=5
2026-02-07 18:42:21 [info     ] RAG query completed
 confidence=0.9340431946749933
sources_cited=5 sources_found=5
Evaluando RAG: ¿Hay noticias sobre vacunas?…
2026-02-07 18:42:22 [info     ] Using LM Studio (local)

2026-02-07 18:42:22 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:42:22 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:42:22 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Hay noticias sobre vacunas?
2026-02-07 18:42:22 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
vacunas? threshold=0.5
2026-02-07 18:42:23 [info     ] Search completed
 results_count=5
2026-02-07 18:42:23 [info     ] Generating RAG response
 articles=5 query=¿Hay noticias sobre
vacunas?
2026-02-07 18:42:40 [info     ] RAG response generated
 answer_length=943
confidence=0.9378020374547982 sources_used=5
```

```
2026-02-07 18:42:40 [info     ] RAG query completed
 confidence=0.9378020374547982
sources_cited=5 sources_found=5
Evaluando RAG: ¿Qué dicen los expertos sobre salud ment…
2026-02-07 18:42:41 [info     ] Using LM Studio (local)

2026-02-07 18:42:41 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:42:41 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:42:41 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué dicen los expertos sobre salud mental?
2026-02-07 18:42:41 [info     ] Searching for articles
 limit=5 query=¿Qué dicen los expertos sobre
salud mental? threshold=0.5
2026-02-07 18:42:42 [info     ] Search completed
 results_count=5
2026-02-07 18:42:42 [info     ] Generating RAG response
 articles=5 query=¿Qué dicen los expertos
sobre salud mental?
2026-02-07 18:43:03 [info     ] RAG response generated
 answer_length=1702
confidence=0.9155151919758029 sources_used=4
2026-02-07 18:43:03 [info     ] RAG query completed
 confidence=0.9155151919758029
sources_cited=4 sources_found=5
Evaluando RAG: ¿Hay avances médicos recientes?…
2026-02-07 18:43:03 [info     ] Using LM Studio (local)

2026-02-07 18:43:03 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:43:03 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
```

```
2026-02-07 18:43:03 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Hay avances médicos recientes?
2026-02-07 18:43:03 [info     ] Searching for articles
 limit=5 query=¿Hay avances médicos
recientes? threshold=0.5
2026-02-07 18:43:04 [info     ] Search completed
 results_count=5
2026-02-07 18:43:04 [info     ] Generating RAG response
 articles=5 query=¿Hay avances médicos
recientes?
2026-02-07 18:43:22 [info     ] RAG response generated
 answer_length=608
confidence=0.8208170301359257 sources_used=2
2026-02-07 18:43:22 [info     ] RAG query completed
 confidence=0.8208170301359257
sources_cited=2 sources_found=5
Evaluando RAG: ¿Qué está pasando en el mundo del deport…
2026-02-07 18:43:23 [info     ] Using LM Studio (local)

2026-02-07 18:43:23 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:43:23 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:43:23 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué está pasando en el mundo del deporte?
2026-02-07 18:43:23 [info     ] Searching for articles
 limit=5 query=¿Qué está pasando en el mundo
del deporte? threshold=0.5
2026-02-07 18:43:25 [info     ] Search completed
 results_count=5
2026-02-07 18:43:25 [info     ] Generating RAG response
 articles=5 query=¿Qué está pasando en el
mundo del deporte?
```

```
2026-02-07 18:43:48 [info     ] RAG response generated
 answer_length=1704
confidence=0.9451813129280625 sources_used=5
2026-02-07 18:43:48 [info     ] RAG query completed
 confidence=0.9451813129280625
sources_cited=5 sources_found=5
Evaluando RAG: ¿Cuáles son los resultados deportivos re…
2026-02-07 18:43:48 [info     ] Using LM Studio (local)

2026-02-07 18:43:48 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:43:48 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:43:48 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Cuáles son los resultados deportivos recientes?
2026-02-07 18:43:48 [info     ] Searching for articles
 limit=5 query=¿Cuáles son los resultados
deportivos recientes? threshold=0.5
2026-02-07 18:43:49 [info     ] Search completed
 results_count=5
2026-02-07 18:43:49 [info     ] Generating RAG response
 articles=5 query=¿Cuáles son los resultados
deportivos recientes?
2026-02-07 18:44:05 [info     ] RAG response generated
 answer_length=309
confidence=0.6765812665071378 sources_used=1
2026-02-07 18:44:05 [info     ] RAG query completed
 confidence=0.6765812665071378
sources_cited=1 sources_found=5
Evaluando RAG: ¿Hay noticias de fútbol?…
2026-02-07 18:44:05 [info     ] Using LM Studio (local)

2026-02-07 18:44:05 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
```

```
2026-02-07 18:44:05 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:44:05 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Hay noticias de fútbol?
2026-02-07 18:44:05 [info     ] Searching for articles
 limit=5 query=¿Hay noticias de fútbol?
threshold=0.5
2026-02-07 18:44:06 [info     ] Search completed
 results_count=5
2026-02-07 18:44:06 [info     ] Generating RAG response
 articles=5 query=¿Hay noticias de
fútbol?
2026-02-07 18:44:23 [info     ] RAG response generated
 answer_length=616
confidence=0.9041615839223826 sources_used=4
2026-02-07 18:44:23 [info     ] RAG query completed
 confidence=0.9041615839223826
sources_cited=4 sources_found=5
Evaluando RAG: ¿Qué fichajes o traspasos hay en el fútb…
2026-02-07 18:44:23 [info     ] Using LM Studio (local)

2026-02-07 18:44:23 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:44:23 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:44:23 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué fichajes o traspasos hay en el fútbol?
2026-02-07 18:44:23 [info     ] Searching for articles
 limit=5 query=¿Qué fichajes o traspasos hay
en el fútbol? threshold=0.5
2026-02-07 18:44:24 [info     ] Search completed
 results_count=5
```

```
2026-02-07 18:44:24 [info     ] Generating RAG response
 articles=5 query=¿Qué fichajes o traspasos
hay en el fútbol?
2026-02-07 18:44:38 [info     ] RAG response generated
 answer_length=88 confidence=0.5
sources_used=0
2026-02-07 18:44:38 [info     ] RAG query completed
 confidence=0.5 sources_cited=3
sources_found=5
Evaluando RAG: ¿Hay noticias sobre ciencia?…
2026-02-07 18:44:38 [info     ] Using LM Studio (local)


2026-02-07 18:44:38 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:44:38 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:44:38 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Hay noticias sobre ciencia?
2026-02-07 18:44:38 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
ciencia? threshold=0.5
2026-02-07 18:44:39 [info     ] Search completed
 results_count=5
2026-02-07 18:44:39 [info     ] Generating RAG response
 articles=5 query=¿Hay noticias sobre
ciencia?
2026-02-07 18:44:57 [info     ] RAG response generated
 answer_length=893
confidence=0.8964573323097484 sources_used=4
2026-02-07 18:44:57 [info     ] RAG query completed
 confidence=0.8964573323097484
sources_cited=4 sources_found=5
Evaluando RAG: ¿Qué descubrimientos científicos hay?…
2026-02-07 18:44:57 [info     ] Using LM Studio (local)
```

```
2026-02-07 18:44:57 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:44:57 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:44:57 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué descubrimientos científicos hay?
2026-02-07 18:44:57 [info     ] Searching for articles
 limit=5 query=¿Qué descubrimientos
científicos hay? threshold=0.5
2026-02-07 18:44:58 [info     ] Search completed
 results_count=5
2026-02-07 18:44:58 [info     ] Generating RAG response
 articles=5 query=¿Qué descubrimientos
científicos hay?
2026-02-07 18:45:18 [info     ] RAG response generated
 answer_length=1163
confidence=0.9002016830190964 sources_used=4
2026-02-07 18:45:18 [info     ] RAG query completed
 confidence=0.9002016830190964
sources_cited=4 sources_found=5
Evaluando RAG: ¿Hay noticias sobre el espacio o astrono…
2026-02-07 18:45:18 [info     ] Using LM Studio (local)

2026-02-07 18:45:18 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:45:18 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:45:18 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Hay noticias sobre el espacio o astronomía?
2026-02-07 18:45:18 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre el
espacio o astronomía? threshold=0.5
```

```
2026-02-07 18:45:19 [info     ] Search completed
 results_count=5
2026-02-07 18:45:19 [info     ] Generating RAG response
 articles=5 query=¿Hay noticias sobre el
espacio o astronomía?
2026-02-07 18:45:37 [info     ] RAG response generated
 answer_length=575
confidence=0.8648284127098108 sources_used=3
2026-02-07 18:45:37 [info     ] RAG query completed
 confidence=0.8648284127098108
sources_cited=3 sources_found=5
Evaluando RAG: ¿Qué noticias hay de entretenimiento?…
2026-02-07 18:45:37 [info     ] Using LM Studio (local)

2026-02-07 18:45:37 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:45:37 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:45:37 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué noticias hay de entretenimiento?
2026-02-07 18:45:37 [info     ] Searching for articles
 limit=5 query=¿Qué noticias hay de
entretenimiento? threshold=0.5
2026-02-07 18:45:38 [info     ] Search completed
 results_count=5
2026-02-07 18:45:38 [info     ] Generating RAG response
 articles=5 query=¿Qué noticias hay de
entretenimiento?
2026-02-07 18:45:54 [info     ] RAG response generated
 answer_length=259
confidence=0.6695019935374266 sources_used=1
2026-02-07 18:45:54 [info     ] RAG query completed
 confidence=0.6695019935374266
sources_cited=1 sources_found=5
Evaluando RAG: ¿Hay estrenos de cine o series?…
```

```
2026-02-07 18:45:54 [info     ] Using LM Studio (local)


2026-02-07 18:45:54 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:45:54 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:45:54 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Hay estrenos de cine o series?
2026-02-07 18:45:54 [info     ] Searching for articles
 limit=5 query=¿Hay estrenos de cine o
series? threshold=0.5
2026-02-07 18:45:55 [info     ] Search completed
 results_count=5
2026-02-07 18:45:55 [info     ] Generating RAG response
 articles=5 query=¿Hay estrenos de cine o
series?
2026-02-07 18:46:16 [info     ] RAG response generated
 answer_length=528
confidence=0.8189765931899606 sources_used=2
2026-02-07 18:46:16 [info     ] RAG query completed
 confidence=0.8189765931899606
sources_cited=2 sources_found=5
Evaluando RAG: ¿Cuáles son las noticias internacionales…
2026-02-07 18:46:16 [info     ] Using LM Studio (local)


2026-02-07 18:46:16 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:46:16 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 18:46:16 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Cuáles son las noticias internacionales más impor
```

```
2026-02-07 18:46:16 [info     ] Searching for articles
 limit=5 query=¿Cuáles son las noticias
internacionales más impor threshold=0.5
2026-02-07 18:46:18 [info     ] Search completed
 results_count=5
2026-02-07 18:46:18 [info     ] Generating RAG response
 articles=5 query=¿Cuáles son las noticias
internacionales más impor
2026-02-07 18:46:35 [info     ] RAG response generated
 answer_length=638
confidence=0.8144537339032467 sources_used=2
2026-02-07 18:46:35 [info     ] RAG query completed
 confidence=0.8144537339032467
sources_cited=2 sources_found=5
```

[9]:

	question	sources_used	\
0	¿Qué está pasando con la economía?	3	
1	¿Cuáles son las perspectivas económicas para 2026?	3	
2	¿Cómo está la inflación actualmente?	2	
3	¿Qué está pasando con los salarios?	1	
4	¿Hay noticias sobre inversiones?	3	
5	¿Qué dicen los analistas económicos?	3	
6	¿Cuáles son las últimas noticias de tecnología?	5	
7	¿Qué avances hay en inteligencia artificial?	5	
8	¿Hay noticias sobre redes sociales?	5	
9	¿Qué está pasando con las empresas tecnológicas?	3	
10	¿Hay noticias sobre ciberseguridad?	1	
11	¿Qué noticias hay sobre salud?	5	
12	¿Hay noticias sobre vacunas?	5	
13	¿Qué dicen los expertos sobre salud mental?	4	
14	¿Hay avances médicos recientes?	2	
15	¿Qué está pasando en el mundo del deporte?	5	
16	¿Cuáles son los resultados deportivos recientes?	1	
17	¿Hay noticias de fútbol?	4	
18	¿Qué fichajes o traspasos hay en el fútbol?	3	
19	¿Hay noticias sobre ciencia?	4	
20	¿Qué descubrimientos científicos hay?	4	
21	¿Hay noticias sobre el espacio o astronomía?	3	
22	¿Qué noticias hay de entretenimiento?	1	
23	¿Hay estrenos de cine o series?	2	
24	¿Cuáles son las noticias internacionales más importantes?	2	

	citations_in_answer	confidence	topic_coverage
0	3	0.861276	0.25

```
1                    3    0.877783            0.50
2                    2    0.824237            0.75
3                    1    0.678461            0.75
4                    3    0.851150            0.25
5                    3    0.865252            0.75
6                    5    0.939006            0.50
7                    5    0.958679            1.00
8                    5    0.937069            0.50
9                    3    0.865510            0.75
10                   1    0.671763            0.25
11                   5    0.934043            0.25
12                   5    0.937802            0.75
13                   4    0.915515            0.50
14                   2    0.820817            0.00
15                   5    0.945181            0.50
16                   1    0.676581            0.75
17                   4    0.904162            0.25
18                   0    0.500000            0.50
19                   4    0.896457            0.75
20                   4    0.900202            1.00
21                   3    0.864828            0.75
22                   1    0.669502            0.25
23                   2    0.818977            1.00
24                   2    0.814454            0.25
```

[10]:
```python
# Resumen de métricas de Generación
print("=" * 50)
print("MÉTRICAS DE GENERACIÓN")
print("=" * 50)
print(f"Fuentes usadas (prom):     {df_generation['sources_used'].mean():.1f}")
print(f"Citas en respuesta (prom): {df_generation['citations_in_answer'].mean():
 ↪.1f}")
print(f"Confianza promedio:        {df_generation['confidence'].mean():.2%}")
print(f"Cobertura de temas (prom): {df_generation['topic_coverage'].mean():.
 ↪2%}")
print(f"Longitud respuesta (prom): {df_generation['answer_length'].mean():.0f}␣
 ↪caracteres")
```

```
==================================================
MÉTRICAS DE GENERACIÓN
==================================================
Fuentes usadas (prom):     3.2
Citas en respuesta (prom): 3.0
Confianza promedio:        83.71%
Cobertura de temas (prom): 55.00%
Longitud respuesta (prom): 920 caracteres
```

[11]:
```
# Ver una respuesta de ejemplo
example = generation_results[0]
print(f"Pregunta: {example['question']}")
print(f"\nRespuesta:")
print(example['answer'])
print(f"\nFuentes: {example['sources_used']}")
print(f"Confianza: {example['confidence']:.2%}")
```

```
Pregunta: ¿Qué está pasando con la economía?

Respuesta:
La economía argentina se encuentra en un proceso de **reconversión productiva**
que busca abrir la economía y diversificar su producción, aunque el camino es
difícil y requiere de intentos sostenidos para evitar fracasos anteriores. Este
proceso se describe como una tarea necesaria pero compleja, con la necesidad de
no repetir los errores del pasado. [1]

En el sector **del conocimiento**, Argentina se ha consolidado como uno de los
motores más dinámicos del país. La exportación de servicios profesionales y
software alcanzó cerca de **US$10 000 millones** el último año, con proyecciones
de llegar a **US$30 000 millones** en la próxima década. El crecimiento se basa
en el desarrollo de nichos vinculados a la inteligencia artificial, aunque
existen restricciones que deben superarse para alcanzar ese objetivo. [2]

En cuanto a la visión de los economistas, se destaca que la economía no debe ser
tratada como una ciencia exacta; las ideas tradicionales sobre el papel del
capital y la modernización están siendo cuestionadas. Se subraya la importancia
de que el capital se emplee en modernizar el país, evitando que solo genere
rentas sin impulsar la productividad. [4]

En síntesis, Argentina está intentando reorientar su economía hacia sectores de
alto valor añadido como el conocimiento y la tecnología, mientras enfrenta
desafíos estructurales y necesita evitar caer en modelos de rentas que no
fomenten la modernización.

Fuentes: 3
Confianza: 86.13%
```

1.6 5. Comparación: Con vs Sin Reranking

[12]:
```
# Evaluar CON reranking
generation_results_rerank = []

for q in EVALUATION_QUESTIONS:
    print(f"Evaluando RAG (con reranking): {q['question'][:40]}...")
    result = evaluate_generation(
        question=q["question"],
```

```
        expected_topics=q["expected_topics"],
        max_sources=5,
        use_reranking=True,  # <-- Con reranking
    )
    generation_results_rerank.append(result)

df_generation_rerank = pd.DataFrame(generation_results_rerank)
```

```
Evaluando RAG (con reranking): ¿Qué está pasando con la economía?…
2026-02-07 18:46:50 [info     ] Using LM Studio (local)


2026-02-07 18:46:50 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:46:50 [info     ] RAG engine initialized

 llm_available=True use_reranking=True
2026-02-07 18:46:50 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Qué está pasando con la economía?
2026-02-07 18:46:50 [info     ] Searching for articles

 limit=10 query=¿Qué está pasando con la

economía? threshold=0.5
2026-02-07 18:46:52 [info     ] Search completed

 results_count=10
2026-02-07 18:46:52 [info     ] Reranking articles

(batch)     candidates=10
2026-02-07 18:47:11 [info     ] Rerank LLM raw response

 response=[1]: 9

[2]: 8

[3]: 4

[4]: 5

[5]: 7

[6]: 8

[7]: 6

[8]: 5

[9]: 4

[10]: 3
```

```
2026-02-07 18:47:11 [info     ] Reranking completed
 scores_parsed=10
top_score=0.804200355106801
2026-02-07 18:47:11 [info     ] Generating RAG response
 articles=5 query=¿Qué está pasando con la
economía?
2026-02-07 18:47:19 [info     ] RAG response generated
 answer_length=1845
confidence=0.9456958081235906 sources_used=5
2026-02-07 18:47:19 [info     ] RAG query completed
 confidence=0.9456958081235906
sources_cited=5 sources_found=5
Evaluando RAG (con reranking): ¿Cuáles son las perspectivas económicas …
2026-02-07 18:47:19 [info     ] Using LM Studio (local)

2026-02-07 18:47:19 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:47:19 [info     ] RAG engine initialized
 llm_available=True use_reranking=True
2026-02-07 18:47:19 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Cuáles son las perspectivas económicas para 2026?
2026-02-07 18:47:19 [info     ] Searching for articles
 limit=10 query=¿Cuáles son las perspectivas
económicas para 2026? threshold=0.5
2026-02-07 18:47:20 [info     ] Search completed
 results_count=10
2026-02-07 18:47:20 [info     ] Reranking articles
(batch)     candidates=10
```

```
2026-02-07 18:47:37 [info     ] Rerank LLM raw response
 response=[1]: 9
[2]: 4
[3]: 8
[4]: 2
[5]: 1
[6]: 7
[7]: 3
[8]: 8
[9]: 2
[10]: 1
2026-02-07 18:47:37 [info     ] Reranking completed
 scores_parsed=10
top_score=0.8460391687283317
2026-02-07 18:47:37 [info     ] Generating RAG response
 articles=5 query=¿Cuáles son las
perspectivas económicas para 2026?
2026-02-07 18:47:45 [info     ] RAG response generated
 answer_length=1487
confidence=0.9104069592600408 sources_used=4
2026-02-07 18:47:45 [info     ] RAG query completed
 confidence=0.9104069592600408
sources_cited=4 sources_found=5
Evaluando RAG (con reranking): ¿Cómo está la inflación actualmente?…
2026-02-07 18:47:46 [info     ] Using LM Studio (local)

2026-02-07 18:47:46 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:47:46 [info     ] RAG engine initialized
 llm_available=True use_reranking=True
2026-02-07 18:47:46 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Cómo está la inflación actualmente?
```

```
2026-02-07 18:47:46 [info     ] Searching for articles
 limit=10 query=¿Cómo está la inflación
actualmente? threshold=0.5
2026-02-07 18:47:47 [info     ] Search completed
 results_count=10
2026-02-07 18:47:47 [info     ] Reranking articles
(batch)     candidates=10
2026-02-07 18:48:01 [info     ] Rerank LLM raw response
 response=[1]: 4
[2]: 9
[3]: 0
[4]: 0
[5]: 1
[6]: 7
[7]: 3
[8]: 0
[9]: 0
[10]: 0
2026-02-07 18:48:01 [info     ] Reranking completed
 scores_parsed=10
top_score=0.8067818912419811
2026-02-07 18:48:01 [info     ] Generating RAG response
 articles=5 query=¿Cómo está la inflación
actualmente?
2026-02-07 18:48:04 [info     ] RAG response generated
 answer_length=331
confidence=0.7013563782483963 sources_used=1
2026-02-07 18:48:04 [info     ] RAG query completed
 confidence=0.7013563782483963
sources_cited=1 sources_found=5
Evaluando RAG (con reranking): ¿Qué está pasando con los salarios?…
2026-02-07 18:48:05 [info     ] Using LM Studio (local)

2026-02-07 18:48:05 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
```

```
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:48:05 [info     ] RAG engine initialized

 llm_available=True use_reranking=True
2026-02-07 18:48:05 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Qué está pasando con los salarios?
2026-02-07 18:48:05 [info     ] Searching for articles

 limit=10 query=¿Qué está pasando con los

salarios? threshold=0.5
2026-02-07 18:48:06 [info     ] Search completed

 results_count=10
2026-02-07 18:48:06 [info     ] Reranking articles

(batch)     candidates=10
2026-02-07 18:48:21 [info     ] Rerank LLM raw response

 response=[1]: 9

[2]: 0

[3]: 0

[4]: 0

[5]: 0

[6]: 7

[7]: 0

[8]: 0

[9]: 0

[10]: 0
2026-02-07 18:48:21 [info     ] Reranking completed

 scores_parsed=10
top_score=0.7961521222034017
2026-02-07 18:48:21 [info     ] Generating RAG response

 articles=5 query=¿Qué está pasando con los

salarios?
2026-02-07 18:48:26 [info     ] RAG response generated

 answer_length=928
confidence=0.8272001940251208 sources_used=2
2026-02-07 18:48:26 [info     ] RAG query completed

 confidence=0.8272001940251208
sources_cited=2 sources_found=5
```

```
Evaluando RAG (con reranking): ¿Hay noticias sobre inversiones?…
2026-02-07 18:48:26 [info     ] Using LM Studio (local)


2026-02-07 18:48:26 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:48:26 [info     ] RAG engine initialized

 llm_available=True use_reranking=True
2026-02-07 18:48:26 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Hay noticias sobre inversiones?
2026-02-07 18:48:26 [info     ] Searching for articles

 limit=10 query=¿Hay noticias sobre

inversiones? threshold=0.5
2026-02-07 18:48:27 [info     ] Search completed

 results_count=10
2026-02-07 18:48:27 [info     ] Reranking articles

(batch)     candidates=10
2026-02-07 18:48:45 [info     ] Rerank LLM raw response

 response=[1]: 7

[2]: 9

[3]: 6

[4]: 8

[5]: 0

[6]: 0

[7]: 0

[8]: 9

[9]: 4

[10]: 1
2026-02-07 18:48:45 [info     ] Reranking completed

 scores_parsed=10
top_score=0.7795532668169513
2026-02-07 18:48:45 [info     ] Generating RAG response

 articles=5 query=¿Hay noticias sobre

inversiones?
```

```
2026-02-07 18:48:49 [info     ] RAG response generated
 answer_length=594
confidence=0.871050789816314 sources_used=3
2026-02-07 18:48:49 [info     ] RAG query completed
 confidence=0.871050789816314
sources_cited=3 sources_found=5
Evaluando RAG (con reranking): ¿Qué dicen los analistas económicos?…
2026-02-07 18:48:49 [info     ] Using LM Studio (local)

2026-02-07 18:48:49 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:48:49 [info     ] RAG engine initialized
 llm_available=True use_reranking=True
2026-02-07 18:48:49 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué dicen los analistas económicos?
2026-02-07 18:48:49 [info     ] Searching for articles
 limit=10 query=¿Qué dicen los analistas
económicos? threshold=0.5
2026-02-07 18:48:50 [info     ] Search completed
 results_count=10
2026-02-07 18:48:50 [info     ] Reranking articles
(batch)     candidates=10
2026-02-07 18:49:09 [info     ] Rerank LLM raw response
 response=[1]: 9
[2]: 7
[3]: 4
[4]: 2
[5]: 6
[6]: 0
[7]: 3
[8]: 5
[9]: 4
[10]: 2
```

```
2026-02-07 18:49:09 [info     ] Reranking completed
 scores_parsed=10
top_score=0.8280586977548614
2026-02-07 18:49:09 [info     ] Generating RAG response
 articles=5 query=¿Qué dicen los analistas
económicos?
2026-02-07 18:49:16 [info     ] RAG response generated
 answer_length=1483
confidence=0.9332399803472898 sources_used=5
2026-02-07 18:49:16 [info     ] RAG query completed
 confidence=0.9332399803472898
sources_cited=5 sources_found=5
Evaluando RAG (con reranking): ¿Cuáles son las últimas noticias de tecn…
2026-02-07 18:49:16 [info     ] Using LM Studio (local)


2026-02-07 18:49:16 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:49:16 [info     ] RAG engine initialized
 llm_available=True use_reranking=True
2026-02-07 18:49:16 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Cuáles son las últimas noticias de tecnología?
2026-02-07 18:49:16 [info     ] Searching for articles
 limit=10 query=¿Cuáles son las últimas
noticias de tecnología? threshold=0.5
2026-02-07 18:49:17 [info     ] Search completed
 results_count=10
2026-02-07 18:49:17 [info     ] Reranking articles
(batch)     candidates=10
```

```
2026-02-07 18:49:34 [info     ] Rerank LLM raw response
 response=[1]: 4
[2]: 7
[3]: 7
[4]: 6
[5]: 3
[6]: 2
[7]: 4
[8]: 8
[9]: 7
[10]: 8
2026-02-07 18:49:34 [info     ] Reranking completed
 scores_parsed=10
top_score=0.7402440935009762
2026-02-07 18:49:34 [info     ] Generating RAG response
 articles=5 query=¿Cuáles son las últimas
noticias de tecnología?
2026-02-07 18:49:42 [info     ] RAG response generated
 answer_length=1625
confidence=0.9038557300290496 sources_used=4
2026-02-07 18:49:42 [info     ] RAG query completed
 confidence=0.9038557300290496
sources_cited=4 sources_found=5
Evaluando RAG (con reranking): ¿Qué avances hay en inteligencia artific…
2026-02-07 18:49:42 [info     ] Using LM Studio (local)

2026-02-07 18:49:42 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:49:42 [info     ] RAG engine initialized
 llm_available=True use_reranking=True
2026-02-07 18:49:42 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué avances hay en inteligencia artificial?
```

```
2026-02-07 18:49:42 [info     ] Searching for articles
 limit=10 query=¿Qué avances hay en
inteligencia artificial? threshold=0.5
2026-02-07 18:49:43 [info     ] Search completed
 results_count=10
2026-02-07 18:49:43 [info     ] Reranking articles
(batch)     candidates=10
2026-02-07 18:49:59 [info     ] Rerank LLM raw response
 response=[1]: 2
[2]: 9
[3]: 8
[4]: 9
[5]: 7
[6]: 8
[7]: 7
[8]: 8
[9]: 6
[10]: 5
2026-02-07 18:49:59 [info     ] Reranking completed
 scores_parsed=10
top_score=0.8546379036661841
2026-02-07 18:49:59 [info     ] Generating RAG response
 articles=5 query=¿Qué avances hay en
inteligencia artificial?
2026-02-07 18:50:06 [info     ] RAG response generated
 answer_length=1286
confidence=0.962383481196574 sources_used=5
2026-02-07 18:50:06 [info     ] RAG query completed
 confidence=0.962383481196574
sources_cited=5 sources_found=5
Evaluando RAG (con reranking): ¿Hay noticias sobre redes sociales?…
2026-02-07 18:50:06 [info     ] Using LM Studio (local)

2026-02-07 18:50:06 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
```

```
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:50:06 [info     ] RAG engine initialized
 llm_available=True use_reranking=True
2026-02-07 18:50:06 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Hay noticias sobre redes sociales?
2026-02-07 18:50:06 [info     ] Searching for articles
 limit=10 query=¿Hay noticias sobre redes
sociales? threshold=0.5
2026-02-07 18:50:07 [info     ] Search completed
 results_count=10
2026-02-07 18:50:07 [info     ] Reranking articles
(batch)     candidates=10
2026-02-07 18:50:23 [info     ] Rerank LLM raw response
 response=[1]: 9
[2]: 8
[3]: 4
[4]: 5
[5]: 7
[6]: 3
[7]: 8
[8]: 2
[9]: 4
[10]: 9
2026-02-07 18:50:23 [info     ] Reranking completed
 scores_parsed=10
top_score=0.7986886398958282
2026-02-07 18:50:23 [info     ] Generating RAG response
 articles=5 query=¿Hay noticias sobre redes
sociales?
2026-02-07 18:50:27 [info     ] RAG response generated
 answer_length=726
confidence=0.8743215226014269 sources_used=3
2026-02-07 18:50:27 [info     ] RAG query completed
 confidence=0.8743215226014269
sources_cited=3 sources_found=5
```

```
Evaluando RAG (con reranking): ¿Qué está pasando con las empresas tecno…
2026-02-07 18:50:27 [info     ] Using LM Studio (local)


2026-02-07 18:50:27 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:50:27 [info     ] RAG engine initialized

 llm_available=True use_reranking=True
2026-02-07 18:50:27 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Qué está pasando con las empresas tecnológicas?
2026-02-07 18:50:27 [info     ] Searching for articles

 limit=10 query=¿Qué está pasando con las

empresas tecnológicas? threshold=0.5
2026-02-07 18:50:28 [info     ] Search completed

 results_count=10
2026-02-07 18:50:28 [info     ] Reranking articles

(batch)     candidates=10
2026-02-07 18:50:47 [info     ] Rerank LLM raw response

 response=[1]: 3

[2]: 8

[3]: 8

[4]: 2

[5]: 4

[6]: 7

[7]: 6

[8]: 5

[9]: 4

[10]: 9
2026-02-07 18:50:47 [info     ] Reranking completed

 scores_parsed=10
top_score=0.8017240141999771
2026-02-07 18:50:47 [info     ] Generating RAG response

 articles=5 query=¿Qué está pasando con las

empresas tecnológicas?
```

```
2026-02-07 18:50:53 [info     ] RAG response generated
 answer_length=1400
confidence=0.9117360593820226 sources_used=4
2026-02-07 18:50:53 [info     ] RAG query completed
 confidence=0.9117360593820226
sources_cited=4 sources_found=5
Evaluando RAG (con reranking): ¿Hay noticias sobre ciberseguridad?…
2026-02-07 18:50:53 [info     ] Using LM Studio (local)


2026-02-07 18:50:53 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:50:53 [info     ] RAG engine initialized
 llm_available=True use_reranking=True
2026-02-07 18:50:53 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Hay noticias sobre ciberseguridad?
2026-02-07 18:50:53 [info     ] Searching for articles
 limit=10 query=¿Hay noticias sobre
ciberseguridad? threshold=0.5
2026-02-07 18:50:54 [info     ] Search completed
 results_count=10
2026-02-07 18:50:54 [info     ] Reranking articles
(batch)     candidates=10
2026-02-07 18:51:10 [info     ] Rerank LLM raw response
 response=[1]: 0
[2]: 10
[3]: 0
[4]: 0
[5]: 1
[6]: 2
[7]: 0
[8]: 0
[9]: 0
[10]: 0
```

```
2026-02-07 18:51:10 [info     ] Reranking completed
 scores_parsed=10
top_score=0.8294071541096528
2026-02-07 18:51:10 [info     ] Generating RAG response
 articles=5 query=¿Hay noticias sobre
ciberseguridad?
2026-02-07 18:51:13 [info     ] RAG response generated
 answer_length=270
confidence=0.7058814308219306 sources_used=1
2026-02-07 18:51:13 [info     ] RAG query completed
 confidence=0.7058814308219306
sources_cited=1 sources_found=5
Evaluando RAG (con reranking): ¿Qué noticias hay sobre salud?…
2026-02-07 18:51:13 [info     ] Using LM Studio (local)

2026-02-07 18:51:13 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:51:13 [info     ] RAG engine initialized
 llm_available=True use_reranking=True
2026-02-07 18:51:13 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué noticias hay sobre salud?
2026-02-07 18:51:13 [info     ] Searching for articles
 limit=10 query=¿Qué noticias hay sobre
salud? threshold=0.5
2026-02-07 18:51:14 [info     ] Search completed
 results_count=10
2026-02-07 18:51:14 [info     ] Reranking articles
(batch)     candidates=10
```

```
2026-02-07 18:51:31 [info     ] Rerank LLM raw response
 response=[1]: 10
[2]: 9
[3]: 8
[4]: 7
[5]: 6
[6]: 6
[7]: 4
[8]: 0
[9]: 9
[10]: 8
2026-02-07 18:51:31 [info     ] Reranking completed
 scores_parsed=10
top_score=0.8464085689565114
2026-02-07 18:51:31 [info     ] Generating RAG response
 articles=5 query=¿Qué noticias hay sobre
salud?
2026-02-07 18:51:40 [info     ] RAG response generated
 answer_length=1953
confidence=0.9545461950583136 sources_used=5
2026-02-07 18:51:40 [info     ] RAG query completed
 confidence=0.9545461950583136
sources_cited=5 sources_found=5
Evaluando RAG (con reranking): ¿Hay noticias sobre vacunas?…
2026-02-07 18:51:40 [info     ] Using LM Studio (local)

2026-02-07 18:51:40 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:51:40 [info     ] RAG engine initialized
 llm_available=True use_reranking=True
2026-02-07 18:51:40 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Hay noticias sobre vacunas?
```

```
2026-02-07 18:51:40 [info     ] Searching for articles
 limit=10 query=¿Hay noticias sobre
vacunas? threshold=0.5
2026-02-07 18:51:41 [info     ] Search completed
 results_count=10
2026-02-07 18:51:41 [info     ] Reranking articles
(batch)     candidates=10
2026-02-07 18:51:58 [info     ] Rerank LLM raw response
 response=[1]: 7
[2]: 8
[3]: 8
[4]: 5
[5]: 7
[6]: 8
[7]: 2
[8]: 2
[9]: 7
[10]: 2
2026-02-07 18:51:58 [info     ] Reranking completed
 scores_parsed=10
top_score=0.746730328385754
2026-02-07 18:51:58 [info     ] Generating RAG response
 articles=5 query=¿Hay noticias sobre
vacunas?
2026-02-07 18:52:04 [info     ] RAG response generated
 answer_length=1157
confidence=0.9445733537275957 sources_used=5
2026-02-07 18:52:04 [info     ] RAG query completed
 confidence=0.9445733537275957
sources_cited=5 sources_found=5
Evaluando RAG (con reranking): ¿Qué dicen los expertos sobre salud ment…
2026-02-07 18:52:04 [info     ] Using LM Studio (local)

2026-02-07 18:52:04 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
```

```
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:52:04 [info     ] RAG engine initialized

 llm_available=True use_reranking=True
2026-02-07 18:52:04 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Qué dicen los expertos sobre salud mental?
2026-02-07 18:52:04 [info     ] Searching for articles

 limit=10 query=¿Qué dicen los expertos

sobre salud mental? threshold=0.5
2026-02-07 18:52:05 [info     ] Search completed

 results_count=10
2026-02-07 18:52:05 [info     ] Reranking articles

(batch)     candidates=10
2026-02-07 18:52:24 [info     ] Rerank LLM raw response

 response=[1]: 6

[2]: 7

[3]: 8

[4]: 9

[5]: 6

[6]: 7

[7]: 8

[8]: 4

[9]: 7

[10]: 6
2026-02-07 18:52:24 [info     ] Reranking completed

 scores_parsed=10
top_score=0.8277916358585201
2026-02-07 18:52:24 [info     ] Generating RAG response

 articles=5 query=¿Qué dicen los expertos

sobre salud mental?
2026-02-07 18:52:32 [info     ] RAG response generated

 answer_length=2109
confidence=0.9536715190457543 sources_used=5
2026-02-07 18:52:32 [info     ] RAG query completed

 confidence=0.9536715190457543
sources_cited=5 sources_found=5
```

```
Evaluando RAG (con reranking): ¿Hay avances médicos recientes?…
2026-02-07 18:52:33 [info     ] Using LM Studio (local)


2026-02-07 18:52:33 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:52:33 [info     ] RAG engine initialized

 llm_available=True use_reranking=True
2026-02-07 18:52:33 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Hay avances médicos recientes?
2026-02-07 18:52:33 [info     ] Searching for articles

 limit=10 query=¿Hay avances médicos

recientes? threshold=0.5
2026-02-07 18:52:34 [info     ] Search completed

 results_count=10
2026-02-07 18:52:34 [info     ] Reranking articles

(batch)     candidates=10
2026-02-07 18:52:50 [info     ] Rerank LLM raw response

 response=[1]: 0

[2]: 1

[3]: 1

[4]: 2

[5]: 0

[6]: 1

[7]: 1

[8]: 1

[9]: 0

[10]: 0
2026-02-07 18:52:50 [info     ] Reranking completed

 scores_parsed=10
top_score=0.45056528140927854
2026-02-07 18:52:50 [info     ] Generating RAG response

 articles=5 query=¿Hay avances médicos

recientes?
```

```
2026-02-07 18:52:57 [info     ] RAG response generated
 answer_length=971
confidence=0.765313912856632 sources_used=2
2026-02-07 18:52:57 [info     ] RAG query completed
 confidence=0.765313912856632
sources_cited=2 sources_found=5
Evaluando RAG (con reranking): ¿Qué está pasando en el mundo del deport…
2026-02-07 18:52:57 [info     ] Using LM Studio (local)


2026-02-07 18:52:57 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:52:57 [info     ] RAG engine initialized
 llm_available=True use_reranking=True
2026-02-07 18:52:57 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué está pasando en el mundo del deporte?
2026-02-07 18:52:57 [info     ] Searching for articles
 limit=10 query=¿Qué está pasando en el
mundo del deporte? threshold=0.5
2026-02-07 18:52:58 [info     ] Search completed
 results_count=10
2026-02-07 18:52:58 [info     ] Reranking articles
(batch)     candidates=10
2026-02-07 18:53:18 [info     ] Rerank LLM raw response
 response=[1]: 9
[2]: 4
[3]: 8
[4]: 6
[5]: 7
[6]: 5
[7]: 3
[8]: 4
[9]: 2
[10]: 7
```

```
2026-02-07 18:53:18 [info     ] Reranking completed
 scores_parsed=10
top_score=0.8228845230604568
2026-02-07 18:53:18 [info     ] Generating RAG response
 articles=5 query=¿Qué está pasando en el
mundo del deporte?
2026-02-07 18:53:25 [info     ] RAG response generated
 answer_length=1262
confidence=0.909689953547235 sources_used=4
2026-02-07 18:53:25 [info     ] RAG query completed
 confidence=0.909689953547235
sources_cited=4 sources_found=5
Evaluando RAG (con reranking): ¿Cuáles son los resultados deportivos re…
2026-02-07 18:53:25 [info     ] Using LM Studio (local)


2026-02-07 18:53:25 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:53:25 [info     ] RAG engine initialized
 llm_available=True use_reranking=True
2026-02-07 18:53:25 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Cuáles son los resultados deportivos recientes?
2026-02-07 18:53:25 [info     ] Searching for articles
 limit=10 query=¿Cuáles son los resultados
deportivos recientes? threshold=0.5
2026-02-07 18:53:27 [info     ] Search completed
 results_count=10
2026-02-07 18:53:27 [info     ] Reranking articles
(batch)     candidates=10
```

```
2026-02-07 18:53:44 [info     ] Rerank LLM raw response
 response=[1]: 0
[2]: 3
[3]: 0
[4]: 7
[5]: 9
[6]: 1
[7]: 0
[8]: 0
[9]: 0
[10]: 0
2026-02-07 18:53:44 [info     ] Reranking completed
 scores_parsed=10
top_score=0.791310848554647
2026-02-07 18:53:44 [info     ] Generating RAG response
 articles=5 query=¿Cuáles son los resultados
deportivos recientes?
2026-02-07 18:53:47 [info     ] RAG response generated
 answer_length=317
confidence=0.8282764014822491 sources_used=2
2026-02-07 18:53:47 [info     ] RAG query completed
 confidence=0.8282764014822491
sources_cited=2 sources_found=5
Evaluando RAG (con reranking): ¿Hay noticias de fútbol?…
2026-02-07 18:53:47 [info     ] Using LM Studio (local)

2026-02-07 18:53:47 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:53:47 [info     ] RAG engine initialized
 llm_available=True use_reranking=True
2026-02-07 18:53:47 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Hay noticias de fútbol?
2026-02-07 18:53:47 [info     ] Searching for articles
 limit=10 query=¿Hay noticias de fútbol?
```

```
threshold=0.5
2026-02-07 18:53:49 [info     ] Search completed

 results_count=10
2026-02-07 18:53:49 [info     ] Reranking articles

(batch)     candidates=10
2026-02-07 18:54:04 [info     ] Rerank LLM raw response

 response=[1]: 8

[2]: 9

[3]: 10

[4]: 9

[5]: 7

[6]: 6

[7]: 10

[8]: 9

[9]: 8

[10]: 10
2026-02-07 18:54:04 [info     ] Reranking completed

 scores_parsed=10
top_score=0.8607637921865204
2026-02-07 18:54:04 [info     ] Generating RAG response

 articles=5 query=¿Hay noticias de

fútbol?
2026-02-07 18:54:09 [info     ] RAG response generated

 answer_length=714
confidence=0.9669264294582053 sources_used=5
2026-02-07 18:54:09 [info     ] RAG query completed

 confidence=0.9669264294582053
sources_cited=5 sources_found=5
Evaluando RAG (con reranking): ¿Qué fichajes o traspasos hay en el fútb…
2026-02-07 18:54:09 [info     ] Using LM Studio (local)


2026-02-07 18:54:09 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:54:09 [info     ] RAG engine initialized

 llm_available=True use_reranking=True
```

```
2026-02-07 18:54:09 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué fichajes o traspasos hay en el fútbol?
2026-02-07 18:54:09 [info     ] Searching for articles
 limit=10 query=¿Qué fichajes o traspasos
hay en el fútbol? threshold=0.5
2026-02-07 18:54:10 [info     ] Search completed
 results_count=10
2026-02-07 18:54:10 [info     ] Reranking articles
(batch)     candidates=10
2026-02-07 18:54:27 [info     ] Rerank LLM raw response
 response=[1]: 1
[2]: 0
[3]: 4
[4]: 0
[5]: 2
[6]: 0
[7]: 9
[8]: 0
[9]: 0
[10]: 0
2026-02-07 18:54:27 [info     ] Reranking completed
 scores_parsed=10
top_score=0.8040796445035829
2026-02-07 18:54:27 [info     ] Generating RAG response
 articles=5 query=¿Qué fichajes o traspasos
hay en el fútbol?
2026-02-07 18:54:31 [info     ] RAG response generated
 answer_length=558 confidence=0.5
sources_used=0
2026-02-07 18:54:31 [info     ] RAG query completed
 confidence=0.5 sources_cited=3
sources_found=5
Evaluando RAG (con reranking): ¿Hay noticias sobre ciencia?…
2026-02-07 18:54:31 [info     ] Using LM Studio (local)
```

```
2026-02-07 18:54:31 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:54:31 [info     ] RAG engine initialized
 llm_available=True use_reranking=True
2026-02-07 18:54:31 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Hay noticias sobre ciencia?
2026-02-07 18:54:31 [info     ] Searching for articles
 limit=10 query=¿Hay noticias sobre
ciencia? threshold=0.5
2026-02-07 18:54:32 [info     ] Search completed
 results_count=10
2026-02-07 18:54:32 [info     ] Reranking articles
(batch)     candidates=10
2026-02-07 18:54:48 [info     ] Rerank LLM raw response
 response=[1]: 9
[2]: 7
[3]: 8
[4]: 6
[5]: 8
[6]: 9
[7]: 7
[8]: 8
[9]: 9
[10]: 5
2026-02-07 18:54:48 [info     ] Reranking completed
 scores_parsed=10
top_score=0.8022133248043066
2026-02-07 18:54:48 [info     ] Generating RAG response
 articles=5 query=¿Hay noticias sobre
ciencia?
2026-02-07 18:54:53 [info     ] RAG response generated
 answer_length=960
confidence=0.953146401120668 sources_used=5
```

```
2026-02-07 18:54:53 [info     ] RAG query completed
 confidence=0.953146401120668
sources_cited=5 sources_found=5
Evaluando RAG (con reranking): ¿Qué descubrimientos científicos hay?…
2026-02-07 18:54:53 [info     ] Using LM Studio (local)


2026-02-07 18:54:53 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:54:53 [info     ] RAG engine initialized
 llm_available=True use_reranking=True
2026-02-07 18:54:53 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué descubrimientos científicos hay?
2026-02-07 18:54:53 [info     ] Searching for articles
 limit=10 query=¿Qué descubrimientos
científicos hay? threshold=0.5
2026-02-07 18:54:54 [info     ] Search completed
 results_count=10
2026-02-07 18:54:54 [info     ] Reranking articles
(batch)     candidates=10
2026-02-07 18:55:13 [info     ] Rerank LLM raw response
 response=[1]: 9
[2]: 1
[3]: 4
[4]: 8
[5]: 2
[6]: 3
[7]: 1
[8]: 4
[9]: 8
[10]: 5
2026-02-07 18:55:13 [info     ] Reranking completed
 scores_parsed=10
top_score=0.8175481208313727
```

```
2026-02-07 18:55:13 [info     ] Generating RAG response
 articles=5 query=¿Qué descubrimientos
científicos hay?
2026-02-07 18:55:19 [info     ] RAG response generated
 answer_length=1357
confidence=0.8906743830837859 sources_used=4
2026-02-07 18:55:19 [info     ] RAG query completed
 confidence=0.8906743830837859
sources_cited=4 sources_found=5
Evaluando RAG (con reranking): ¿Hay noticias sobre el espacio o astrono…
2026-02-07 18:55:19 [info     ] Using LM Studio (local)


2026-02-07 18:55:19 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:55:19 [info     ] RAG engine initialized
 llm_available=True use_reranking=True
2026-02-07 18:55:19 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Hay noticias sobre el espacio o astronomía?
2026-02-07 18:55:19 [info     ] Searching for articles
 limit=10 query=¿Hay noticias sobre el
espacio o astronomía? threshold=0.5
2026-02-07 18:55:20 [info     ] Search completed
 results_count=10
2026-02-07 18:55:20 [info     ] Reranking articles
(batch)     candidates=10
```

```
2026-02-07 18:55:39 [info     ] Rerank LLM raw response
 response=[1]: 8
[2]: 7
[3]: 0
[4]: 6
[5]: 9
[6]: 8
[7]: 0
[8]: 9
[9]: 7
[10]: 8
2026-02-07 18:55:39 [info     ] Reranking completed
 scores_parsed=10
top_score=0.8077487638296196
2026-02-07 18:55:39 [info     ] Generating RAG response
 articles=5 query=¿Hay noticias sobre el
espacio o astronomía?
2026-02-07 18:55:44 [info     ] RAG response generated
 answer_length=790
confidence=0.9142205458210876 sources_used=4
2026-02-07 18:55:44 [info     ] RAG query completed
 confidence=0.9142205458210876
sources_cited=4 sources_found=5
Evaluando RAG (con reranking): ¿Qué noticias hay de entretenimiento?…
2026-02-07 18:55:44 [info     ] Using LM Studio (local)

2026-02-07 18:55:44 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:55:44 [info     ] RAG engine initialized
 llm_available=True use_reranking=True
2026-02-07 18:55:44 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué noticias hay de entretenimiento?
```

```
2026-02-07 18:55:44 [info     ] Searching for articles
 limit=10 query=¿Qué noticias hay de
entretenimiento? threshold=0.5
2026-02-07 18:55:45 [info     ] Search completed
 results_count=10
2026-02-07 18:55:45 [info     ] Reranking articles
(batch)     candidates=10
2026-02-07 18:56:02 [info     ] Rerank LLM raw response
 response=[1]: 0
[2]: 3
[3]: 0
[4]: 3
[5]: 7
[6]: 0
[7]: 8
[8]: 0
[9]: 3
[10]: 4
2026-02-07 18:56:02 [info     ] Reranking completed
 scores_parsed=10
top_score=0.7227886865030329
2026-02-07 18:56:02 [info     ] Generating RAG response
 articles=5 query=¿Qué noticias hay de
entretenimiento?
2026-02-07 18:56:08 [info     ] RAG response generated
 answer_length=853
confidence=0.81965436703466 sources_used=2
2026-02-07 18:56:08 [info     ] RAG query completed
 confidence=0.81965436703466
sources_cited=2 sources_found=5
Evaluando RAG (con reranking): ¿Hay estrenos de cine o series?…
2026-02-07 18:56:08 [info     ] Using LM Studio (local)

2026-02-07 18:56:08 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
```

```
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:56:08 [info     ] RAG engine initialized

 llm_available=True use_reranking=True
2026-02-07 18:56:08 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Hay estrenos de cine o series?
2026-02-07 18:56:08 [info     ] Searching for articles

 limit=10 query=¿Hay estrenos de cine o

series? threshold=0.5
2026-02-07 18:56:09 [info     ] Search completed

 results_count=10
2026-02-07 18:56:09 [info     ] Reranking articles

(batch)     candidates=10
2026-02-07 18:56:29 [info     ] Rerank LLM raw response

 response=[1]: 10

[2]: 5

[3]: 8

[4]: 7

[5]: 3

[6]: 0

[7]: 0

[8]: 2

[9]: 0

[10]: 4
2026-02-07 18:56:29 [info     ] Reranking completed

 scores_parsed=10
top_score=0.8599555158509463
2026-02-07 18:56:29 [info     ] Generating RAG response

 articles=5 query=¿Hay estrenos de cine o

series?
2026-02-07 18:56:34 [info     ] RAG response generated

 answer_length=620
confidence=0.8397645135418853 sources_used=2
2026-02-07 18:56:34 [info     ] RAG query completed

 confidence=0.8397645135418853
sources_cited=2 sources_found=5
```

```
Evaluando RAG (con reranking): ¿Cuáles son las noticias internacionales…
2026-02-07 18:56:34 [info     ] Using LM Studio (local)


2026-02-07 18:56:34 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 18:56:34 [info     ] RAG engine initialized

 llm_available=True use_reranking=True
2026-02-07 18:56:34 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Cuáles son las noticias internacionales más impor
2026-02-07 18:56:34 [info     ] Searching for articles

 limit=10 query=¿Cuáles son las noticias

internacionales más impor threshold=0.5
2026-02-07 18:56:35 [info     ] Search completed

 results_count=10
2026-02-07 18:56:35 [info     ] Reranking articles

(batch)     candidates=10
2026-02-07 18:56:52 [info     ] Rerank LLM raw response

 response=[1]: 3

[2]: 4

[3]: 2

[4]: 8

[5]: 9

[6]: 7

[7]: 6

[8]: 2

[9]: 6

[10]: 5
2026-02-07 18:56:52 [info     ] Reranking completed

 scores_parsed=10
top_score=0.778212408838036
2026-02-07 18:56:52 [info     ] Generating RAG response

 articles=5 query=¿Cuáles son las noticias

internacionales más impor
```

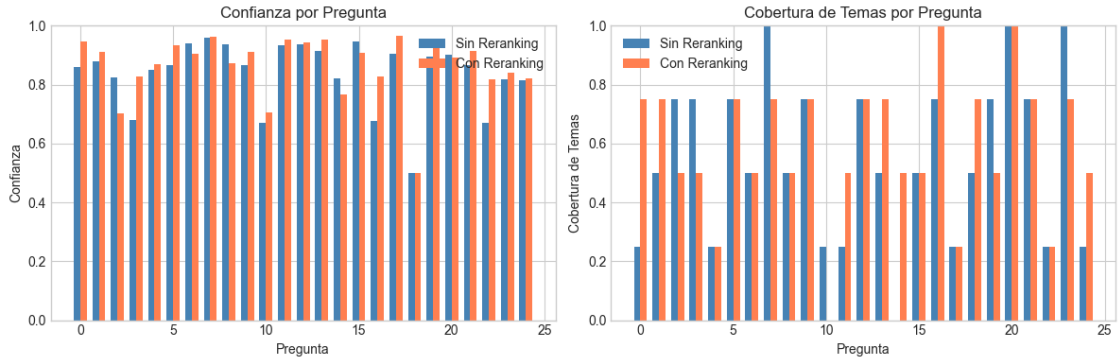
```
2026-02-07 18:56:57 [info     ] RAG response generated
 answer_length=682
confidence=0.8206639616011344 sources_used=2
2026-02-07 18:56:57 [info     ] RAG query completed
 confidence=0.8206639616011344
sources_cited=2 sources_found=5
```

[17]:
```
# Comparación lado a lado
comparison = pd.DataFrame({
    'Métrica': ['Confianza', 'Cobertura Temas', 'Citas Usadas', 'Longitud␣
 ↪Respuesta'],
    'Sin Reranking': [
        f"{df_generation['confidence'].mean():.2%}",
        f"{df_generation['topic_coverage'].mean():.2%}",
        f"{df_generation['citations_in_answer'].mean():.1f}",
        f"{df_generation['answer_length'].mean():.0f}",
    ],
    'Con Reranking': [
        f"{df_generation_rerank['confidence'].mean():.2%}",
        f"{df_generation_rerank['topic_coverage'].mean():.2%}",
        f"{df_generation_rerank['citations_in_answer'].mean():.1f}",
        f"{df_generation_rerank['answer_length'].mean():.0f}",
    ],
})

print("\nCOMPARACIÓN: Sin Reranking vs Con Reranking")
print("=" * 50)
print(comparison.to_string(index=False))
```

```
COMPARACIÓN: Sin Reranking vs Con Reranking
==================================================
           Métrica Sin Reranking Con Reranking
         Confianza        83.71%        86.43%
   Cobertura Temas        55.00%        59.00%
      Citas Usadas           3.0           3.4
Longitud Respuesta           920          1051
```

[14]:
```
# Visualización de comparación
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

# Confianza
x = range(len(df_generation))
width = 0.35
axes[0].bar([i - width/2 for i in x], df_generation['confidence'], width,␣
 ↪label='Sin Reranking', color='steelblue')
```

```
axes[0].bar([i + width/2 for i in x], df_generation_rerank['confidence'],␣
↪width, label='Con Reranking', color='coral')
axes[0].set_xlabel('Pregunta')
axes[0].set_ylabel('Confianza')
axes[0].set_title('Confianza por Pregunta')
axes[0].legend()
axes[0].set_ylim(0, 1)

# Topic Coverage
axes[1].bar([i - width/2 for i in x], df_generation['topic_coverage'], width,␣
↪label='Sin Reranking', color='steelblue')
axes[1].bar([i + width/2 for i in x], df_generation_rerank['topic_coverage'],␣
↪width, label='Con Reranking', color='coral')
axes[1].set_xlabel('Pregunta')
axes[1].set_ylabel('Cobertura de Temas')
axes[1].set_title('Cobertura de Temas por Pregunta')
axes[1].legend()
axes[1].set_ylim(0, 1)

plt.tight_layout()
plt.savefig('../docs/images/reranking_comparison.png', dpi=150,␣
↪bbox_inches='tight')
plt.show()
```



1.7 6. Evaluación de Faithfulness (LLM-as-Judge)

Faithfulness mide si la respuesta generada está basada fielmente en los documentos recuperados (no alucina).

[18]:
```
import re

def evaluate_faithfulness(question: str, answer: str, articles: list) -> dict:
    """
```

```
    Evalúa si la respuesta es fiel a las fuentes (no alucina).
    Usa LLM-as-Judge para evaluar.

    Args:
        question: La pregunta original
        answer: La respuesta generada
        articles: Lista de tuplas (Article, score) con contenido completo
    """
    llm = get_available_client()
    if not llm:
        return {"faithfulness_score": None, "explanation": "No LLM available"}

    # Incluir contenido de los artículos, no solo títulos
    sources_text = ""
    for i, (article, score) in enumerate(articles):
        content = article.content or article.description or ""
        content_preview = content[:500] + "..." if len(content) > 500 else␣
 ↪content
        sources_text += f"\n[{i+1}] {article.title}\n"
        sources_text += f"    Fuente: {article.source_name}\n"
        sources_text += f"    Contenido: {content_preview}\n"

    prompt = f"""Evalúa si la siguiente respuesta está basada fielmente en las␣
 ↪fuentes proporcionadas.

PREGUNTA: {question}

FUENTES:
{sources_text}

RESPUESTA: {answer}

Evalúa del 1 al 10:
- 10: Completamente fiel, todo está en las fuentes
- 5: Parcialmente fiel, algunas cosas no están en las fuentes
- 1: No es fiel, hay información inventada

Responde SOLO con el número (1-10):"""

    try:
        response = llm.generate(prompt, max_tokens=150, temperature=0)
        raw_response = response.content.strip()

        print(f"  LLM response: '{raw_response}'")

        numbers = re.findall(r'\b(\d+)\b', raw_response)
        if numbers:
```

```
            score = min(int(numbers[0]), 10) / 10.0
            print(f"  Score extraído: {score:.0%}")
            return {"faithfulness_score": score}
        else:
            print(f"  No se encontró número en la respuesta")
            return {"faithfulness_score": None}

    except Exception as e:
        print(f"  Error: {e}")
        return {"faithfulness_score": None}

print("  Función evaluate_faithfulness definida (con contenido de artículos)")
```

```
  Función evaluate_faithfulness definida (con contenido de artículos)
```

[19]:
```
# Evaluar faithfulness para las respuestas generadas por LM Studio
# Ahora usando contenido completo de los artículos (no solo títulos)
faithfulness_scores = []

for i, result in enumerate(generation_results):
    print(f"Evaluando faithfulness {i+1}/{len(generation_results)}...")

    with get_db() as db:
        engine = RAGEngine(db, use_reranking=False)
        # Obtener artículos CON CONTENIDO
        articles = engine.get_context_only(
            question=EVALUATION_QUESTIONS[i]['question'],
            max_sources=5
        )

        faith_result = evaluate_faithfulness(
            question=EVALUATION_QUESTIONS[i]['question'],
            answer=result['answer'],
            articles=articles,
        )
        faithfulness_scores.append(faith_result['faithfulness_score'])

df_generation['faithfulness'] = faithfulness_scores
print(f"\n{'='*50}")
print(f"Faithfulness LM Studio promedio: {df_generation['faithfulness'].mean():.
 ↪2%}")
```

```
Evaluando faithfulness 1/25…
2026-02-07 19:22:08 [info     ] Using LM Studio (local)


2026-02-07 19:22:08 [info     ] Initialized
```

LMStudioEmbeddingGenerator

```
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:22:08 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:22:08 [info     ] Searching for articles

 limit=5 query=¿Qué está pasando con la

economía? threshold=0.5
2026-02-07 19:22:08 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:22:08 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:22:08 [info     ] Searching for articles

 limit=5 query=¿Qué está pasando con la

economía? threshold=0.5
2026-02-07 19:22:10 [info     ] Search completed

 results_count=5
2026-02-07 19:22:10 [info     ] Using LM Studio (local)


  LLM response: '5'
  Score extraído: 50%
Evaluando faithfulness 2/25…
2026-02-07 19:22:28 [info     ] Using LM Studio (local)


2026-02-07 19:22:28 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:22:28 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:22:28 [info     ] Searching for articles

 limit=5 query=¿Cuáles son las perspectivas

económicas para 2026? threshold=0.5
2026-02-07 19:22:30 [info     ] Search completed

 results_count=5
2026-02-07 19:22:30 [info     ] Using LM Studio (local)


  LLM response: '8'
```

```
  Score extraído: 80%
Evaluando faithfulness 3/25…
2026-02-07 19:22:44 [info     ] Using LM Studio (local)


2026-02-07 19:22:44 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:22:44 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:22:44 [info     ] Searching for articles

 limit=5 query=¿Cómo está la inflación

actualmente? threshold=0.5
2026-02-07 19:22:45 [info     ] Search completed

 results_count=5
2026-02-07 19:22:45 [info     ] Using LM Studio (local)


  LLM response: '7'
  Score extraído: 70%
Evaluando faithfulness 4/25…
2026-02-07 19:23:00 [info     ] Using LM Studio (local)


2026-02-07 19:23:00 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:23:00 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:23:00 [info     ] Searching for articles

 limit=5 query=¿Qué está pasando con los

salarios? threshold=0.5
2026-02-07 19:23:01 [info     ] Search completed

 results_count=5
2026-02-07 19:23:01 [info     ] Using LM Studio (local)


  LLM response: ''
  No se encontró número en la respuesta
Evaluando faithfulness 5/25…
```

```
2026-02-07 19:23:21 [info     ] Using LM Studio (local)

2026-02-07 19:23:21 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:23:21 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:23:21 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
inversiones? threshold=0.5
2026-02-07 19:23:22 [info     ] Search completed
 results_count=5
2026-02-07 19:23:23 [info     ] Using LM Studio (local)

  LLM response: '5'
  Score extraído: 50%
Evaluando faithfulness 6/25…
2026-02-07 19:23:37 [info     ] Using LM Studio (local)

2026-02-07 19:23:37 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:23:37 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:23:37 [info     ] Searching for articles
 limit=5 query=¿Qué dicen los analistas
económicos? threshold=0.5
2026-02-07 19:23:38 [info     ] Search completed
 results_count=5
2026-02-07 19:23:38 [info     ] Using LM Studio (local)

  LLM response: '8'
  Score extraído: 80%
Evaluando faithfulness 7/25…
2026-02-07 19:23:52 [info     ] Using LM Studio (local)
```

```
2026-02-07 19:23:52 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:23:52 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:23:52 [info     ] Searching for articles
 limit=5 query=¿Cuáles son las últimas
noticias de tecnología? threshold=0.5
2026-02-07 19:23:53 [info     ] Search completed
 results_count=5
2026-02-07 19:23:53 [info     ] Using LM Studio (local)


  LLM response: '10'
  Score extraído: 100%
Evaluando faithfulness 8/25…
2026-02-07 19:24:10 [info     ] Using LM Studio (local)


2026-02-07 19:24:10 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:24:10 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:24:10 [info     ] Searching for articles
 limit=5 query=¿Qué avances hay en
inteligencia artificial? threshold=0.5
2026-02-07 19:24:10 [info     ] Search completed
 results_count=5
2026-02-07 19:24:10 [info     ] Using LM Studio (local)


  LLM response: '10'
  Score extraído: 100%
Evaluando faithfulness 9/25…
2026-02-07 19:24:25 [info     ] Using LM Studio (local)


2026-02-07 19:24:25 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
```

```
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:24:25 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:24:25 [info     ] Searching for articles

 limit=5 query=¿Hay noticias sobre redes

sociales? threshold=0.5
2026-02-07 19:24:26 [info     ] Search completed

 results_count=5
2026-02-07 19:24:26 [info     ] Using LM Studio (local)


  LLM response: '10'
  Score extraído: 100%
Evaluando faithfulness 10/25…
2026-02-07 19:24:43 [info     ] Using LM Studio (local)


2026-02-07 19:24:43 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:24:43 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:24:43 [info     ] Searching for articles

 limit=5 query=¿Qué está pasando con las

empresas tecnológicas? threshold=0.5
2026-02-07 19:24:43 [info     ] Search completed

 results_count=5
2026-02-07 19:24:43 [info     ] Using LM Studio (local)


  LLM response: '10'
  Score extraído: 100%
Evaluando faithfulness 11/25…
2026-02-07 19:24:57 [info     ] Using LM Studio (local)


2026-02-07 19:24:57 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:24:57 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
```

```
2026-02-07 19:24:57 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
ciberseguridad? threshold=0.5
2026-02-07 19:24:58 [info     ] Search completed
 results_count=5
2026-02-07 19:24:58 [info     ] Using LM Studio (local)


  LLM response: '5'
  Score extraído: 50%
Evaluando faithfulness 12/25…
2026-02-07 19:25:14 [info     ] Using LM Studio (local)


2026-02-07 19:25:14 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:25:14 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:25:14 [info     ] Searching for articles
 limit=5 query=¿Qué noticias hay sobre
salud? threshold=0.5
2026-02-07 19:25:15 [info     ] Search completed
 results_count=5
2026-02-07 19:25:15 [info     ] Using LM Studio (local)


  LLM response: '2'
  Score extraído: 20%
Evaluando faithfulness 13/25…
2026-02-07 19:25:31 [info     ] Using LM Studio (local)


2026-02-07 19:25:31 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:25:31 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:25:31 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
vacunas? threshold=0.5
```

```
2026-02-07 19:25:32 [info     ] Search completed
 results_count=5
2026-02-07 19:25:32 [info     ] Using LM Studio (local)


  LLM response: '10'
  Score extraído: 100%
Evaluando faithfulness 14/25…
2026-02-07 19:25:46 [info     ] Using LM Studio (local)


2026-02-07 19:25:46 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:25:46 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:25:46 [info     ] Searching for articles
 limit=5 query=¿Qué dicen los expertos sobre
salud mental? threshold=0.5
2026-02-07 19:25:47 [info     ] Search completed
 results_count=5
2026-02-07 19:25:47 [info     ] Using LM Studio (local)


  LLM response: '3'
  Score extraído: 30%
Evaluando faithfulness 15/25…
2026-02-07 19:26:04 [info     ] Using LM Studio (local)


2026-02-07 19:26:04 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:26:04 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:26:04 [info     ] Searching for articles
 limit=5 query=¿Hay avances médicos
recientes? threshold=0.5
2026-02-07 19:26:05 [info     ] Search completed
 results_count=5
```

```
2026-02-07 19:26:05 [info     ] Using LM Studio (local)

  LLM response: '10'
  Score extraído: 100%
Evaluando faithfulness 16/25…
2026-02-07 19:26:23 [info     ] Using LM Studio (local)

2026-02-07 19:26:23 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:26:23 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:26:23 [info     ] Searching for articles
 limit=5 query=¿Qué está pasando en el mundo
del deporte? threshold=0.5
2026-02-07 19:26:24 [info     ] Search completed
 results_count=5
2026-02-07 19:26:24 [info     ] Using LM Studio (local)

  LLM response: '10'
  Score extraído: 100%
Evaluando faithfulness 17/25…
2026-02-07 19:26:40 [info     ] Using LM Studio (local)

2026-02-07 19:26:40 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:26:40 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:26:40 [info     ] Searching for articles
 limit=5 query=¿Cuáles son los resultados
deportivos recientes? threshold=0.5
2026-02-07 19:26:41 [info     ] Search completed
 results_count=5
2026-02-07 19:26:41 [info     ] Using LM Studio (local)

  LLM response: '10'
```

```
  Score extraído: 100%
Evaluando faithfulness 18/25…
2026-02-07 19:26:57 [info     ] Using LM Studio (local)


2026-02-07 19:26:57 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:26:57 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:26:57 [info     ] Searching for articles

 limit=5 query=¿Hay noticias de fútbol?
threshold=0.5
2026-02-07 19:26:58 [info     ] Search completed

 results_count=5
2026-02-07 19:26:58 [info     ] Using LM Studio (local)


  LLM response: '10'
  Score extraído: 100%
Evaluando faithfulness 19/25…
2026-02-07 19:27:14 [info     ] Using LM Studio (local)


2026-02-07 19:27:14 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:27:14 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:27:14 [info     ] Searching for articles

 limit=5 query=¿Qué fichajes o traspasos hay

en el fútbol? threshold=0.5
2026-02-07 19:27:15 [info     ] Search completed

 results_count=5
2026-02-07 19:27:15 [info     ] Using LM Studio (local)


  LLM response: '10'
  Score extraído: 100%
Evaluando faithfulness 20/25…
2026-02-07 19:27:33 [info     ] Using LM Studio (local)
```

```
2026-02-07 19:27:33 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:27:33 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:27:33 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
ciencia? threshold=0.5
2026-02-07 19:27:34 [info     ] Search completed
 results_count=5
2026-02-07 19:27:34 [info     ] Using LM Studio (local)


  LLM response: '10'
  Score extraído: 100%
Evaluando faithfulness 21/25…
2026-02-07 19:27:49 [info     ] Using LM Studio (local)


2026-02-07 19:27:49 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:27:49 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:27:49 [info     ] Searching for articles
 limit=5 query=¿Qué descubrimientos
científicos hay? threshold=0.5
2026-02-07 19:27:50 [info     ] Search completed
 results_count=5
2026-02-07 19:27:50 [info     ] Using LM Studio (local)


  LLM response: '10'
  Score extraído: 100%
Evaluando faithfulness 22/25…
2026-02-07 19:28:07 [info     ] Using LM Studio (local)


2026-02-07 19:28:07 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
```

```
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:28:07 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:28:07 [info     ] Searching for articles

 limit=5 query=¿Hay noticias sobre el

espacio o astronomía? threshold=0.5
2026-02-07 19:28:08 [info     ] Search completed

 results_count=5
2026-02-07 19:28:08 [info     ] Using LM Studio (local)


  LLM response: '10'
  Score extraído: 100%
Evaluando faithfulness 23/25…
2026-02-07 19:28:24 [info     ] Using LM Studio (local)


2026-02-07 19:28:24 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:28:24 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:28:24 [info     ] Searching for articles

 limit=5 query=¿Qué noticias hay de

entretenimiento? threshold=0.5
2026-02-07 19:28:25 [info     ] Search completed

 results_count=5
2026-02-07 19:28:25 [info     ] Using LM Studio (local)


  LLM response: '2'
  Score extraído: 20%
Evaluando faithfulness 24/25…
2026-02-07 19:28:43 [info     ] Using LM Studio (local)


2026-02-07 19:28:43 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:28:43 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
```

```
2026-02-07 19:28:43 [info     ] Searching for articles
 limit=5 query=¿Hay estrenos de cine o
series? threshold=0.5
2026-02-07 19:28:44 [info     ] Search completed
 results_count=5
2026-02-07 19:28:44 [info     ] Using LM Studio (local)


  LLM response: '5'
  Score extraído: 50%
Evaluando faithfulness 25/25…
2026-02-07 19:29:03 [info     ] Using LM Studio (local)


2026-02-07 19:29:03 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:29:03 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:29:03 [info     ] Searching for articles
 limit=5 query=¿Cuáles son las noticias
internacionales más impor threshold=0.5
2026-02-07 19:29:04 [info     ] Search completed
 results_count=5
2026-02-07 19:29:04 [info     ] Using LM Studio (local)


  LLM response: '7'
  Score extraído: 70%

==================================================
Faithfulness LM Studio promedio: 77.92%
```

1.7.1 6.1 Comparación de Evaluadores: LM Studio vs DeepSeek

Comparamos cómo evalúan faithfulness ambos modelos para ver si son consistentes.

[20]:
```python
# Función para evaluar faithfulness con un proveedor específico
from src.llm.factory import get_llm_client
from src.config.settings import LLMProvider, settings

def evaluate_faithfulness_with_provider(
    question: str,
    answer: str,
```

```
    articles: list,
    provider: LLMProvider
) -> dict:
    """
    Evalúa faithfulness usando un proveedor LLM específico.

    Args:
        question: La pregunta original
        answer: La respuesta generada
        articles: Lista de tuplas (Article, score) con contenido completo
        provider: Proveedor LLM a usar como juez
    """
    try:
        llm = get_llm_client(provider)
    except Exception as e:
        print(f"  Error inicializando {provider}: {e}")
        return {"faithfulness_score": None, "provider": provider.value}

    # Incluir contenido de los artículos, no solo títulos
    sources_text = ""
    for i, (article, score) in enumerate(articles):
        content = article.content or article.description or ""
        content_preview = content[:500] + "..." if len(content) > 500 else␣
 ↪content
        sources_text += f"\n[{i+1}] {article.title}\n"
        sources_text += f"    Fuente: {article.source_name}\n"
        sources_text += f"    Contenido: {content_preview}\n"

    prompt = f"""Evalúa si la siguiente respuesta está basada fielmente en las␣
 ↪fuentes proporcionadas.

PREGUNTA: {question}

FUENTES:
{sources_text}

RESPUESTA: {answer}

Evalúa del 1 al 10:
- 10: Completamente fiel, todo está en las fuentes
- 5: Parcialmente fiel, algunas cosas no están en las fuentes
- 1: No es fiel, hay información inventada

Responde SOLO con el número (1-10):"""

    try:
        response = llm.generate(prompt, max_tokens=150, temperature=0)
```

```
        raw_response = response.content.strip()

        print(f"  [{provider.value}] Response: '{raw_response}'")

        numbers = re.findall(r'\b(\d+)\b', raw_response)
        if numbers:
            score = min(int(numbers[0]), 10) / 10.0
            return {"faithfulness_score": score, "provider": provider.value}
        else:
            return {"faithfulness_score": None, "provider": provider.value}

    except Exception as e:
        print(f"  [{provider.value}] Error: {e}")
        return {"faithfulness_score": None, "provider": provider.value}

# Verificar configuración
print("Configuración:")
print(f"  DeepSeek API Key: {'  Configurado' if settings.deepseek_api_key else␣
 ↪'  NO CONFIGURADO'}")
print("\n  Función evaluate_faithfulness_with_provider definida (con contenido␣
 ↪de artículos)")
```

```
Configuración:
  DeepSeek API Key:   Configurado

  Función evaluate_faithfulness_with_provider definida (con contenido de
artículos)
```

[21]:
```
# Comparar evaluadores: LM Studio vs DeepSeek
# Evaluamos las MISMAS respuestas (generadas por LM Studio) con ambos␣
 ↪evaluadores
# Ahora usando contenido completo de los artículos

comparison_results = []

for i, result in enumerate(generation_results):
    question = EVALUATION_QUESTIONS[i]['question']
    print(f"\n{'='*50}")
    print(f"Pregunta {i+1}: {question[:40]}...")

    with get_db() as db:
        engine = RAGEngine(db, use_reranking=False)
        # Obtener artículos CON CONTENIDO
        articles = engine.get_context_only(
            question=question,
            max_sources=5
        )
```

```
        # Evaluar con LM Studio
        print("Evaluando con LM Studio...")
        lm_result = evaluate_faithfulness_with_provider(
            question, result['answer'], articles, LLMProvider.LM_STUDIO
        )

        # Evaluar con DeepSeek (si está configurado)
        ds_result = {"faithfulness_score": None, "provider": "deepseek"}
        if settings.deepseek_api_key:
            print("Evaluando con DeepSeek...")
            ds_result = evaluate_faithfulness_with_provider(
                question, result['answer'], articles, LLMProvider.DEEPSEEK
            )

        comparison_results.append({
            "question": question,
            "lm_studio_score": lm_result['faithfulness_score'],
            "deepseek_score": ds_result['faithfulness_score'],
        })

# Mostrar resultados
df_comparison = pd.DataFrame(comparison_results)
print("\n" + "="*60)
print("COMPARACIÓN DE EVALUADORES DE FAITHFULNESS")
print("="*60)
print(df_comparison.to_string(index=False))

# Resumen
lm_scores = [r['lm_studio_score'] for r in comparison_results if␣
 ↪r['lm_studio_score'] is not None]
ds_scores = [r['deepseek_score'] for r in comparison_results if␣
 ↪r['deepseek_score'] is not None]

print(f"\nLM Studio promedio:  {np.mean(lm_scores):.1%}" if lm_scores else␣
 ↪"\nLM Studio: No data")
print(f"DeepSeek promedio:   {np.mean(ds_scores):.1%}" if ds_scores else␣
 ↪"DeepSeek: No configurado")
```

```
==================================================
Pregunta 1: ¿Qué está pasando con la economía?…
2026-02-07 19:30:22 [info     ] Using LM Studio (local)


2026-02-07 19:30:22 [info     ] Initialized
```

LMStudioEmbeddingGenerator

```
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:30:22 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:30:22 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:30:22 [info     ] Searching for articles

 limit=5 query=¿Qué está pasando con la

economía? threshold=0.5
2026-02-07 19:30:24 [info     ] Search completed

 results_count=5
Evaluando con LM Studio…
2026-02-07 19:30:24 [info     ] Initializing LM Studio

client  model=openai/gpt-oss-20b
  [lm_studio] Response: '5'
Evaluando con DeepSeek…
2026-02-07 19:30:41 [info     ] Initializing DeepSeek

client   model=deepseek-chat
  [deepseek] Response: '5'

==================================================
Pregunta 2: ¿Cuáles son las perspectivas económicas …
2026-02-07 19:30:43 [info     ] Using LM Studio (local)


2026-02-07 19:30:44 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:30:44 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:30:44 [info     ] Searching for articles

 limit=5 query=¿Cuáles son las perspectivas

económicas para 2026? threshold=0.5
2026-02-07 19:30:45 [info     ] Search completed

 results_count=5
Evaluando con LM Studio…
2026-02-07 19:30:45 [info     ] Initializing LM Studio

client  model=openai/gpt-oss-20b
```

```
  [lm_studio] Response: '8'
Evaluando con DeepSeek…
2026-02-07 19:31:02 [info     ] Initializing DeepSeek

client   model=deepseek-chat
  [deepseek] Response: '5'

==================================================
Pregunta 3: ¿Cómo está la inflación actualmente?…
2026-02-07 19:31:04 [info     ] Using LM Studio (local)


2026-02-07 19:31:04 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:31:04 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:31:04 [info     ] Searching for articles

 limit=5 query=¿Cómo está la inflación

actualmente? threshold=0.5
2026-02-07 19:31:05 [info     ] Search completed

 results_count=5
Evaluando con LM Studio…
2026-02-07 19:31:05 [info     ] Initializing LM Studio

client  model=openai/gpt-oss-20b
  [lm_studio] Response: '7'
Evaluando con DeepSeek…
2026-02-07 19:31:24 [info     ] Initializing DeepSeek

client   model=deepseek-chat
  [deepseek] Response: '5'

==================================================
Pregunta 4: ¿Qué está pasando con los salarios?…
2026-02-07 19:31:26 [info     ] Using LM Studio (local)


2026-02-07 19:31:26 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:31:26 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
```

```
2026-02-07 19:31:26 [info     ] Searching for articles
 limit=5 query=¿Qué está pasando con los
salarios? threshold=0.5
2026-02-07 19:31:28 [info     ] Search completed
 results_count=5
Evaluando con LM Studio…
2026-02-07 19:31:28 [info     ] Initializing LM Studio
client  model=openai/gpt-oss-20b
  [lm_studio] Error: LM Studio API error: Error code: 400 - {'error': 'The model
has crashed without additional information. (Exit code: null)'}
Evaluando con DeepSeek…
2026-02-07 19:31:45 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '5'

==================================================
Pregunta 5: ¿Hay noticias sobre inversiones?…
2026-02-07 19:31:47 [info     ] Using LM Studio (local)


2026-02-07 19:31:47 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:31:47 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:31:47 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
inversiones? threshold=0.5
2026-02-07 19:31:47 [info     ] Search completed
 results_count=5
Evaluando con LM Studio…
2026-02-07 19:31:47 [info     ] Initializing LM Studio
client  model=openai/gpt-oss-20b
  [lm_studio] Response: '5'
Evaluando con DeepSeek…
2026-02-07 19:32:05 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '5'

==================================================
```

```
Pregunta 6: ¿Qué dicen los analistas económicos?…
2026-02-07 19:32:06 [info     ] Using LM Studio (local)


2026-02-07 19:32:06 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:32:06 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:32:06 [info     ] Searching for articles

 limit=5 query=¿Qué dicen los analistas

económicos? threshold=0.5
2026-02-07 19:32:07 [info     ] Search completed

 results_count=5
Evaluando con LM Studio…
2026-02-07 19:32:07 [info     ] Initializing LM Studio

client  model=openai/gpt-oss-20b
  [lm_studio] Response: '8'
Evaluando con DeepSeek…
2026-02-07 19:32:24 [info     ] Initializing DeepSeek

client   model=deepseek-chat
  [deepseek] Response: '5'

==================================================
Pregunta 7: ¿Cuáles son las últimas noticias de tecn…
2026-02-07 19:32:25 [info     ] Using LM Studio (local)


2026-02-07 19:32:25 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:32:25 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:32:25 [info     ] Searching for articles

 limit=5 query=¿Cuáles son las últimas

noticias de tecnología? threshold=0.5
2026-02-07 19:32:26 [info     ] Search completed

 results_count=5
Evaluando con LM Studio…
```

```
2026-02-07 19:32:26 [info     ] Initializing LM Studio
client  model=openai/gpt-oss-20b
  [lm_studio] Response: '10'
Evaluando con DeepSeek…
2026-02-07 19:32:46 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '5'

==================================================
Pregunta 8: ¿Qué avances hay en inteligencia artific…
2026-02-07 19:32:47 [info     ] Using LM Studio (local)


2026-02-07 19:32:47 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:32:47 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:32:47 [info     ] Searching for articles
 limit=5 query=¿Qué avances hay en
inteligencia artificial? threshold=0.5
2026-02-07 19:32:48 [info     ] Search completed
 results_count=5
Evaluando con LM Studio…
2026-02-07 19:32:48 [info     ] Initializing LM Studio
client  model=openai/gpt-oss-20b
  [lm_studio] Response: '10'
Evaluando con DeepSeek…
2026-02-07 19:33:02 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '5'

==================================================
Pregunta 9: ¿Hay noticias sobre redes sociales?…
2026-02-07 19:33:04 [info     ] Using LM Studio (local)


2026-02-07 19:33:04 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
```

```
2026-02-07 19:33:04 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:33:04 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre redes
sociales? threshold=0.5
2026-02-07 19:33:05 [info     ] Search completed
 results_count=5
Evaluando con LM Studio…
2026-02-07 19:33:05 [info     ] Initializing LM Studio
client  model=openai/gpt-oss-20b
  [lm_studio] Response: '10'
Evaluando con DeepSeek…
2026-02-07 19:33:21 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '8'

==================================================
Pregunta 10: ¿Qué está pasando con las empresas tecno…
2026-02-07 19:33:23 [info     ] Using LM Studio (local)


2026-02-07 19:33:23 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:33:23 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:33:23 [info     ] Searching for articles
 limit=5 query=¿Qué está pasando con las
empresas tecnológicas? threshold=0.5
2026-02-07 19:33:24 [info     ] Search completed
 results_count=5
Evaluando con LM Studio…
2026-02-07 19:33:24 [info     ] Initializing LM Studio
client  model=openai/gpt-oss-20b
  [lm_studio] Response: '10'
Evaluando con DeepSeek…
2026-02-07 19:33:40 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '5'
```

```
==================================================
Pregunta 11: ¿Hay noticias sobre ciberseguridad?…
2026-02-07 19:33:42 [info     ] Using LM Studio (local)


2026-02-07 19:33:42 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:33:42 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:33:42 [info     ] Searching for articles

 limit=5 query=¿Hay noticias sobre

ciberseguridad? threshold=0.5
2026-02-07 19:33:42 [info     ] Search completed

 results_count=5
Evaluando con LM Studio…
2026-02-07 19:33:42 [info     ] Initializing LM Studio

client  model=openai/gpt-oss-20b
  [lm_studio] Response: '5'
Evaluando con DeepSeek…
2026-02-07 19:34:00 [info     ] Initializing DeepSeek

client   model=deepseek-chat
  [deepseek] Response: '5'

==================================================
Pregunta 12: ¿Qué noticias hay sobre salud?…
2026-02-07 19:34:01 [info     ] Using LM Studio (local)


2026-02-07 19:34:01 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:34:01 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:34:01 [info     ] Searching for articles

 limit=5 query=¿Qué noticias hay sobre

salud? threshold=0.5
2026-02-07 19:34:02 [info     ] Search completed

 results_count=5
Evaluando con LM Studio…
```

```
2026-02-07 19:34:02 [info     ] Initializing LM Studio
client  model=openai/gpt-oss-20b
  [lm_studio] Response: '2'
Evaluando con DeepSeek…
2026-02-07 19:34:18 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '7'

==================================================
Pregunta 13: ¿Hay noticias sobre vacunas?…
2026-02-07 19:34:20 [info     ] Using LM Studio (local)


2026-02-07 19:34:20 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:34:20 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:34:20 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
vacunas? threshold=0.5
2026-02-07 19:34:21 [info     ] Search completed
 results_count=5
Evaluando con LM Studio…
2026-02-07 19:34:21 [info     ] Initializing LM Studio
client  model=openai/gpt-oss-20b
  [lm_studio] Response: '10'
Evaluando con DeepSeek…
2026-02-07 19:34:36 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '5'

==================================================
Pregunta 14: ¿Qué dicen los expertos sobre salud ment…
2026-02-07 19:34:38 [info     ] Using LM Studio (local)


2026-02-07 19:34:38 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
```

```
2026-02-07 19:34:38 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:34:38 [info     ] Searching for articles
 limit=5 query=¿Qué dicen los expertos sobre
salud mental? threshold=0.5
2026-02-07 19:34:39 [info     ] Search completed
 results_count=5
Evaluando con LM Studio…
2026-02-07 19:34:39 [info     ] Initializing LM Studio
client  model=openai/gpt-oss-20b
  [lm_studio] Response: '3'
Evaluando con DeepSeek…
2026-02-07 19:34:57 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '5'

==================================================
Pregunta 15: ¿Hay avances médicos recientes?…
2026-02-07 19:34:58 [info     ] Using LM Studio (local)


2026-02-07 19:34:58 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:34:58 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:34:58 [info     ] Searching for articles
 limit=5 query=¿Hay avances médicos
recientes? threshold=0.5
2026-02-07 19:34:59 [info     ] Search completed
 results_count=5
Evaluando con LM Studio…
2026-02-07 19:34:59 [info     ] Initializing LM Studio
client  model=openai/gpt-oss-20b
  [lm_studio] Response: '10'
Evaluando con DeepSeek…
2026-02-07 19:35:14 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '5'
```

```
==================================================
Pregunta 16: ¿Qué está pasando en el mundo del deport…
2026-02-07 19:35:16 [info     ] Using LM Studio (local)


2026-02-07 19:35:16 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:35:16 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:35:16 [info     ] Searching for articles

 limit=5 query=¿Qué está pasando en el mundo

del deporte? threshold=0.5
2026-02-07 19:35:17 [info     ] Search completed

 results_count=5
Evaluando con LM Studio…
2026-02-07 19:35:17 [info     ] Initializing LM Studio

client  model=openai/gpt-oss-20b
  [lm_studio] Response: '10'
Evaluando con DeepSeek…
2026-02-07 19:35:32 [info     ] Initializing DeepSeek

client   model=deepseek-chat
  [deepseek] Response: '10'

==================================================
Pregunta 17: ¿Cuáles son los resultados deportivos re…
2026-02-07 19:35:33 [info     ] Using LM Studio (local)


2026-02-07 19:35:33 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:35:33 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:35:33 [info     ] Searching for articles

 limit=5 query=¿Cuáles son los resultados

deportivos recientes? threshold=0.5
2026-02-07 19:35:34 [info     ] Search completed

 results_count=5
Evaluando con LM Studio…
```

```
2026-02-07 19:35:34 [info     ] Initializing LM Studio
client  model=openai/gpt-oss-20b
  [lm_studio] Response: '10'
Evaluando con DeepSeek…
2026-02-07 19:35:50 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '5'

==================================================
Pregunta 18: ¿Hay noticias de fútbol?…
2026-02-07 19:35:52 [info     ] Using LM Studio (local)


2026-02-07 19:35:52 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:35:52 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:35:52 [info     ] Searching for articles
 limit=5 query=¿Hay noticias de fútbol?
threshold=0.5
2026-02-07 19:35:53 [info     ] Search completed
 results_count=5
Evaluando con LM Studio…
2026-02-07 19:35:53 [info     ] Initializing LM Studio
client  model=openai/gpt-oss-20b
  [lm_studio] Response: '10'
Evaluando con DeepSeek…
2026-02-07 19:36:09 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '10'

==================================================
Pregunta 19: ¿Qué fichajes o traspasos hay en el fútb…
2026-02-07 19:36:10 [info     ] Using LM Studio (local)


2026-02-07 19:36:10 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
```

```
2026-02-07 19:36:10 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:36:10 [info     ] Searching for articles
 limit=5 query=¿Qué fichajes o traspasos hay
en el fútbol? threshold=0.5
2026-02-07 19:36:12 [info     ] Search completed
 results_count=5
Evaluando con LM Studio…
2026-02-07 19:36:12 [info     ] Initializing LM Studio
client  model=openai/gpt-oss-20b
  [lm_studio] Response: '10'
Evaluando con DeepSeek…
2026-02-07 19:36:27 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '10'

==================================================
Pregunta 20: ¿Hay noticias sobre ciencia?…
2026-02-07 19:36:28 [info     ] Using LM Studio (local)


2026-02-07 19:36:28 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:36:28 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:36:28 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
ciencia? threshold=0.5
2026-02-07 19:36:29 [info     ] Search completed
 results_count=5
Evaluando con LM Studio…
2026-02-07 19:36:29 [info     ] Initializing LM Studio
client  model=openai/gpt-oss-20b
  [lm_studio] Response: '10'
Evaluando con DeepSeek…
2026-02-07 19:36:45 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '5'
```

```
==================================================
Pregunta 21: ¿Qué descubrimientos científicos hay?…
2026-02-07 19:36:47 [info     ] Using LM Studio (local)


2026-02-07 19:36:47 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:36:47 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:36:47 [info     ] Searching for articles

 limit=5 query=¿Qué descubrimientos

científicos hay? threshold=0.5
2026-02-07 19:36:48 [info     ] Search completed

 results_count=5
Evaluando con LM Studio…
2026-02-07 19:36:48 [info     ] Initializing LM Studio

client  model=openai/gpt-oss-20b
  [lm_studio] Response: '10'
Evaluando con DeepSeek…
2026-02-07 19:37:04 [info     ] Initializing DeepSeek

client   model=deepseek-chat
  [deepseek] Response: '5'

==================================================
Pregunta 22: ¿Hay noticias sobre el espacio o astrono…
2026-02-07 19:37:05 [info     ] Using LM Studio (local)


2026-02-07 19:37:05 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:37:05 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:37:05 [info     ] Searching for articles

 limit=5 query=¿Hay noticias sobre el

espacio o astronomía? threshold=0.5
2026-02-07 19:37:06 [info     ] Search completed

 results_count=5
Evaluando con LM Studio…
```

```
2026-02-07 19:37:06 [info     ] Initializing LM Studio
client  model=openai/gpt-oss-20b
  [lm_studio] Response: '10'
Evaluando con DeepSeek…
2026-02-07 19:37:23 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '7'

==================================================
Pregunta 23: ¿Qué noticias hay de entretenimiento?…
2026-02-07 19:37:25 [info     ] Using LM Studio (local)


2026-02-07 19:37:25 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:37:25 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:37:25 [info     ] Searching for articles
 limit=5 query=¿Qué noticias hay de
entretenimiento? threshold=0.5
2026-02-07 19:37:26 [info     ] Search completed
 results_count=5
Evaluando con LM Studio…
2026-02-07 19:37:26 [info     ] Initializing LM Studio
client  model=openai/gpt-oss-20b
  [lm_studio] Response: '2'
Evaluando con DeepSeek…
2026-02-07 19:37:43 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '5'

==================================================
Pregunta 24: ¿Hay estrenos de cine o series?…
2026-02-07 19:37:44 [info     ] Using LM Studio (local)


2026-02-07 19:37:44 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
```

```
2026-02-07 19:37:44 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:37:44 [info     ] Searching for articles
 limit=5 query=¿Hay estrenos de cine o
series? threshold=0.5
2026-02-07 19:37:46 [info     ] Search completed
 results_count=5
Evaluando con LM Studio…
2026-02-07 19:37:46 [info     ] Initializing LM Studio
client  model=openai/gpt-oss-20b
  [lm_studio] Response: '5'
Evaluando con DeepSeek…
2026-02-07 19:38:04 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '5'

==================================================
Pregunta 25: ¿Cuáles son las noticias internacionales…
2026-02-07 19:38:06 [info     ] Using LM Studio (local)


2026-02-07 19:38:06 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:38:06 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:38:06 [info     ] Searching for articles
 limit=5 query=¿Cuáles son las noticias
internacionales más impor threshold=0.5
2026-02-07 19:38:07 [info     ] Search completed
 results_count=5
Evaluando con LM Studio…
2026-02-07 19:38:07 [info     ] Initializing LM Studio
client  model=openai/gpt-oss-20b
  [lm_studio] Response: '7'
Evaluando con DeepSeek…
2026-02-07 19:38:23 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '5'
```

```
============================================================
COMPARACIÓN DE EVALUADORES DE FAITHFULNESS
============================================================
                                                question  lm_studio_score
deepseek_score
                        ¿Qué está pasando con la economía?              0.5
0.5
       ¿Cuáles son las perspectivas económicas para 2026?              0.8
0.5
                      ¿Cómo está la inflación actualmente?              0.7
0.5
                       ¿Qué está pasando con los salarios?              NaN
0.5
                          ¿Hay noticias sobre inversiones?              0.5
0.5
                      ¿Qué dicen los analistas económicos?              0.8
0.5
          ¿Cuáles son las últimas noticias de tecnología?              1.0
0.5
             ¿Qué avances hay en inteligencia artificial?              1.0
0.5
                       ¿Hay noticias sobre redes sociales?              1.0
0.8
         ¿Qué está pasando con las empresas tecnológicas?              1.0
0.5
                       ¿Hay noticias sobre ciberseguridad?              0.5
0.5
                            ¿Qué noticias hay sobre salud?              0.2
0.7
                              ¿Hay noticias sobre vacunas?              1.0
0.5
              ¿Qué dicen los expertos sobre salud mental?              0.3
0.5
                           ¿Hay avances médicos recientes?              1.0
0.5
               ¿Qué está pasando en el mundo del deporte?              1.0
1.0
         ¿Cuáles son los resultados deportivos recientes?              1.0
0.5
                                  ¿Hay noticias de fútbol?              1.0
1.0
              ¿Qué fichajes o traspasos hay en el fútbol?              1.0
1.0
                              ¿Hay noticias sobre ciencia?              1.0
0.5
                    ¿Qué descubrimientos científicos hay?              1.0
0.5
             ¿Hay noticias sobre el espacio o astronomía?              1.0
```

```
0.7
                    ¿Qué noticias hay de entretenimiento?              0.2
0.5
                          ¿Hay estrenos de cine o series?              0.5
0.5
¿Cuáles son las noticias internacionales más importantes?              0.7
0.5

LM Studio promedio:  77.9%
DeepSeek promedio:   58.8%
```

1.7.2 6.2 Comparación de Generación RAG: LM Studio vs DeepSeek

Ahora comparamos las **respuestas RAG generadas** por cada modelo para ver cuál produce menos alucinaciones.

[22]:
```
# Comparar generación RAG con diferentes modelos
# Esta celda genera respuestas RAG usando DeepSeek y las evalúa
import re

if not settings.deepseek_api_key:
    print("  DeepSeek no configurado. Salta esta celda.")
else:
    deepseek_generation_results = []

    for q in EVALUATION_QUESTIONS:
        print(f"\nGenerando con DeepSeek: {q['question'][:40]}...")

        with get_db() as db:
            # Crear RAG Engine con DeepSeek
            deepseek_client = get_llm_client(LLMProvider.DEEPSEEK)
            engine = RAGEngine(db, llm_client=deepseek_client,␣
 ↪use_reranking=False)

            response = engine.ask(
                question=q["question"],
                max_sources=5,
            )

            # Métricas
            citations = len(re.findall(r'\[\d+\]', response.answer))
            answer_lower = response.answer.lower()
            topics_found = [t for t in q["expected_topics"] if t.lower() in␣
 ↪answer_lower]

            deepseek_generation_results.append({
                "question": q["question"],
                "answer": response.answer,
```

```
                "sources_used": len(response.sources),
                "citations": citations,
                "confidence": response.confidence,
                "topic_coverage": len(topics_found) / len(q["expected_topics"]),
            })

            print(f"  Confianza: {response.confidence:.1%}, Fuentes:␣
↪{len(response.sources)}")

    df_deepseek_gen = pd.DataFrame(deepseek_generation_results)
    print("\n" + "="*50)
    print("GENERACIÓN CON DEEPSEEK")
    print("="*50)
    print(df_deepseek_gen[['question', 'sources_used', 'citations',␣
↪'confidence', 'topic_coverage']].to_string(index=False))
```

```
Generando con DeepSeek: ¿Qué está pasando con la economía?…
2026-02-07 19:40:31 [info     ] Initializing DeepSeek

client   model=deepseek-chat
2026-02-07 19:40:31 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:40:31 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:40:31 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Qué está pasando con la economía?
2026-02-07 19:40:31 [info     ] Searching for articles

 limit=5 query=¿Qué está pasando con la

economía? threshold=0.5
2026-02-07 19:40:33 [info     ] Search completed

 results_count=5
2026-02-07 19:40:33 [info     ] Generating RAG response

 articles=5 query=¿Qué está pasando con la

economía?
2026-02-07 19:40:41 [info     ] RAG response generated

 answer_length=1128
confidence=0.8605775974337163 sources_used=3
2026-02-07 19:40:41 [info     ] RAG query completed

 confidence=0.8605775974337163
```

```
sources_cited=3 sources_found=5
  Confianza: 86.1%, Fuentes: 3

Generando con DeepSeek: ¿Cuáles son las perspectivas económicas …
2026-02-07 19:40:41 [info     ] Initializing DeepSeek

client   model=deepseek-chat
2026-02-07 19:40:41 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:40:41 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:40:41 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Cuáles son las perspectivas económicas para 2026?
2026-02-07 19:40:41 [info     ] Searching for articles

 limit=5 query=¿Cuáles son las perspectivas

económicas para 2026? threshold=0.5
2026-02-07 19:40:41 [info     ] Search completed

 results_count=5
2026-02-07 19:40:41 [info     ] Generating RAG response

 articles=5 query=¿Cuáles son las

perspectivas económicas para 2026?
2026-02-07 19:40:50 [info     ] RAG response generated

 answer_length=1189
confidence=0.9559694278903662 sources_used=5
2026-02-07 19:40:50 [info     ] RAG query completed

 confidence=0.9559694278903662
sources_cited=5 sources_found=5
  Confianza: 95.6%, Fuentes: 5

Generando con DeepSeek: ¿Cómo está la inflación actualmente?…
2026-02-07 19:40:50 [info     ] Initializing DeepSeek

client   model=deepseek-chat
2026-02-07 19:40:50 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:40:50 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
```

```
2026-02-07 19:40:50 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Cómo está la inflación actualmente?
2026-02-07 19:40:50 [info     ] Searching for articles
 limit=5 query=¿Cómo está la inflación
actualmente? threshold=0.5
2026-02-07 19:40:50 [info     ] Search completed
 results_count=5
2026-02-07 19:40:50 [info     ] Generating RAG response
 articles=5 query=¿Cómo está la inflación
actualmente?
2026-02-07 19:40:56 [info     ] RAG response generated
 answer_length=964
confidence=0.8242365580995433 sources_used=2
2026-02-07 19:40:56 [info     ] RAG query completed
 confidence=0.8242365580995433
sources_cited=2 sources_found=5
  Confianza: 82.4%, Fuentes: 2

Generando con DeepSeek: ¿Qué está pasando con los salarios?…
2026-02-07 19:40:56 [info     ] Initializing DeepSeek
client   model=deepseek-chat
2026-02-07 19:40:56 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:40:56 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:40:56 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué está pasando con los salarios?
2026-02-07 19:40:56 [info     ] Searching for articles
 limit=5 query=¿Qué está pasando con los
salarios? threshold=0.5
2026-02-07 19:40:56 [info     ] Search completed
 results_count=5
```

```
2026-02-07 19:40:56 [info     ] Generating RAG response
 articles=5 query=¿Qué está pasando con los
salarios?
2026-02-07 19:41:01 [info     ] RAG response generated
 answer_length=586
confidence=0.6784608488813607 sources_used=1
2026-02-07 19:41:01 [info     ] RAG query completed
 confidence=0.6784608488813607
sources_cited=1 sources_found=5
  Confianza: 67.8%, Fuentes: 1

Generando con DeepSeek: ¿Hay noticias sobre inversiones?…
2026-02-07 19:41:01 [info     ] Initializing DeepSeek
client   model=deepseek-chat
2026-02-07 19:41:01 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:41:01 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:41:01 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Hay noticias sobre inversiones?
2026-02-07 19:41:01 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
inversiones? threshold=0.5
2026-02-07 19:41:01 [info     ] Search completed
 results_count=5
2026-02-07 19:41:01 [info     ] Generating RAG response
 articles=5 query=¿Hay noticias sobre
inversiones?
2026-02-07 19:41:07 [info     ] RAG response generated
 answer_length=752
confidence=0.8495847586181656 sources_used=3
2026-02-07 19:41:07 [info     ] RAG query completed
 confidence=0.8495847586181656
sources_cited=3 sources_found=5
  Confianza: 85.0%, Fuentes: 3
```

```
Generando con DeepSeek: ¿Qué dicen los analistas económicos?…
2026-02-07 19:41:07 [info     ] Initializing DeepSeek

client   model=deepseek-chat
2026-02-07 19:41:07 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:41:07 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:41:07 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Qué dicen los analistas económicos?
2026-02-07 19:41:07 [info     ] Searching for articles

 limit=5 query=¿Qué dicen los analistas

económicos? threshold=0.5
2026-02-07 19:41:07 [info     ] Search completed

 results_count=5
2026-02-07 19:41:07 [info     ] Generating RAG response

 articles=5 query=¿Qué dicen los analistas

económicos?
2026-02-07 19:41:13 [info     ] RAG response generated

 answer_length=897
confidence=0.8642502708557653 sources_used=3
2026-02-07 19:41:13 [info     ] RAG query completed

 confidence=0.8642502708557653
sources_cited=3 sources_found=5
  Confianza: 86.4%, Fuentes: 3

Generando con DeepSeek: ¿Cuáles son las últimas noticias de tecn…
2026-02-07 19:41:13 [info     ] Initializing DeepSeek

client   model=deepseek-chat
2026-02-07 19:41:13 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:41:13 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:41:13 [info     ] Processing RAG query

 category=None max_sources=5
```

```
question=¿Cuáles son las últimas noticias de tecnología?
2026-02-07 19:41:13 [info     ] Searching for articles
 limit=5 query=¿Cuáles son las últimas
noticias de tecnología? threshold=0.5
2026-02-07 19:41:14 [info     ] Search completed
 results_count=5
2026-02-07 19:41:14 [info     ] Generating RAG response
 articles=5 query=¿Cuáles son las últimas
noticias de tecnología?
2026-02-07 19:41:22 [info     ] RAG response generated
 answer_length=1229
confidence=0.9390062660764192 sources_used=5
2026-02-07 19:41:22 [info     ] RAG query completed
 confidence=0.9390062660764192
sources_cited=5 sources_found=5
  Confianza: 93.9%, Fuentes: 5

Generando con DeepSeek: ¿Qué avances hay en inteligencia artific…
2026-02-07 19:41:22 [info     ] Initializing DeepSeek
client   model=deepseek-chat
2026-02-07 19:41:22 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:41:22 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:41:22 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué avances hay en inteligencia artificial?
2026-02-07 19:41:22 [info     ] Searching for articles
 limit=5 query=¿Qué avances hay en
inteligencia artificial? threshold=0.5
2026-02-07 19:41:22 [info     ] Search completed
 results_count=5
2026-02-07 19:41:22 [info     ] Generating RAG response
 articles=5 query=¿Qué avances hay en
inteligencia artificial?
```

```
2026-02-07 19:41:30 [info     ] RAG response generated
 answer_length=1261
confidence=0.9586791687982389 sources_used=5
2026-02-07 19:41:30 [info     ] RAG query completed
 confidence=0.9586791687982389
sources_cited=5 sources_found=5
  Confianza: 95.9%, Fuentes: 5

Generando con DeepSeek: ¿Hay noticias sobre redes sociales?…
2026-02-07 19:41:30 [info     ] Initializing DeepSeek
client   model=deepseek-chat
2026-02-07 19:41:30 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:41:30 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:41:30 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Hay noticias sobre redes sociales?
2026-02-07 19:41:30 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre redes
sociales? threshold=0.5
2026-02-07 19:41:30 [info     ] Search completed
 results_count=5
2026-02-07 19:41:30 [info     ] Generating RAG response
 articles=5 query=¿Hay noticias sobre redes
sociales?
2026-02-07 19:41:37 [info     ] RAG response generated
 answer_length=1058
confidence=0.8971191947402255 sources_used=4
2026-02-07 19:41:37 [info     ] RAG query completed
 confidence=0.8971191947402255
sources_cited=4 sources_found=5
  Confianza: 89.7%, Fuentes: 4

Generando con DeepSeek: ¿Qué está pasando con las empresas tecno…
2026-02-07 19:41:37 [info     ] Initializing DeepSeek
client   model=deepseek-chat
```

```
2026-02-07 19:41:37 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:41:37 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:41:37 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué está pasando con las empresas tecnológicas?
2026-02-07 19:41:37 [info     ] Searching for articles
 limit=5 query=¿Qué está pasando con las
empresas tecnológicas? threshold=0.5
2026-02-07 19:41:37 [info     ] Search completed
 results_count=5
2026-02-07 19:41:37 [info     ] Generating RAG response
 articles=5 query=¿Qué está pasando con las
empresas tecnológicas?
2026-02-07 19:41:45 [info     ] RAG response generated
 answer_length=1231
confidence=0.9049075650198416 sources_used=4
2026-02-07 19:41:45 [info     ] RAG query completed
 confidence=0.9049075650198416
sources_cited=4 sources_found=5
  Confianza: 90.5%, Fuentes: 4

Generando con DeepSeek: ¿Hay noticias sobre ciberseguridad?…
2026-02-07 19:41:45 [info     ] Initializing DeepSeek
client   model=deepseek-chat
2026-02-07 19:41:45 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:41:45 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:41:45 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Hay noticias sobre ciberseguridad?
```

```
2026-02-07 19:41:45 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
ciberseguridad? threshold=0.5
2026-02-07 19:41:46 [info     ] Search completed
 results_count=5
2026-02-07 19:41:46 [info     ] Generating RAG response
 articles=5 query=¿Hay noticias sobre
ciberseguridad?
2026-02-07 19:41:50 [info     ] RAG response generated
 answer_length=433
confidence=0.6717628616438612 sources_used=1
2026-02-07 19:41:50 [info     ] RAG query completed
 confidence=0.6717628616438612
sources_cited=1 sources_found=5
  Confianza: 67.2%, Fuentes: 1

Generando con DeepSeek: ¿Qué noticias hay sobre salud?…
2026-02-07 19:41:50 [info     ] Initializing DeepSeek
client   model=deepseek-chat
2026-02-07 19:41:50 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:41:50 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:41:50 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué noticias hay sobre salud?
2026-02-07 19:41:50 [info     ] Searching for articles
 limit=5 query=¿Qué noticias hay sobre
salud? threshold=0.5
2026-02-07 19:41:50 [info     ] Search completed
 results_count=5
2026-02-07 19:41:50 [info     ] Generating RAG response
 articles=5 query=¿Qué noticias hay sobre
salud?
2026-02-07 19:41:58 [info     ] RAG response generated
 answer_length=1159
```

```
confidence=0.9340431946749933 sources_used=5
2026-02-07 19:41:58 [info     ] RAG query completed

 confidence=0.9340431946749933
sources_cited=5 sources_found=5
  Confianza: 93.4%, Fuentes: 5

Generando con DeepSeek: ¿Hay noticias sobre vacunas?…
2026-02-07 19:41:58 [info     ] Initializing DeepSeek

client   model=deepseek-chat
2026-02-07 19:41:58 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:41:58 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:41:58 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Hay noticias sobre vacunas?
2026-02-07 19:41:58 [info     ] Searching for articles

 limit=5 query=¿Hay noticias sobre

vacunas? threshold=0.5
2026-02-07 19:41:58 [info     ] Search completed

 results_count=5
2026-02-07 19:41:58 [info     ] Generating RAG response

 articles=5 query=¿Hay noticias sobre

vacunas?
2026-02-07 19:42:06 [info     ] RAG response generated

 answer_length=1027
confidence=0.8980745650299216 sources_used=4
2026-02-07 19:42:06 [info     ] RAG query completed

 confidence=0.8980745650299216
sources_cited=4 sources_found=5
  Confianza: 89.8%, Fuentes: 4

Generando con DeepSeek: ¿Qué dicen los expertos sobre salud ment…
2026-02-07 19:42:06 [info     ] Initializing DeepSeek

client   model=deepseek-chat
2026-02-07 19:42:06 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
```

```
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:42:06 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:42:06 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Qué dicen los expertos sobre salud mental?
2026-02-07 19:42:06 [info     ] Searching for articles

 limit=5 query=¿Qué dicen los expertos sobre

salud mental? threshold=0.5
2026-02-07 19:42:06 [info     ] Search completed

 results_count=5
2026-02-07 19:42:06 [info     ] Generating RAG response

 articles=5 query=¿Qué dicen los expertos

sobre salud mental?
2026-02-07 19:42:16 [info     ] RAG response generated

 answer_length=1422
confidence=0.9150781133421969 sources_used=4
2026-02-07 19:42:16 [info     ] RAG query completed

 confidence=0.9150781133421969
sources_cited=4 sources_found=5
  Confianza: 91.5%, Fuentes: 4

Generando con DeepSeek: ¿Hay avances médicos recientes?…
2026-02-07 19:42:16 [info     ] Initializing DeepSeek

client   model=deepseek-chat
2026-02-07 19:42:16 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:42:16 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:42:16 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Hay avances médicos recientes?
2026-02-07 19:42:16 [info     ] Searching for articles

 limit=5 query=¿Hay avances médicos

recientes? threshold=0.5
2026-02-07 19:42:16 [info     ] Search completed

 results_count=5
```

```
2026-02-07 19:42:16 [info     ] Generating RAG response
 articles=5 query=¿Hay avances médicos
recientes?
2026-02-07 19:42:20 [info     ] RAG response generated
 answer_length=381
confidence=0.9412362831647072 sources_used=5
2026-02-07 19:42:20 [info     ] RAG query completed
 confidence=0.9412362831647072
sources_cited=5 sources_found=5
  Confianza: 94.1%, Fuentes: 5

Generando con DeepSeek: ¿Qué está pasando en el mundo del deport…
2026-02-07 19:42:20 [info     ] Initializing DeepSeek
client   model=deepseek-chat
2026-02-07 19:42:20 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:42:20 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:42:20 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué está pasando en el mundo del deporte?
2026-02-07 19:42:20 [info     ] Searching for articles
 limit=5 query=¿Qué está pasando en el mundo
del deporte? threshold=0.5
2026-02-07 19:42:20 [info     ] Search completed
 results_count=5
2026-02-07 19:42:20 [info     ] Generating RAG response
 articles=5 query=¿Qué está pasando en el
mundo del deporte?
2026-02-07 19:42:27 [info     ] RAG response generated
 answer_length=1046
confidence=0.9451813129280625 sources_used=5
2026-02-07 19:42:27 [info     ] RAG query completed
 confidence=0.9451813129280625
sources_cited=5 sources_found=5
  Confianza: 94.5%, Fuentes: 5
```

```
Generando con DeepSeek: ¿Cuáles son los resultados deportivos re…
2026-02-07 19:42:27 [info     ] Initializing DeepSeek

client   model=deepseek-chat
2026-02-07 19:42:27 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:42:27 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:42:27 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Cuáles son los resultados deportivos recientes?
2026-02-07 19:42:27 [info     ] Searching for articles

 limit=5 query=¿Cuáles son los resultados

deportivos recientes? threshold=0.5
2026-02-07 19:42:27 [info     ] Search completed

 results_count=5
2026-02-07 19:42:27 [info     ] Generating RAG response

 articles=5 query=¿Cuáles son los resultados

deportivos recientes?
2026-02-07 19:42:32 [info     ] RAG response generated

 answer_length=586
confidence=0.8165528029644983 sources_used=2
2026-02-07 19:42:32 [info     ] RAG query completed

 confidence=0.8165528029644983
sources_cited=2 sources_found=5
  Confianza: 81.7%, Fuentes: 2

Generando con DeepSeek: ¿Hay noticias de fútbol?…
2026-02-07 19:42:32 [info     ] Initializing DeepSeek

client   model=deepseek-chat
2026-02-07 19:42:32 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:42:32 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:42:32 [info     ] Processing RAG query

 category=None max_sources=5
```

```
question=¿Hay noticias de fútbol?
2026-02-07 19:42:32 [info     ] Searching for articles

 limit=5 query=¿Hay noticias de fútbol?
threshold=0.5
2026-02-07 19:42:32 [info     ] Search completed

 results_count=5
2026-02-07 19:42:32 [info     ] Generating RAG response

 articles=5 query=¿Hay noticias de

fútbol?
2026-02-07 19:42:39 [info     ] RAG response generated

 answer_length=954
confidence=0.9041615839223826 sources_used=4
2026-02-07 19:42:39 [info     ] RAG query completed

 confidence=0.9041615839223826
sources_cited=4 sources_found=5
  Confianza: 90.4%, Fuentes: 4

Generando con DeepSeek: ¿Qué fichajes o traspasos hay en el fútb…
2026-02-07 19:42:39 [info     ] Initializing DeepSeek

client   model=deepseek-chat
2026-02-07 19:42:39 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:42:39 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:42:39 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Qué fichajes o traspasos hay en el fútbol?
2026-02-07 19:42:39 [info     ] Searching for articles

 limit=5 query=¿Qué fichajes o traspasos hay

en el fútbol? threshold=0.5
2026-02-07 19:42:39 [info     ] Search completed

 results_count=5
2026-02-07 19:42:39 [info     ] Generating RAG response

 articles=5 query=¿Qué fichajes o traspasos

hay en el fútbol?
2026-02-07 19:42:43 [info     ] RAG response generated

 answer_length=389 confidence=0.2
```

```
sources_used=1
2026-02-07 19:42:43 [info     ] RAG query completed

 confidence=0.2 sources_cited=1
sources_found=5
  Confianza: 20.0%, Fuentes: 1

Generando con DeepSeek: ¿Hay noticias sobre ciencia?…
2026-02-07 19:42:43 [info     ] Initializing DeepSeek

client   model=deepseek-chat
2026-02-07 19:42:43 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:42:43 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:42:43 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Hay noticias sobre ciencia?
2026-02-07 19:42:43 [info     ] Searching for articles

 limit=5 query=¿Hay noticias sobre

ciencia? threshold=0.5
2026-02-07 19:42:43 [info     ] Search completed

 results_count=5
2026-02-07 19:42:43 [info     ] Generating RAG response

 articles=5 query=¿Hay noticias sobre

ciencia?
2026-02-07 19:42:51 [info     ] RAG response generated

 answer_length=1266
confidence=0.9361255410096275 sources_used=5
2026-02-07 19:42:51 [info     ] RAG query completed

 confidence=0.9361255410096275
sources_cited=5 sources_found=5
  Confianza: 93.6%, Fuentes: 5

Generando con DeepSeek: ¿Qué descubrimientos científicos hay?…
2026-02-07 19:42:51 [info     ] Initializing DeepSeek

client   model=deepseek-chat
2026-02-07 19:42:51 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
```

```
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:42:51 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:42:51 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Qué descubrimientos científicos hay?
2026-02-07 19:42:51 [info     ] Searching for articles

 limit=5 query=¿Qué descubrimientos

científicos hay? threshold=0.5
2026-02-07 19:42:51 [info     ] Search completed

 results_count=5
2026-02-07 19:42:51 [info     ] Generating RAG response

 articles=5 query=¿Qué descubrimientos

científicos hay?
2026-02-07 19:42:59 [info     ] RAG response generated

 answer_length=1221
confidence=0.9395166125400706 sources_used=5
2026-02-07 19:42:59 [info     ] RAG query completed

 confidence=0.9395166125400706
sources_cited=5 sources_found=5
  Confianza: 94.0%, Fuentes: 5

Generando con DeepSeek: ¿Hay noticias sobre el espacio o astrono…
2026-02-07 19:42:59 [info     ] Initializing DeepSeek

client   model=deepseek-chat
2026-02-07 19:42:59 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:42:59 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:42:59 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Hay noticias sobre el espacio o astronomía?
2026-02-07 19:42:59 [info     ] Searching for articles

 limit=5 query=¿Hay noticias sobre el

espacio o astronomía? threshold=0.5
2026-02-07 19:43:00 [info     ] Search completed

 results_count=5
```

```
2026-02-07 19:43:00 [info     ] Generating RAG response
 articles=5 query=¿Hay noticias sobre el
espacio o astronomía?
2026-02-07 19:43:06 [info     ] RAG response generated
 answer_length=952
confidence=0.9043998690446889 sources_used=4
2026-02-07 19:43:06 [info     ] RAG query completed
 confidence=0.9043998690446889
sources_cited=4 sources_found=5
  Confianza: 90.4%, Fuentes: 4

Generando con DeepSeek: ¿Qué noticias hay de entretenimiento?…
2026-02-07 19:43:06 [info     ] Initializing DeepSeek
client   model=deepseek-chat
2026-02-07 19:43:06 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:43:06 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:43:06 [info     ] Processing RAG query
 category=None max_sources=5
question=¿Qué noticias hay de entretenimiento?
2026-02-07 19:43:06 [info     ] Searching for articles
 limit=5 query=¿Qué noticias hay de
entretenimiento? threshold=0.5
2026-02-07 19:43:07 [info     ] Search completed
 results_count=5
2026-02-07 19:43:07 [info     ] Generating RAG response
 articles=5 query=¿Qué noticias hay de
entretenimiento?
2026-02-07 19:43:12 [info     ] RAG response generated
 answer_length=719
confidence=0.9309473476425174 sources_used=5
2026-02-07 19:43:12 [info     ] RAG query completed
 confidence=0.9309473476425174
sources_cited=5 sources_found=5
  Confianza: 93.1%, Fuentes: 5
```

```
Generando con DeepSeek: ¿Hay estrenos de cine o series?…
2026-02-07 19:43:12 [info     ] Initializing DeepSeek

client   model=deepseek-chat
2026-02-07 19:43:12 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:43:12 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:43:12 [info     ] Processing RAG query

 category=None max_sources=5
question=¿Hay estrenos de cine o series?
2026-02-07 19:43:12 [info     ] Searching for articles

 limit=5 query=¿Hay estrenos de cine o

series? threshold=0.5
2026-02-07 19:43:12 [info     ] Search completed

 results_count=5
2026-02-07 19:43:12 [info     ] Generating RAG response

 articles=5 query=¿Hay estrenos de cine o

series?
2026-02-07 19:43:19 [info     ] RAG response generated

 answer_length=848
confidence=0.9367703018429272 sources_used=5
2026-02-07 19:43:19 [info     ] RAG query completed

 confidence=0.9367703018429272
sources_cited=5 sources_found=5
  Confianza: 93.7%, Fuentes: 5

Generando con DeepSeek: ¿Cuáles son las noticias internacionales…
2026-02-07 19:43:19 [info     ] Initializing DeepSeek

client   model=deepseek-chat
2026-02-07 19:43:19 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:43:19 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:43:19 [info     ] Processing RAG query

 category=None max_sources=5
```

```
question=¿Cuáles son las noticias internacionales más impor
2026-02-07 19:43:19 [info     ] Searching for articles

 limit=5 query=¿Cuáles son las noticias

internacionales más impor threshold=0.5
2026-02-07 19:43:19 [info     ] Search completed

 results_count=5
2026-02-07 19:43:19 [info     ] Generating RAG response

 articles=5 query=¿Cuáles son las noticias

internacionales más impor
2026-02-07 19:43:26 [info     ] RAG response generated

 answer_length=1050
confidence=0.8548983581615576 sources_used=3
2026-02-07 19:43:26 [info     ] RAG query completed

 confidence=0.8548983581615576
sources_cited=3 sources_found=5
  Confianza: 85.5%, Fuentes: 3

==================================================
GENERACIÓN CON DEEPSEEK
==================================================
                                                  question  sources_used
citations  confidence  topic_coverage
                       ¿Qué está pasando con la economía?             3
3    0.860578            0.50
       ¿Cuáles son las perspectivas económicas para 2026?             5
5    0.955969            0.75
                     ¿Cómo está la inflación actualmente?             2
4    0.824237            0.75
                      ¿Qué está pasando con los salarios?             1
3    0.678461            0.75
                         ¿Hay noticias sobre inversiones?             3
6    0.849585            0.25
                     ¿Qué dicen los analistas económicos?             3
3    0.864250            0.75
          ¿Cuáles son las últimas noticias de tecnología?             5
5    0.939006            0.50
              ¿Qué avances hay en inteligencia artificial?             5
5    0.958679            0.75
                      ¿Hay noticias sobre redes sociales?             4
4    0.897119            0.75
         ¿Qué está pasando con las empresas tecnológicas?             4
4    0.904908            0.50
                      ¿Hay noticias sobre ciberseguridad?             1
2    0.671763            0.25
```

```
                          ¿Qué noticias hay sobre salud?            5
5    0.934043            0.25
                             ¿Hay noticias sobre vacunas?            4
4    0.898075            0.75
              ¿Qué dicen los expertos sobre salud mental?            4
4    0.915078            0.25
                           ¿Hay avances médicos recientes?            5
5    0.941236            0.25
               ¿Qué está pasando en el mundo del deporte?            5
5    0.945181            0.50
         ¿Cuáles son los resultados deportivos recientes?            2
2    0.816553            1.00
                                   ¿Hay noticias de fútbol?            4
4    0.904162            0.50
              ¿Qué fichajes o traspasos hay en el fútbol?            1
1    0.200000            1.00
                             ¿Hay noticias sobre ciencia?            5
5    0.936126            0.75
                    ¿Qué descubrimientos científicos hay?            5
5    0.939517            1.00
             ¿Hay noticias sobre el espacio o astronomía?            4
4    0.904400            0.75
                    ¿Qué noticias hay de entretenimiento?            5
6    0.930947            0.25
                           ¿Hay estrenos de cine o series?            5
5    0.936770            1.00
¿Cuáles son las noticias internacionales más importantes?            3
3    0.854898            0.50
```

[23]:
```python
# Evaluar faithfulness de respuestas generadas por DeepSeek
# Usamos DeepSeek también como evaluador para consistencia
# Ahora pasando contenido completo de los artículos

if settings.deepseek_api_key and 'deepseek_generation_results' in dir():
    deepseek_faithfulness = []

    for i, result in enumerate(deepseek_generation_results):
        print(f"\nEvaluando faithfulness DeepSeek→DeepSeek {i+1}/
 ↪{len(deepseek_generation_results)}...")

        with get_db() as db:
            engine = RAGEngine(db, use_reranking=False)
            # Obtener artículos CON CONTENIDO
            articles = engine.get_context_only(
                question=EVALUATION_QUESTIONS[i]['question'],
                max_sources=5
            )
```

```
            faith_result = evaluate_faithfulness_with_provider(
                EVALUATION_QUESTIONS[i]['question'],
                result['answer'],
                articles,
                LLMProvider.DEEPSEEK
            )
            deepseek_faithfulness.append(faith_result['faithfulness_score'])

    df_deepseek_gen['faithfulness'] = deepseek_faithfulness

    # Comparación final
    print("\n" + "="*60)
    print("COMPARACIÓN FINAL: GENERACIÓN RAG")
    print("="*60)

    lm_conf = f"{df_generation['confidence'].mean():.1%}"
    ds_conf = f"{df_deepseek_gen['confidence'].mean():.1%}"
    lm_topic = f"{df_generation['topic_coverage'].mean():.1%}"
    ds_topic = f"{df_deepseek_gen['topic_coverage'].mean():.1%}"
    lm_citas = f"{df_generation['citations_in_answer'].mean():.1f}"
    ds_citas = f"{df_deepseek_gen['citations'].mean():.1f}"

    lm_faith = df_generation['faithfulness'].mean() if 'faithfulness' in␣
↪df_generation.columns else 0
    ds_faith = df_deepseek_gen['faithfulness'].mean() if deepseek_faithfulness␣
↪else 0
    lm_faith_str = f"{lm_faith:.1%}" if lm_faith else "N/A"
    ds_faith_str = f"{ds_faith:.1%}" if ds_faith else "N/A"

    print(f"\n{'Métrica':<25} {'LM Studio':<15} {'DeepSeek':<15}")
    print("-"*55)
    print(f"{'Confianza promedio':<25} {lm_conf:<15} {ds_conf:<15}")
    print(f"{'Cobertura temas':<25} {lm_topic:<15} {ds_topic:<15}")
    print(f"{'Citas usadas':<25} {lm_citas:<15} {ds_citas:<15}")
    print(f"{'Faithfulness':<25} {lm_faith_str:<15} {ds_faith_str:<15}")

    print("\n" + "="*60)
    if ds_faith and lm_faith and ds_faith > lm_faith + 0.1:
        print("  DeepSeek genera respuestas más fieles a las fuentes")
    elif ds_faith and lm_faith and lm_faith > ds_faith + 0.1:
        print("  LM Studio genera respuestas más fieles a las fuentes")
    else:
        print("  Ambos modelos tienen faithfulness similar")
else:
    print("  Ejecuta primero la celda de generación con DeepSeek")
```

```
Evaluando faithfulness DeepSeek→DeepSeek 1/25…
2026-02-07 19:44:14 [info     ] Using LM Studio (local)


2026-02-07 19:44:14 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:14 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:44:14 [info     ] Searching for articles

 limit=5 query=¿Qué está pasando con la

economía? threshold=0.5
2026-02-07 19:44:14 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:14 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:44:14 [info     ] Searching for articles

 limit=5 query=¿Qué está pasando con la

economía? threshold=0.5
2026-02-07 19:44:14 [info     ] Search completed

 results_count=5
2026-02-07 19:44:14 [info     ] Initializing DeepSeek

client   model=deepseek-chat
  [deepseek] Response: '5'


Evaluando faithfulness DeepSeek→DeepSeek 2/25…
2026-02-07 19:44:15 [info     ] Using LM Studio (local)


2026-02-07 19:44:15 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:15 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
```

```
2026-02-07 19:44:15 [info     ] Searching for articles
 limit=5 query=¿Cuáles son las perspectivas
económicas para 2026? threshold=0.5
2026-02-07 19:44:15 [info     ] Search completed
 results_count=5
2026-02-07 19:44:15 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '5'

Evaluando faithfulness DeepSeek→DeepSeek 3/25…
2026-02-07 19:44:17 [info     ] Using LM Studio (local)


2026-02-07 19:44:17 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:17 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:44:17 [info     ] Searching for articles
 limit=5 query=¿Cómo está la inflación
actualmente? threshold=0.5
2026-02-07 19:44:17 [info     ] Search completed
 results_count=5
2026-02-07 19:44:17 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '5'

Evaluando faithfulness DeepSeek→DeepSeek 4/25…
2026-02-07 19:44:18 [info     ] Using LM Studio (local)


2026-02-07 19:44:18 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:18 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:44:18 [info     ] Searching for articles
 limit=5 query=¿Qué está pasando con los
salarios? threshold=0.5
```

```
2026-02-07 19:44:18 [info     ] Search completed
 results_count=5
2026-02-07 19:44:18 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '5'

Evaluando faithfulness DeepSeek→DeepSeek 5/25…
2026-02-07 19:44:20 [info     ] Using LM Studio (local)


2026-02-07 19:44:20 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:20 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:44:20 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
inversiones? threshold=0.5
2026-02-07 19:44:20 [info     ] Search completed
 results_count=5
2026-02-07 19:44:20 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '8'

Evaluando faithfulness DeepSeek→DeepSeek 6/25…
2026-02-07 19:44:21 [info     ] Using LM Studio (local)


2026-02-07 19:44:21 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:21 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:44:21 [info     ] Searching for articles
 limit=5 query=¿Qué dicen los analistas
económicos? threshold=0.5
2026-02-07 19:44:21 [info     ] Search completed
 results_count=5
```

```
2026-02-07 19:44:21 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '5'

Evaluando faithfulness DeepSeek→DeepSeek 7/25…
2026-02-07 19:44:22 [info     ] Using LM Studio (local)


2026-02-07 19:44:22 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:22 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:44:22 [info     ] Searching for articles
 limit=5 query=¿Cuáles son las últimas
noticias de tecnología? threshold=0.5
2026-02-07 19:44:22 [info     ] Search completed
 results_count=5
2026-02-07 19:44:22 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '8'

Evaluando faithfulness DeepSeek→DeepSeek 8/25…
2026-02-07 19:44:24 [info     ] Using LM Studio (local)


2026-02-07 19:44:24 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:24 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:44:24 [info     ] Searching for articles
 limit=5 query=¿Qué avances hay en
inteligencia artificial? threshold=0.5
2026-02-07 19:44:24 [info     ] Search completed
 results_count=5
2026-02-07 19:44:24 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '8'
```

```
Evaluando faithfulness DeepSeek→DeepSeek 9/25…
2026-02-07 19:44:25 [info     ] Using LM Studio (local)


2026-02-07 19:44:25 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:25 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:44:25 [info     ] Searching for articles

 limit=5 query=¿Hay noticias sobre redes

sociales? threshold=0.5
2026-02-07 19:44:25 [info     ] Search completed

 results_count=5
2026-02-07 19:44:25 [info     ] Initializing DeepSeek

client   model=deepseek-chat
  [deepseek] Response: '8'

Evaluando faithfulness DeepSeek→DeepSeek 10/25…
2026-02-07 19:44:27 [info     ] Using LM Studio (local)


2026-02-07 19:44:27 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:27 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:44:27 [info     ] Searching for articles

 limit=5 query=¿Qué está pasando con las

empresas tecnológicas? threshold=0.5
2026-02-07 19:44:27 [info     ] Search completed

 results_count=5
2026-02-07 19:44:27 [info     ] Initializing DeepSeek

client   model=deepseek-chat
  [deepseek] Response: '8'

Evaluando faithfulness DeepSeek→DeepSeek 11/25…
```

```
2026-02-07 19:44:28 [info     ] Using LM Studio (local)


2026-02-07 19:44:28 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:28 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:44:28 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
ciberseguridad? threshold=0.5
2026-02-07 19:44:28 [info     ] Search completed
 results_count=5
2026-02-07 19:44:28 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '8'

Evaluando faithfulness DeepSeek→DeepSeek 12/25…
2026-02-07 19:44:30 [info     ] Using LM Studio (local)


2026-02-07 19:44:30 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:30 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:44:30 [info     ] Searching for articles
 limit=5 query=¿Qué noticias hay sobre
salud? threshold=0.5
2026-02-07 19:44:30 [info     ] Search completed
 results_count=5
2026-02-07 19:44:30 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '10'

Evaluando faithfulness DeepSeek→DeepSeek 13/25…
2026-02-07 19:44:32 [info     ] Using LM Studio (local)
```

```
2026-02-07 19:44:32 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:32 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:44:32 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
vacunas? threshold=0.5
2026-02-07 19:44:32 [info     ] Search completed
 results_count=5
2026-02-07 19:44:32 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '8'

Evaluando faithfulness DeepSeek→DeepSeek 14/25…
2026-02-07 19:44:33 [info     ] Using LM Studio (local)


2026-02-07 19:44:33 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:33 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:44:33 [info     ] Searching for articles
 limit=5 query=¿Qué dicen los expertos sobre
salud mental? threshold=0.5
2026-02-07 19:44:33 [info     ] Search completed
 results_count=5
2026-02-07 19:44:33 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '8'

Evaluando faithfulness DeepSeek→DeepSeek 15/25…
2026-02-07 19:44:34 [info     ] Using LM Studio (local)


2026-02-07 19:44:34 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
```

```
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:34 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:44:34 [info     ] Searching for articles

 limit=5 query=¿Hay avances médicos

recientes? threshold=0.5
2026-02-07 19:44:34 [info     ] Search completed

 results_count=5
2026-02-07 19:44:34 [info     ] Initializing DeepSeek

client   model=deepseek-chat
  [deepseek] Response: '10'

Evaluando faithfulness DeepSeek→DeepSeek 16/25…
2026-02-07 19:44:36 [info     ] Using LM Studio (local)


2026-02-07 19:44:36 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:36 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:44:36 [info     ] Searching for articles

 limit=5 query=¿Qué está pasando en el mundo

del deporte? threshold=0.5
2026-02-07 19:44:36 [info     ] Search completed

 results_count=5
2026-02-07 19:44:36 [info     ] Initializing DeepSeek

client   model=deepseek-chat
  [deepseek] Response: '10'

Evaluando faithfulness DeepSeek→DeepSeek 17/25…
2026-02-07 19:44:38 [info     ] Using LM Studio (local)


2026-02-07 19:44:38 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:38 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
```

```
2026-02-07 19:44:38 [info     ] Searching for articles
 limit=5 query=¿Cuáles son los resultados
deportivos recientes? threshold=0.5
2026-02-07 19:44:38 [info     ] Search completed
 results_count=5
2026-02-07 19:44:38 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '5'

Evaluando faithfulness DeepSeek→DeepSeek 18/25…
2026-02-07 19:44:39 [info     ] Using LM Studio (local)


2026-02-07 19:44:39 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:39 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:44:39 [info     ] Searching for articles
 limit=5 query=¿Hay noticias de fútbol?
threshold=0.5
2026-02-07 19:44:39 [info     ] Search completed
 results_count=5
2026-02-07 19:44:39 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '10'

Evaluando faithfulness DeepSeek→DeepSeek 19/25…
2026-02-07 19:44:41 [info     ] Using LM Studio (local)


2026-02-07 19:44:41 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:41 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:44:41 [info     ] Searching for articles
 limit=5 query=¿Qué fichajes o traspasos hay
en el fútbol? threshold=0.5
```

```
2026-02-07 19:44:41 [info     ] Search completed
 results_count=5
2026-02-07 19:44:41 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '5'

Evaluando faithfulness DeepSeek→DeepSeek 20/25…
2026-02-07 19:44:42 [info     ] Using LM Studio (local)


2026-02-07 19:44:42 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:42 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:44:42 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
ciencia? threshold=0.5
2026-02-07 19:44:42 [info     ] Search completed
 results_count=5
2026-02-07 19:44:42 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '10'

Evaluando faithfulness DeepSeek→DeepSeek 21/25…
2026-02-07 19:44:44 [info     ] Using LM Studio (local)


2026-02-07 19:44:44 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:44 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:44:44 [info     ] Searching for articles
 limit=5 query=¿Qué descubrimientos
científicos hay? threshold=0.5
2026-02-07 19:44:44 [info     ] Search completed
 results_count=5
```

```
2026-02-07 19:44:44 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '5'

Evaluando faithfulness DeepSeek→DeepSeek 22/25…
2026-02-07 19:44:45 [info     ] Using LM Studio (local)


2026-02-07 19:44:45 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:45 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:44:45 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre el
espacio o astronomía? threshold=0.5
2026-02-07 19:44:45 [info     ] Search completed
 results_count=5
2026-02-07 19:44:45 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '10'

Evaluando faithfulness DeepSeek→DeepSeek 23/25…
2026-02-07 19:44:47 [info     ] Using LM Studio (local)


2026-02-07 19:44:47 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:47 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:44:47 [info     ] Searching for articles
 limit=5 query=¿Qué noticias hay de
entretenimiento? threshold=0.5
2026-02-07 19:44:47 [info     ] Search completed
 results_count=5
2026-02-07 19:44:47 [info     ] Initializing DeepSeek
client   model=deepseek-chat
  [deepseek] Response: '10'
```

```
Evaluando faithfulness DeepSeek→DeepSeek 24/25…
2026-02-07 19:44:48 [info     ] Using LM Studio (local)


2026-02-07 19:44:48 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:48 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:44:48 [info     ] Searching for articles

 limit=5 query=¿Hay estrenos de cine o

series? threshold=0.5
2026-02-07 19:44:48 [info     ] Search completed

 results_count=5
2026-02-07 19:44:48 [info     ] Initializing DeepSeek

client   model=deepseek-chat
  [deepseek] Response: '10'

Evaluando faithfulness DeepSeek→DeepSeek 25/25…
2026-02-07 19:44:50 [info     ] Using LM Studio (local)


2026-02-07 19:44:50 [info     ] Initialized

LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:44:50 [info     ] RAG engine initialized

 llm_available=True use_reranking=False
2026-02-07 19:44:50 [info     ] Searching for articles

 limit=5 query=¿Cuáles son las noticias

internacionales más impor threshold=0.5
2026-02-07 19:44:50 [info     ] Search completed

 results_count=5
2026-02-07 19:44:50 [info     ] Initializing DeepSeek

client   model=deepseek-chat
  [deepseek] Response: '5'

============================================================
COMPARACIÓN FINAL: GENERACIÓN RAG
============================================================
```

```
Métrica                    LM Studio       DeepSeek
-------------------------------------------------------
Confianza promedio         83.7%           85.8%
Cobertura temas            55.0%           61.0%
Citas usadas               3.0             4.1
Faithfulness               77.9%           75.6%

============================================================
 Ambos modelos tienen faithfulness similar
```

1.7.3 6.3 Fact-Checking: Verificación por Claims

Implementamos una evaluación más rigurosa siguiendo las recomendaciones de Gilhuly y Shahzad (2025):

1. **Descomponer** la respuesta en afirmaciones individuales (claims)
2. **Verificar** cada claim contra las fuentes originales
3. **Calcular** la tasa de alucinación (claims sin respaldo)

[24]:
```python
# Funciones de Fact-Checking (MEJORADAS)

def extract_claims(answer: str, llm_client=None) -> list[str]:
    """
    Descompone una respuesta en afirmaciones factuales individuales.
    """
    if llm_client is None:
        llm_client = get_available_client()

    prompt = """Extrae SOLO las afirmaciones factuales PRINCIPALES del␣
 ↪siguiente texto.
NO extraigas detalles menores o inferencias. Solo hechos concretos y␣
 ↪verificables.
Devuelve UNA afirmación por línea, sin números ni viñetas.
Máximo 5-7 afirmaciones principales.

TEXTO:
{text}

AFIRMACIONES PRINCIPALES (una por línea):"""

    try:
        response = llm_client.generate(prompt.format(text=answer),␣
 ↪max_tokens=500, temperature=0)
        claims = [line.strip() for line in response.content.strip().split('\n')␣
 ↪if line.strip()]
        # Filtrar líneas que no parecen claims (muy cortas o son headers)
        claims = [c for c in claims if len(c) > 20 and not c.endswith(':')]
```

```
        return claims[:7]  # Máximo 7 claims
    except Exception as e:
        print(f"  Error extrayendo claims: {e}")
        return []


def verify_claim(claim: str, sources_text: str, llm_client=None) -> dict:
    """
    Verifica si un claim está respaldado por las fuentes.
    IMPORTANTE: sources_text debe incluir el CONTENIDO de los artículos.
    """
    if llm_client is None:
        llm_client = get_available_client()

    prompt = """Determina si la siguiente AFIRMACIÓN está respaldada por la␣
 ↪información en las FUENTES.

AFIRMACIÓN: {claim}

FUENTES (títulos y contenido):
{sources}

Instrucciones:
- RESPALDADO: La información está claramente en las fuentes (puede ser␣
 ↪parafraseada)
- PARCIAL: La idea general está en las fuentes pero faltan detalles específicos
- NO RESPALDADO: La información NO aparece en las fuentes

Tu respuesta (una sola palabra - RESPALDADO, PARCIAL, o NO RESPALDADO):"""

    try:
        response = llm_client.generate(
            prompt.format(claim=claim, sources=sources_text),
            max_tokens=50,
            temperature=0
        )
        result = response.content.strip().upper()

        if 'RESPALDADO' in result and 'NO' not in result:
            return {'supported': True, 'status': 'RESPALDADO'}
        elif 'PARCIAL' in result:
            return {'supported': True, 'status': 'PARCIAL'}  # Contamos parcial␣
 ↪como OK
        else:
            return {'supported': False, 'status': 'NO RESPALDADO'}
    except Exception as e:
        print(f"  Error verificando claim: {e}")
```

```
        return {'supported': False, 'status': 'ERROR'}


def fact_check_answer(question: str, answer: str, articles: list,␣
 ↪llm_client=None) -> dict:
    """
    Realiza fact-checking completo de una respuesta.

    Args:
        question: La pregunta original
        answer: La respuesta generada
        articles: Lista de tuplas (Article, score) con el contenido completo
        llm_client: Cliente LLM para evaluación

    Returns:
        dict con claims, verificaciones y tasa de alucinación
    """
    if llm_client is None:
        llm_client = get_available_client()

    # 1. Extraer claims (máximo 7)
    claims = extract_claims(answer, llm_client)

    if not claims:
        return {
            'total_claims': 0,
            'supported_claims': 0,
            'hallucination_rate': None,
            'claims_detail': []
        }

    # 2. Preparar texto de fuentes CON CONTENIDO
    sources_text = ""
    for i, (article, score) in enumerate(articles):
        # Incluir título + descripción/contenido (primeros 500 chars)
        content = article.content or article.description or ""
        content_preview = content[:500] + "..." if len(content) > 500 else␣
 ↪content
        sources_text += f"\n[{i+1}] {article.title}\n"
        sources_text += f"    Fuente: {article.source_name}\n"
        sources_text += f"    Contenido: {content_preview}\n"

    # 3. Verificar cada claim
    claims_detail = []
    supported_count = 0

    for claim in claims:
```

```
        result = verify_claim(claim, sources_text, llm_client)
        claims_detail.append({
            'claim': claim,
            'supported': result['supported'],
            'status': result['status']
        })
        if result['supported']:
            supported_count += 1

    # 4. Calcular tasa de alucinación
    hallucination_rate = 1 - (supported_count / len(claims))

    return {
        'total_claims': len(claims),
        'supported_claims': supported_count,
        'unsupported_claims': len(claims) - supported_count,
        'hallucination_rate': hallucination_rate,
        'claims_detail': claims_detail
    }

print("Funciones de fact-checking definidas")
```

```
Funciones de fact-checking definidas
```

[25]:
```
# Ejecutar Fact-Checking MEJORADO con DeepSeek
# Ahora pasamos el CONTENIDO de los artículos, no solo los títulos

if not settings.deepseek_api_key:
    print("  DeepSeek no configurado. Usando LM Studio.")
    fact_check_llm = get_available_client()
else:
    fact_check_llm = get_llm_client(LLMProvider.DEEPSEEK)
    print("  Usando DeepSeek para fact-checking")

fact_check_results = []

# Usar las respuestas de DeepSeek (o LM Studio si no hay)
responses_to_check = deepseek_generation_results if␣
 ↪'deepseek_generation_results' in dir() else generation_results
print(f"  Verificando respuestas de: {'DeepSeek' if␣
 ↪'deepseek_generation_results' in dir() else 'LM Studio'}")
print(f"  Total preguntas a evaluar: {len(responses_to_check)}")

for i, result in enumerate(responses_to_check):
    question = EVALUATION_QUESTIONS[i]['question']
    print(f"\n{'='*60}")
```

```
    print(f"[{i+1}/{len(responses_to_check)}] Fact-checking: {question[:50]}...
↪")

    with get_db() as db:
        engine = RAGEngine(db, use_reranking=False)
        # Obtener artículos CON CONTENIDO (no solo SourceCitation)
        articles = engine.get_context_only(question=question, max_sources=5)

        # Fact-check pasando los artículos completos
        fc_result = fact_check_answer(
            question=question,
            answer=result['answer'],
            articles=articles,  # Ahora son tuplas (Article, score) con␣
↪contenido
            llm_client=fact_check_llm
        )

        fact_check_results.append({
            'question': question,
            **fc_result
        })

        print(f"  Claims totales: {fc_result['total_claims']}")
        print(f"  Respaldados: {fc_result['supported_claims']}")
        print(f"  Sin respaldo: {fc_result['unsupported_claims']}")
        if fc_result['hallucination_rate'] is not None:
            print(f"  Tasa alucinación: {fc_result['hallucination_rate']:.1%}")

# Crear DataFrame
df_factcheck = pd.DataFrame(fact_check_results)
print("\n" + "="*60)
print("RESUMEN FACT-CHECKING (CON CONTENIDO DE ARTÍCULOS)")
print("="*60)
print(df_factcheck[['question', 'total_claims', 'supported_claims',␣
↪'unsupported_claims', 'hallucination_rate']].to_string(index=False))
```

```
2026-02-07 19:45:52 [info     ] Initializing DeepSeek
client   model=deepseek-chat
 Usando DeepSeek para fact-checking
 Verificando respuestas de: DeepSeek
 Total preguntas a evaluar: 25

============================================================
[1/25] Fact-checking: ¿Qué está pasando con la economía?…
2026-02-07 19:45:52 [info     ] Using LM Studio (local)
```

```
2026-02-07 19:45:52 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:45:52 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:45:52 [info     ] Searching for articles
 limit=5 query=¿Qué está pasando con la
economía? threshold=0.5
2026-02-07 19:45:52 [info     ] Search completed
 results_count=5
  Claims totales: 5
  Respaldados: 3
  Sin respaldo: 2
  Tasa alucinación: 40.0%

============================================================
[2/25] Fact-checking: ¿Cuáles son las perspectivas económicas para 2026?…
2026-02-07 19:46:03 [info     ] Using LM Studio (local)


2026-02-07 19:46:03 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:46:03 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:46:03 [info     ] Searching for articles
 limit=5 query=¿Cuáles son las perspectivas
económicas para 2026? threshold=0.5
2026-02-07 19:46:03 [info     ] Search completed
 results_count=5
  Claims totales: 7
  Respaldados: 5
  Sin respaldo: 2
  Tasa alucinación: 28.6%

============================================================
[3/25] Fact-checking: ¿Cómo está la inflación actualmente?…
2026-02-07 19:46:16 [info     ] Using LM Studio (local)
```

```
2026-02-07 19:46:16 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:46:16 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:46:16 [info     ] Searching for articles
 limit=5 query=¿Cómo está la inflación
actualmente? threshold=0.5
2026-02-07 19:46:16 [info     ] Search completed
 results_count=5
  Claims totales: 5
  Respaldados: 3
  Sin respaldo: 2
  Tasa alucinación: 40.0%

============================================================
[4/25] Fact-checking: ¿Qué está pasando con los salarios?…
2026-02-07 19:46:27 [info     ] Using LM Studio (local)


2026-02-07 19:46:27 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:46:27 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:46:27 [info     ] Searching for articles
 limit=5 query=¿Qué está pasando con los
salarios? threshold=0.5
2026-02-07 19:46:27 [info     ] Search completed
 results_count=5
  Claims totales: 5
  Respaldados: 1
  Sin respaldo: 4
  Tasa alucinación: 80.0%

============================================================
[5/25] Fact-checking: ¿Hay noticias sobre inversiones?…
2026-02-07 19:46:38 [info     ] Using LM Studio (local)
```

```
2026-02-07 19:46:38 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:46:38 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:46:38 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
inversiones? threshold=0.5
2026-02-07 19:46:38 [info     ] Search completed
 results_count=5
  Claims totales: 5
  Respaldados: 4
  Sin respaldo: 1
  Tasa alucinación: 20.0%

============================================================
[6/25] Fact-checking: ¿Qué dicen los analistas económicos?…
2026-02-07 19:46:50 [info     ] Using LM Studio (local)


2026-02-07 19:46:50 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:46:50 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:46:50 [info     ] Searching for articles
 limit=5 query=¿Qué dicen los analistas
económicos? threshold=0.5
2026-02-07 19:46:50 [info     ] Search completed
 results_count=5
  Claims totales: 6
  Respaldados: 4
  Sin respaldo: 2
  Tasa alucinación: 33.3%

============================================================
[7/25] Fact-checking: ¿Cuáles son las últimas noticias de tecnología?…
2026-02-07 19:47:02 [info     ] Using LM Studio (local)
```

```
2026-02-07 19:47:02 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:47:02 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:47:02 [info     ] Searching for articles
 limit=5 query=¿Cuáles son las últimas
noticias de tecnología? threshold=0.5
2026-02-07 19:47:03 [info     ] Search completed
 results_count=5
  Claims totales: 7
  Respaldados: 4
  Sin respaldo: 3
  Tasa alucinación: 42.9%

============================================================
[8/25] Fact-checking: ¿Qué avances hay en inteligencia artificial?…
2026-02-07 19:47:16 [info     ] Using LM Studio (local)


2026-02-07 19:47:16 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:47:16 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:47:16 [info     ] Searching for articles
 limit=5 query=¿Qué avances hay en
inteligencia artificial? threshold=0.5
2026-02-07 19:47:17 [info     ] Search completed
 results_count=5
  Claims totales: 6
  Respaldados: 6
  Sin respaldo: 0
  Tasa alucinación: 0.0%

=============================================================
[9/25] Fact-checking: ¿Hay noticias sobre redes sociales?…
2026-02-07 19:47:29 [info     ] Using LM Studio (local)
```

```
2026-02-07 19:47:29 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:47:29 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:47:29 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre redes
sociales? threshold=0.5
2026-02-07 19:47:29 [info     ] Search completed
 results_count=5
  Claims totales: 4
  Respaldados: 4
  Sin respaldo: 0
  Tasa alucinación: 0.0%

============================================================
[10/25] Fact-checking: ¿Qué está pasando con las empresas tecnológicas?…
2026-02-07 19:47:38 [info     ] Using LM Studio (local)


2026-02-07 19:47:38 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:47:38 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:47:38 [info     ] Searching for articles
 limit=5 query=¿Qué está pasando con las
empresas tecnológicas? threshold=0.5
2026-02-07 19:47:38 [info     ] Search completed
 results_count=5
  Claims totales: 5
  Respaldados: 4
  Sin respaldo: 1
  Tasa alucinación: 20.0%

============================================================
[11/25] Fact-checking: ¿Hay noticias sobre ciberseguridad?…
2026-02-07 19:47:49 [info     ] Using LM Studio (local)
```

```
2026-02-07 19:47:49 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:47:49 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:47:49 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
ciberseguridad? threshold=0.5
2026-02-07 19:47:49 [info     ] Search completed
 results_count=5
  Claims totales: 4
  Respaldados: 2
  Sin respaldo: 2
  Tasa alucinación: 50.0%

============================================================
[12/25] Fact-checking: ¿Qué noticias hay sobre salud?…
2026-02-07 19:47:58 [info     ] Using LM Studio (local)


2026-02-07 19:47:58 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:47:58 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:47:58 [info     ] Searching for articles
 limit=5 query=¿Qué noticias hay sobre
salud? threshold=0.5
2026-02-07 19:47:58 [info     ] Search completed
 results_count=5
  Claims totales: 5
  Respaldados: 5
  Sin respaldo: 0
  Tasa alucinación: 0.0%

============================================================
[13/25] Fact-checking: ¿Hay noticias sobre vacunas?…
2026-02-07 19:48:10 [info     ] Using LM Studio (local)
```

```
2026-02-07 19:48:10 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:48:10 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:48:10 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
vacunas? threshold=0.5
2026-02-07 19:48:10 [info     ] Search completed
 results_count=5
  Claims totales: 6
  Respaldados: 5
  Sin respaldo: 1
  Tasa alucinación: 16.7%

============================================================
[14/25] Fact-checking: ¿Qué dicen los expertos sobre salud mental?…
2026-02-07 19:48:23 [info     ] Using LM Studio (local)


2026-02-07 19:48:23 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:48:23 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:48:23 [info     ] Searching for articles
 limit=5 query=¿Qué dicen los expertos sobre
salud mental? threshold=0.5
2026-02-07 19:48:23 [info     ] Search completed
 results_count=5
  Claims totales: 5
  Respaldados: 3
  Sin respaldo: 2
  Tasa alucinación: 40.0%

============================================================
[15/25] Fact-checking: ¿Hay avances médicos recientes?…
2026-02-07 19:48:33 [info     ] Using LM Studio (local)
```

```
2026-02-07 19:48:33 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:48:33 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:48:33 [info     ] Searching for articles
 limit=5 query=¿Hay avances médicos
recientes? threshold=0.5
2026-02-07 19:48:34 [info     ] Search completed
 results_count=5
  Claims totales: 6
  Respaldados: 5
  Sin respaldo: 1
  Tasa alucinación: 16.7%

============================================================
[16/25] Fact-checking: ¿Qué está pasando en el mundo del deporte?…
2026-02-07 19:48:45 [info     ] Using LM Studio (local)


2026-02-07 19:48:45 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:48:45 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:48:45 [info     ] Searching for articles
 limit=5 query=¿Qué está pasando en el mundo
del deporte? threshold=0.5
2026-02-07 19:48:45 [info     ] Search completed
 results_count=5
  Claims totales: 5
  Respaldados: 5
  Sin respaldo: 0
  Tasa alucinación: 0.0%

============================================================
[17/25] Fact-checking: ¿Cuáles son los resultados deportivos recientes?…
2026-02-07 19:48:55 [info     ] Using LM Studio (local)
```

```
2026-02-07 19:48:55 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:48:55 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:48:55 [info     ] Searching for articles
 limit=5 query=¿Cuáles son los resultados
deportivos recientes? threshold=0.5
2026-02-07 19:48:55 [info     ] Search completed
 results_count=5
  Claims totales: 5
  Respaldados: 4
  Sin respaldo: 1
  Tasa alucinación: 20.0%

============================================================
[18/25] Fact-checking: ¿Hay noticias de fútbol?…
2026-02-07 19:49:06 [info     ] Using LM Studio (local)


2026-02-07 19:49:06 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:49:06 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:49:06 [info     ] Searching for articles
 limit=5 query=¿Hay noticias de fútbol?
threshold=0.5
2026-02-07 19:49:06 [info     ] Search completed
 results_count=5
  Claims totales: 6
  Respaldados: 5
  Sin respaldo: 1
  Tasa alucinación: 16.7%

============================================================
[19/25] Fact-checking: ¿Qué fichajes o traspasos hay en el fútbol?…
2026-02-07 19:49:19 [info     ] Using LM Studio (local)
```

```
2026-02-07 19:49:19 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:49:19 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:49:19 [info     ] Searching for articles
 limit=5 query=¿Qué fichajes o traspasos hay
en el fútbol? threshold=0.5
2026-02-07 19:49:19 [info     ] Search completed
 results_count=5
  Claims totales: 6
  Respaldados: 2
  Sin respaldo: 4
  Tasa alucinación: 66.7%

============================================================
[20/25] Fact-checking: ¿Hay noticias sobre ciencia?…
2026-02-07 19:49:31 [info     ] Using LM Studio (local)


2026-02-07 19:49:31 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:49:31 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:49:31 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre
ciencia? threshold=0.5
2026-02-07 19:49:31 [info     ] Search completed
 results_count=5
  Claims totales: 5
  Respaldados: 3
  Sin respaldo: 2
  Tasa alucinación: 40.0%

============================================================
[21/25] Fact-checking: ¿Qué descubrimientos científicos hay?…
2026-02-07 19:49:43 [info     ] Using LM Studio (local)
```

```
2026-02-07 19:49:43 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:49:43 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:49:43 [info     ] Searching for articles
 limit=5 query=¿Qué descubrimientos
científicos hay? threshold=0.5
2026-02-07 19:49:43 [info     ] Search completed
 results_count=5
  Claims totales: 5
  Respaldados: 5
  Sin respaldo: 0
  Tasa alucinación: 0.0%

============================================================
[22/25] Fact-checking: ¿Hay noticias sobre el espacio o astronomía?…
2026-02-07 19:49:53 [info     ] Using LM Studio (local)


2026-02-07 19:49:53 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:49:53 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:49:53 [info     ] Searching for articles
 limit=5 query=¿Hay noticias sobre el
espacio o astronomía? threshold=0.5
2026-02-07 19:49:53 [info     ] Search completed
 results_count=5
  Claims totales: 5
  Respaldados: 5
  Sin respaldo: 0
  Tasa alucinación: 0.0%

============================================================
[23/25] Fact-checking: ¿Qué noticias hay de entretenimiento?…
2026-02-07 19:50:04 [info     ] Using LM Studio (local)
```

```
2026-02-07 19:50:04 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:50:04 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:50:04 [info     ] Searching for articles
 limit=5 query=¿Qué noticias hay de
entretenimiento? threshold=0.5
2026-02-07 19:50:04 [info     ] Search completed
 results_count=5
  Claims totales: 6
  Respaldados: 5
  Sin respaldo: 1
  Tasa alucinación: 16.7%

============================================================
[24/25] Fact-checking: ¿Hay estrenos de cine o series?…
2026-02-07 19:50:16 [info     ] Using LM Studio (local)


2026-02-07 19:50:16 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:50:16 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:50:16 [info     ] Searching for articles
 limit=5 query=¿Hay estrenos de cine o
series? threshold=0.5
2026-02-07 19:50:16 [info     ] Search completed
 results_count=5
  Claims totales: 6
  Respaldados: 6
  Sin respaldo: 0
  Tasa alucinación: 0.0%

============================================================
[25/25] Fact-checking: ¿Cuáles son las noticias internacionales más impor…
2026-02-07 19:50:28 [info     ] Using LM Studio (local)
```

```
2026-02-07 19:50:28 [info     ] Initialized
LMStudioEmbeddingGenerator
base_url=http://192.168.68.54:1234/v1
model=text-embedding-nomic-embed-text-v1.5
2026-02-07 19:50:28 [info     ] RAG engine initialized
 llm_available=True use_reranking=False
2026-02-07 19:50:28 [info     ] Searching for articles
 limit=5 query=¿Cuáles son las noticias
internacionales más impor threshold=0.5
2026-02-07 19:50:28 [info     ] Search completed
 results_count=5
  Claims totales: 5
  Respaldados: 4
  Sin respaldo: 1
  Tasa alucinación: 20.0%

============================================================
RESUMEN FACT-CHECKING (CON CONTENIDO DE ARTÍCULOS)
============================================================
                                                  question  total_claims
supported_claims  unsupported_claims  hallucination_rate
                       ¿Qué está pasando con la economía?             5
3                   2            0.400000
       ¿Cuáles son las perspectivas económicas para 2026?             7
5                   2            0.285714
                      ¿Cómo está la inflación actualmente?             5
3                   2            0.400000
                        ¿Qué está pasando con los salarios?             5
1                   4            0.800000
                           ¿Hay noticias sobre inversiones?             5
4                   1            0.200000
                      ¿Qué dicen los analistas económicos?             6
4                   2            0.333333
          ¿Cuáles son las últimas noticias de tecnología?             7
4                   3            0.428571
             ¿Qué avances hay en inteligencia artificial?             6
6                   0            0.000000
                        ¿Hay noticias sobre redes sociales?             4
4                   0            0.000000
         ¿Qué está pasando con las empresas tecnológicas?             5
4                   1            0.200000
                        ¿Hay noticias sobre ciberseguridad?             4
2                   2            0.500000
                              ¿Qué noticias hay sobre salud?             5
5                   0            0.000000
```

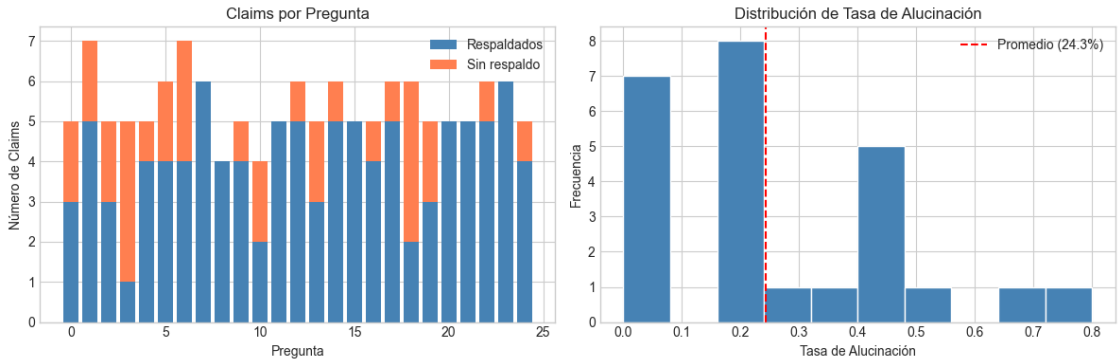
```
                              ¿Hay noticias sobre vacunas?            6
5                   1            0.166667
               ¿Qué dicen los expertos sobre salud mental?            5
3                   2            0.400000
                         ¿Hay avances médicos recientes?            6
5                   1            0.166667
                ¿Qué está pasando en el mundo del deporte?            5
5                   0            0.000000
          ¿Cuáles son los resultados deportivos recientes?            5
4                   1            0.200000
                                 ¿Hay noticias de fútbol?            6
5                   1            0.166667
               ¿Qué fichajes o traspasos hay en el fútbol?            6
2                   4            0.666667
                              ¿Hay noticias sobre ciencia?            5
3                   2            0.400000
                    ¿Qué descubrimientos científicos hay?            5
5                   0            0.000000
              ¿Hay noticias sobre el espacio o astronomía?            5
5                   0            0.000000
                    ¿Qué noticias hay de entretenimiento?            6
5                   1            0.166667
                         ¿Hay estrenos de cine o series?            6
6                   0            0.000000
¿Cuáles son las noticias internacionales más importantes?            5
4                   1            0.200000
```

[26]:
```python
# Análisis de Fact-Checking
if len(df_factcheck) > 0 and df_factcheck['hallucination_rate'].notna().any():
    avg_hallucination = df_factcheck['hallucination_rate'].mean()
    avg_claims = df_factcheck['total_claims'].mean()
    total_claims = df_factcheck['total_claims'].sum()
    total_supported = df_factcheck['supported_claims'].sum()

    print("=" * 60)
    print("ANÁLISIS DE FACT-CHECKING")
    print("=" * 60)
    print(f"\nMuestra evaluada: {len(df_factcheck)} respuestas")
    print(f"Total claims analizados: {total_claims}")
    print(f"Claims promedio por respuesta: {avg_claims:.1f}")
    print()
    print(f"Claims respaldados: {total_supported} ({total_supported/
 ↪total_claims:.1%})")
    print(f"Claims sin respaldo: {total_claims - total_supported}␣
 ↪({(total_claims - total_supported)/total_claims:.1%})")
    print()
    print(f"{'='*40}")
```

```
    print(f"TASA DE ALUCINACIÓN PROMEDIO: {avg_hallucination:.1%}")
    print(f"{'='*40}")
    print()

    # Interpretación
    if avg_hallucination < 0.1:
        print("  Excelente: Menos del 10% de alucinación")
    elif avg_hallucination < 0.25:
        print("  Bueno: Entre 10-25% de alucinación")
    elif avg_hallucination < 0.40:
        print("  Moderado: Entre 25-40% de alucinación")
    else:
        print("  Alto: Más del 40% de alucinación - requiere mejoras")

    # Visualización
    fig, axes = plt.subplots(1, 2, figsize=(12, 4))

    # Gráfico de barras por pregunta
    x = range(len(df_factcheck))
    axes[0].bar(x, df_factcheck['supported_claims'], label='Respaldados',␣
 ↪color='steelblue')
    axes[0].bar(x, df_factcheck['unsupported_claims'],␣
 ↪bottom=df_factcheck['supported_claims'],
                label='Sin respaldo', color='coral')
    axes[0].set_xlabel('Pregunta')
    axes[0].set_ylabel('Número de Claims')
    axes[0].set_title('Claims por Pregunta')
    axes[0].legend()

    # Distribución de tasa de alucinación
    axes[1].hist(df_factcheck['hallucination_rate'].dropna(), bins=10,␣
 ↪color='steelblue', edgecolor='white')
    axes[1].axvline(x=avg_hallucination, color='red', linestyle='--',␣
 ↪label=f'Promedio ({avg_hallucination:.1%})')
    axes[1].set_xlabel('Tasa de Alucinación')
    axes[1].set_ylabel('Frecuencia')
    axes[1].set_title('Distribución de Tasa de Alucinación')
    axes[1].legend()

    plt.tight_layout()
    plt.savefig('../docs/images/factcheck_analysis.png', dpi=150,␣
 ↪bbox_inches='tight')
    plt.show()
else:
    print("  No hay datos de fact-checking para analizar")
```

==========

```
ANÁLISIS DE FACT-CHECKING
============================================================

Muestra evaluada: 25 respuestas
Total claims analizados: 135
Claims promedio por respuesta: 5.4

Claims respaldados: 102 (75.6%)
Claims sin respaldo: 33 (24.4%)

========================================
TASA DE ALUCINACIÓN PROMEDIO: 24.3%
========================================

  Bueno: Entre 10-25% de alucinación
```



1.8 7. Resumen Final

[28]:
```python
# Resumen final de todas las métricas
print("=" * 70)
print("RESUMEN DE EVALUACIÓN DEL SISTEMA RAG")
print("=" * 70)
print(f"\nDataset: {len(EVALUATION_QUESTIONS)} preguntas de prueba")
print(f"Base de datos: {total_articles} artículos")
print()

# Retrieval (independiente del LLM)
print("MÉTRICAS DE RETRIEVAL")
print("-" * 50)
print(f"  Precision@5:        {df_retrieval['precision_at_k'].mean():.2%}")
print(f"  Hit Rate:           {df_retrieval['hit_rate'].mean():.2%}")
print(f"  MRR:                {df_retrieval['mrr'].mean():.3f}")
print(f"  Avg Similarity:     {df_retrieval['avg_similarity'].mean():.3f}")
print()
```

```python
# Generación - Comparación LM Studio vs DeepSeek
print("MÉTRICAS DE GENERACIÓN (LM Studio vs DeepSeek)")
print("-" * 50)
print(f"{'Métrica':<25} {'LM Studio':<15} {'DeepSeek':<15}")
print("-" * 50)

# Confianza
lm_conf = df_generation['confidence'].mean()
ds_conf = df_deepseek_gen['confidence'].mean() if 'df_deepseek_gen' in dir()␣
↪else None
lm_conf_str = f"{lm_conf:.1%}"
ds_conf_str = f"{ds_conf:.1%}" if ds_conf else "N/A"
print(f"{'Confianza':<25} {lm_conf_str:<15} {ds_conf_str:<15}")

# Cobertura temas
lm_topic = df_generation['topic_coverage'].mean()
ds_topic = df_deepseek_gen['topic_coverage'].mean() if 'df_deepseek_gen' in␣
↪dir() else None
lm_topic_str = f"{lm_topic:.1%}"
ds_topic_str = f"{ds_topic:.1%}" if ds_topic else "N/A"
print(f"{'Cobertura Temas':<25} {lm_topic_str:<15} {ds_topic_str:<15}")

# Citas usadas
lm_citas = df_generation['citations_in_answer'].mean()
ds_citas = df_deepseek_gen['citations'].mean() if 'df_deepseek_gen' in dir()␣
↪else None
lm_citas_str = f"{lm_citas:.1f}"
ds_citas_str = f"{ds_citas:.1f}" if ds_citas else "N/A"
print(f"{'Citas Usadas':<25} {lm_citas_str:<15} {ds_citas_str:<15}")

# Faithfulness (LLM-as-Judge)
lm_faith = df_generation['faithfulness'].mean() if␣
↪df_generation['faithfulness'].notna().any() else None
ds_faith = df_deepseek_gen['faithfulness'].mean() if 'df_deepseek_gen' in dir()␣
↪and 'faithfulness' in df_deepseek_gen.columns else None
lm_faith_str = f"{lm_faith:.1%}" if lm_faith else "N/A"
ds_faith_str = f"{ds_faith:.1%}" if ds_faith else "N/A"
print(f"{'Faithfulness (Judge)':<25} {lm_faith_str:<15} {ds_faith_str:<15}")

print()
print("COMPARACIÓN RERANKING (LM Studio)")
print("-" * 50)
print(f"  Sin Reranking:      {df_generation['confidence'].mean():.2%}␣
↪confianza")
```

```
print(f"  Con Reranking:      {df_generation_rerank['confidence'].mean():.2%}␣
 ↪confianza")
print(f"  Mejora:             +{(df_generation_rerank['confidence'].mean() -␣
 ↪df_generation['confidence'].mean()):.2%}")

# Fact-Checking (si está disponible)
if 'df_factcheck' in dir() and len(df_factcheck) > 0 and␣
 ↪df_factcheck['hallucination_rate'].notna().any():
    print()
    print("FACT-CHECKING (Verificación por Claims)")
    print("-" * 50)
    avg_hallucination = df_factcheck['hallucination_rate'].mean()
    total_claims = df_factcheck['total_claims'].sum()
    total_supported = df_factcheck['supported_claims'].sum()
    print(f"  Muestra evaluada:   {len(df_factcheck)} respuestas")
    print(f"  Total claims:       {total_claims}")
    print(f"  Claims respaldados: {total_supported} ({total_supported/
 ↪total_claims:.1%})")
    print(f"  Tasa alucinación:   {avg_hallucination:.1%}")

print()
print("=" * 70)
# Conclusión
if ds_faith and lm_faith:
    if ds_faith > lm_faith + 0.1:
        print("  DeepSeek genera respuestas más fieles a las fuentes")
        print(f"  (Faithfulness: {ds_faith:.1%} vs {lm_faith:.1%}, diferencia␣
 ↪de {ds_faith - lm_faith:.1%})")
    elif lm_faith > ds_faith + 0.1:
        print("  LM Studio genera respuestas más fieles a las fuentes")
    else:
        print("  Ambos modelos tienen faithfulness similar")
print("=" * 70)
```

```
======================================================================
RESUMEN DE EVALUACIÓN DEL SISTEMA RAG
======================================================================

Dataset: 25 preguntas de prueba
Base de datos: 1688 artículos

MÉTRICAS DE RETRIEVAL
--------------------------------------------------
  Precision@5:        100.00%
  Hit Rate:           100.00%
  MRR:                1.000
  Avg Similarity:     0.702
```

```
MÉTRICAS DE GENERACIÓN (LM Studio vs DeepSeek)
--------------------------------------------------
Métrica                    LM Studio       DeepSeek
--------------------------------------------------
Confianza                  83.7%           85.8%
Cobertura Temas            55.0%           61.0%
Citas Usadas               3.0             4.1
Faithfulness (Judge)       77.9%           75.6%

COMPARACIÓN RERANKING (LM Studio)
--------------------------------------------------
  Sin Reranking:      83.71% confianza
  Con Reranking:      86.43% confianza
  Mejora:             +2.72%

FACT-CHECKING (Verificación por Claims)
--------------------------------------------------
  Muestra evaluada:   25 respuestas
  Total claims:       135
  Claims respaldados: 102 (75.6%)
  Tasa alucinación:   24.3%

======================================================================
 Ambos modelos tienen faithfulness similar
======================================================================
```

[29]:
```
# Guardar resultados para el TFM
results_summary = {
    'timestamp': datetime.now().isoformat(),
    'dataset_size': len(EVALUATION_QUESTIONS),
    'db_articles': total_articles,
    'retrieval': {
        'precision_at_5': float(df_retrieval['precision_at_k'].mean()),
        'hit_rate': float(df_retrieval['hit_rate'].mean()),
        'mrr': float(df_retrieval['mrr'].mean()),
        'avg_similarity': float(df_retrieval['avg_similarity'].mean()),
    },
    'generation': {
        'lm_studio': {
            'confidence': float(df_generation['confidence'].mean()),
            'topic_coverage': float(df_generation['topic_coverage'].mean()),
            'citations_used': float(df_generation['citations_in_answer'].
 ↪mean()),
            'faithfulness': float(df_generation['faithfulness'].mean()) if␣
 ↪df_generation['faithfulness'].notna().any() else None,
        },
```

```
        'deepseek': {
            'confidence': float(df_deepseek_gen['confidence'].mean()) if␣
↪'df_deepseek_gen' in dir() else None,
            'topic_coverage': float(df_deepseek_gen['topic_coverage'].mean())␣
↪if 'df_deepseek_gen' in dir() else None,
            'citations_used': float(df_deepseek_gen['citations'].mean()) if␣
↪'df_deepseek_gen' in dir() else None,
            'faithfulness': float(df_deepseek_gen['faithfulness'].mean()) if␣
↪'df_deepseek_gen' in dir() and 'faithfulness' in df_deepseek_gen.columns␣
↪else None,
        } if 'df_deepseek_gen' in dir() else None,
    },
    'reranking_comparison': {
        'without': float(df_generation['confidence'].mean()),
        'with': float(df_generation_rerank['confidence'].mean()),
        'improvement': float(df_generation_rerank['confidence'].mean() -␣
↪df_generation['confidence'].mean()),
    },
    'fact_checking': None
}

# Agregar fact-checking si está disponible
if 'df_factcheck' in dir() and len(df_factcheck) > 0 and␣
↪df_factcheck['hallucination_rate'].notna().any():
    results_summary['fact_checking'] = {
        'sample_size': len(df_factcheck),
        'total_claims': int(df_factcheck['total_claims'].sum()),
        'supported_claims': int(df_factcheck['supported_claims'].sum()),
        'unsupported_claims': int(df_factcheck['total_claims'].sum() -␣
↪df_factcheck['supported_claims'].sum()),
        'avg_hallucination_rate': float(df_factcheck['hallucination_rate'].
↪mean()),
        'claims_per_response': float(df_factcheck['total_claims'].mean()),
    }

import json
with open('../docs/rag_evaluation_results.json', 'w') as f:
    json.dump(results_summary, f, indent=2, default=str)

print(" Resultados guardados en docs/rag_evaluation_results.json")
print("\nContenido del archivo:")
print(json.dumps(results_summary, indent=2, default=str))
```

```
 Resultados guardados en docs/rag_evaluation_results.json

Contenido del archivo:
{
```

```
  "timestamp": "2026-02-07T19:55:38.825739",
  "dataset_size": 25,
  "db_articles": 1688,
  "retrieval": {
    "precision_at_5": 1.0,
    "hit_rate": 1.0,
    "mrr": 1.0,
    "avg_similarity": 0.7020484250318778
  },
  "generation": {
    "lm_studio": {
      "confidence": 0.8371483089000817,
      "topic_coverage": 0.55,
      "citations_used": 3.04,
      "faithfulness": 0.7791666666666667
    },
    "deepseek": {
      "confidence": 0.8584616161730261,
      "topic_coverage": 0.61,
      "citations_used": 4.08,
      "faithfulness": 0.7559999999999999
    }
  },
  "reranking_comparison": {
    "without": 0.8371483089000817,
    "with": 0.8643300108492386,
    "improvement": 0.02718170194915681
  },
  "fact_checking": {
    "sample_size": 25,
    "total_claims": 135,
    "supported_claims": 102,
    "unsupported_claims": 33,
    "avg_hallucination_rate": 0.24323809523809525,
    "claims_per_response": 5.4
  }
}
```